Structure of Atom

Question1

Given below are two statements :

Statement I : When an electric discharge is passed through gaseous hydrogen, the hydrogen molecules dissociate and the energetically excited hydrogen atoms produce electromagnetic radiation of discrete frequencies.

Statement II : The frequency of second line of Balmer series obtained from He^{+} is equal to that of first line of Lyman series obtained from hydrogen atom.

In the light of the above statements, choose the correct answer from the options given below :

JEE Main 2026 (Online) 21st January Morning Shift

Options:

A.

Both Statement I and Statement II are true

B.

Statement I is true but Statement II is false

C.

Statement I is false but Statement II is true

D.

Both Statement I and Statement II are false

Answer: A

Solution:

Statement I: True.

When electric discharge is passed through gaseous hydrogen, H_2 molecules dissociate into hydrogen atoms. These atoms get excited and, on returning to lower energy levels, emit radiation of **discrete frequencies**, giving a **line spectrum** (as per NCERT description of atomic spectra).

Statement II: True.

Use the Rydberg formula (NCERT) for hydrogen-like species:

$\tilde{\nu} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

(a) Second line of Balmer series of He^+

Balmer series means $n_1 = 2$.

Second line means transition $n_2 = 4 \rightarrow n_1 = 2$.

For He^+, $Z = 2$.

$\tilde{\nu}_{\mathrm{He}^+} = R(2)^2 \left(\frac{1}{2^2} - \frac{1}{4^2}\right) = 4R \left(\frac{1}{4} - \frac{1}{16}\right) = 4R \left(\frac{3}{16}\right) = \frac{3R}{4}$

(b) First line of Lyman series of hydrogen atom

Lyman series means $n_1 = 1$.

First line means transition $n_2 = 2 \rightarrow n_1 = 1$.

For hydrogen, $Z = 1$.

$\tilde{\nu}_H = R(1)^2 \left(\frac{1}{1^2} - \frac{1}{2^2}\right) = R \left(1 - \frac{1}{4}\right) = \frac{3R}{4}$

So, $\tilde{\nu}_{\mathrm{He}^+} = \tilde{\nu}_H$, hence their **frequencies are equal**.

Correct option: A

Both Statement I and Statement II are true.

Question2

Consider the following spectral lines for atomic hydrogen :

A. First line of Paschen series

B. Second line of Balmer series

C. Third line of Paschen series

D. Fourth line of Bracket series

The correct arrangement of the above lines in ascending order of energy is :

JEE Main 2026 (Online) 21st January Evening Shift

Options:

A.

D < C < A < B

B.

A < B < C < D

C.

D < A < C < B

D.

C < D < B < A

Answer: C

Solution:

For hydrogen spectral lines (emission),

$E = \frac{hc}{\lambda} \quad \Rightarrow \quad E \propto \frac{1}{\lambda}$

Using Rydberg formula (NCERT):

$\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \quad (n_2 > n_1)$

So photon energy is proportional to:

$\Delta = \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

Now identify each line:

- **Paschen series**: $n_1 = 3$

- **A (first line)**: $n_2 = 4 \rightarrow 3$
- **C (third line)**: $n_2 = 6 \rightarrow 3$
- **Balmer series**: $n_1 = 2$
- **B (second line)**: $n_2 = 4 \rightarrow 2$
- **Brackett series**: $n_1 = 4$
- **D (fourth line)**: $n_2 = 8 \rightarrow 4$

Compute Δ for each:

D: $8 \rightarrow 4$

$\Delta_D = \frac{1}{4^2} - \frac{1}{8^2} = \frac{1}{16} - \frac{1}{64} = \frac{3}{64} = 0.046875$

A: $4 \rightarrow 3$

$\Delta_A = \frac{1}{3^2} - \frac{1}{4^2} = \frac{1}{9} - \frac{1}{16} = \frac{7}{144} \approx 0.04861$

C: $6 \rightarrow 3$

$\Delta_C = \frac{1}{3^2} - \frac{1}{6^2} = \frac{1}{9} - \frac{1}{36} = \frac{1}{12} \approx 0.08333$

B: $4 \rightarrow 2$

$\Delta_B = \frac{1}{2^2} - \frac{1}{4^2} = \frac{1}{4} - \frac{1}{16} = \frac{3}{16} = 0.1875$

Ascending order of energy (smallest Δ to largest Δ):

$D < A < C < B$

Correct option: C

Question3

The energy required by electrons, present in the first Bohr orbit of hydrogen atom to be excited to second Bohr orbit is ____ $\mathrm{Jmol^{-1}}$.

Given: $R_H = 2.18 \times 10^{-11}$ergs.

JEE Main 2026 (Online) 22nd January Morning Shift

Options:

A.

9.835×10^{12}

B.

1.635×10^{-11}

C.

1.635×10^{-18}

D.

9.835×10^{5}

Answer: D

Solution:

$$E_n = -R_H \times \frac{Z^2}{n^2}$$
$$\Delta E = 2.18 \times 10^{-11} \times 10^{-7} \times 1^2 \left[\frac{1}{1^2} - \frac{1}{2^2}\right]$$
$$= 1.635 \times 10^{-18} \text{Joule/ atom}$$
$$= 1.635 \times 10^{-18} \times 6.02 \times 10^{23} \text{ Joule / mole}$$
$$= 9.835 \times 10^{5} \text{ Joule / mole}$$

Question4

The energy of first (lowest) Balmer line of H atom is x J. The energy (in J) of second Balmer line of H atom is :

JEE Main 2026 (Online) 22nd January Evening Shift

Options:

A.

$\frac{x}{1.35}$

B.

$1.35x$

C.

x^2

D.

$2x$

Answer: B

Solution:

Balmer series means transitions **ending at** $n = 2$.

Photon energy emitted in a transition $n \to 2$ is

$E = 13.6\,\mathrm{eV}\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$

First (lowest) Balmer line

This is the **first line of Balmer series**: $3 \to 2$

$E_1 = 13.6\left(\frac{1}{4} - \frac{1}{9}\right) = 13.6\left(\frac{5}{36}\right)$

Given $E_1 = x$ J.

Second Balmer line

This is $4 \to 2$

$E_2 = 13.6\left(\frac{1}{4} - \frac{1}{16}\right) = 13.6\left(\frac{3}{16}\right)$

Ratio

$\frac{E_2}{E_1} = \frac{\left(\frac{3}{16}\right)}{\left(\frac{5}{36}\right)} = \frac{3}{16} \cdot \frac{36}{5} = \frac{108}{80} = 1.35$

So,

$E_2 = 1.35\,x$

Correct option: B) $1.35x$

Question5

Which of the following statements regarding the energy of the stationary state is true in the following one - electron systems ?

JEE Main 2026 (Online) 23rd January Morning Shift

Options:

A.

$+2.18 \times 10^{-18}$ J for second orbit of He^{+}ion

B.

$+8.72 \times 10^{-18}$ J for first orbit of He^{+}ion

C.

-1.09×10^{-18} J for second orbit of H atom.

D.

-2.18×10^{-18} J for third orbit of Li^{2+} ion

Answer: D

Solution:

For one–electron (hydrogen-like) systems, the energy of the n^{th} stationary state is (NCERT):

$E_n = -2.18 \times 10^{-18} \frac{Z^2}{n^2}$ J

where Z is the atomic number and n is the orbit number.

Check each option

Option A: Second orbit of He^{+}

Here $Z = 2,\ n = 2$

$E_2 = -2.18 \times 10^{-18} \frac{2^2}{2^2} = -2.18 \times 10^{-18}$ J

Given is $+2.18 \times 10^{-18}$ **J** (positive), so **false**.

Option B: First orbit of He^{+}

$Z = 2,\ n = 1$

$E_1 = -2.18 \times 10^{-18} \frac{4}{1} = -8.72 \times 10^{-18}$ J

Given is $+8.72 \times 10^{-18}$ **J**, so **false**.

Option C: Second orbit of H atom

$Z = 1,\ n = 2$

$E_2 = -2.18 \times 10^{-18} \frac{1}{4} = -0.545 \times 10^{-18}$ J

Given is -1.09×10^{-18} **J**, so **false**.

Option D: Third orbit of Li^{2+}

$Z = 3,\ n = 3$

$E_3 = -2.18 \times 10^{-18} \frac{9}{9} = -2.18 \times 10^{-18}$ J

This matches the given value, so **true**.

Correct option: D

Question6

Given,

(A) $\mathrm{n} = 5,\ \mathrm{m}_1 = -1$

(B) $\mathrm{n} = 3, \mathrm{l} = 2,\ \mathrm{m}_1 = -1,\ \mathrm{m}_\mathrm{s} = +\frac{1}{2}$

The maximum number of electron(s) in an atom that can have the quantum numbers as given in (A) and (B) respectively are :

JEE Main 2026 (Online) 23rd January Morning Shift

Options:

A.

4 and 1

B.

2 and 4

C.

26 and 1

D.

8 and 1

Answer: D

Solution:

(A) $n = 5$. So we look at all subshells for this value of n, i.e. $l = 0, 1, 2, 3, 4$, and check where the given magnetic quantum number $m_l = -1$ is possible.

For $l = 0$ $m_l = 0$, i.e. only one orbital is possible and it has $m_l = 0$, so here we do not get any orbital with $m_l = -1$.

For $l = 1$ $m_l = -1, 0, 1 \Rightarrow$ 2 electrons can have $m_l = -1$ in this subshell, because there is exactly one orbital with $m_l = -1$, and one orbital can hold 2 electrons (with opposite spins).

For $l = 2$ $m_l = -2, -1, 0, 1, 2 \Rightarrow$ 2 electrons can again have $m_l = -1$, because here also there is one orbital with $m_l = -1$, which can contain 2 electrons.

For $l = 3$ $m_l = -3, -2, -1, 0, 1, 2, 3 \Rightarrow$ 2 electrons can have $m_l = -1$ for the same reason: one orbital with this m_l value, accommodating 2 electrons.

For $l = 4$ $m_l = -4, -3, -2, 1, 0, 1, 2, 3, 4 \Rightarrow$ 2 electrons can have the given magnetic quantum number in this subshell as well (one such orbital $\Rightarrow$ 2 electrons).

Total number of electrons $= 8$, because there are 4 such orbitals (for $l = 1, 2, 3, 4$), each holding 2 electrons.

(B) $n = 3, l = 2,\ m_l = -1,\ m_s = +1/2 \Rightarrow$ only 1 electron is possible, because all four quantum numbers are completely specified. By Pauli's exclusion principle, no two electrons in an atom can have the same set of four quantum numbers, so at most one electron can have this exact combination.

Question7

Identify the INCORRECT statements from the following :

A. Notation ${}^{24}_{12}\mathrm{Mg}$ represents 24 protons and 12 neutrons.

B. Wavelength of a radiation of frequency $4.5 \times 10^{15}\ \mathrm{s}^{-1}$ is 6.7×10^{-8} m.

C. One radiation has wavelength $= \lambda_1(900\ \mathrm{nm})$ and energy $= \mathrm{E}_1$. Other radiation has wavelength $= \lambda_2(300\ \mathrm{nm})$ and energy $= \mathrm{E}_2 \cdot \mathrm{E}_1 : \mathrm{E}_2 = 3 : 1$.

D. Number of photons of light of wavelength 2000 pm that provides 1 J of energy is 1.006×10^{16}.

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 23rd January Evening Shift

Options:

A.

A and D Only

B.

A and C Only

C.

A and B Only

D.

B and C Only

Answer: B

Solution:

Statement A

For $^{24}_{12}\mathrm{Mg}$:

- Atomic number $Z = 12 \Rightarrow$ number of protons $= 12$
- Mass number $A = 24 \Rightarrow$ protons + neutrons $= 24$

So, neutrons $= 24 - 12 = 12$.

Given statement says **24 protons and 12 neutrons**, which is wrong.

✅ **A is incorrect**

Statement B

Wavelength λ is given by

$\lambda = \frac{c}{\nu}$

Here $c = 3 \times 10^8 \text{ m s}^{-1}$ and $\nu = 4.5 \times 10^{15} \text{ s}^{-1}$

$\lambda = \frac{3\times10^8}{4.5\times10^{15}} = 0.6667 \times 10^{-7} = 6.67 \times 10^{-8} \text{ m}$

Matches the given value.

✅ **B is correct**

Statement C

Energy of radiation:

$E = \frac{hc}{\lambda} \Rightarrow E \propto \frac{1}{\lambda}$

So,

$\frac{E_1}{E_2} = \frac{\lambda_2}{\lambda_1} = \frac{300}{900} = \frac{1}{3}$

Therefore,

$E_1 : E_2 = 1 : 3$

But given is $E_1 : E_2 = 3 : 1$, which is wrong.

✅ **C is incorrect**

Statement D

Energy per photon:

$E = \frac{hc}{\lambda}$

Given $\lambda = 2000 \text{ pm} = 2000 \times 10^{-12} = 2 \times 10^{-9} \text{ m}$

$E = \frac{(6.626\times10^{-34})(3\times10^{8})}{2\times10^{-9}} = 9.939 \times 10^{-17} \text{ J}$

Number of photons for 1 J:

$N = \frac{1}{9.939\times10^{-17}} = 1.006 \times 10^{16}$

Matches the given value.

✅ **D is correct**

Incorrect statements: A and C only

Correct option: Option B

Question8

The work functions of two metals $(\mathrm{M_A}$ and $\mathrm{M_B})$ are in the $1 : 2$ ratio. When these metals are exposed to photons of energy 6 eV , the kinetic energy of liberated electrons of $M_A : M_B$ is in the ratio of $2.642 : 1$. The work functions (in eV) of M_A and M_B are respectively.

JEE Main 2026 (Online) 23rd January Evening Shift

Options:

A.

$3.1, 6.2$

B.

$1.5, 3.0$

C.

$2.3, 4.6$

D.

$1.4, 2.8$

Answer: C

Solution:

Let the work functions be ϕ_A and ϕ_B.

Given ratio of work functions:

$\phi_A : \phi_B = 1 : 2 \;\Rightarrow\; \phi_B = 2\phi_A$

Photon energy is $h\nu = 6\,\text{eV}$.

By Einstein's photoelectric equation:

$K_{\text{max}} = h\nu - \phi$

So,

$K_A = 6 - \phi_A$

$K_B = 6 - \phi_B = 6 - 2\phi_A$

Given,

$\frac{K_A}{K_B} = \frac{2.642}{1} = 2.642$

So,

$\frac{6-\phi_A}{6-2\phi_A} = 2.642$

Cross-multiplying:

$6 - \phi_A = 2.642(6 - 2\phi_A)$

$6 - \phi_A = 15.852 - 5.284\phi_A$

Bring like terms together:

$5.284\phi_A - \phi_A = 15.852 - 6$

$4.284\phi_A = 9.852$

$\phi_A = \frac{9.852}{4.284} \approx 2.3\,\text{eV}$

Then,

$\phi_B = 2\phi_A \approx 4.6\,\text{eV}$

Answer: $\boxed{2.3\,\text{eV},\ 4.6\,\text{eV}}$ **(Option C)**

Question9

The wavelength of spectral line obtained in the spectrum of Li^{2+} ion, when the transition takes place between two levels whose sum is 4 and difference is 2 , is

JEE Main 2026 (Online) 24th January Evening Shift

Options:

A.

1.14×10^{-6} cm

B.

2.28×10^{-6} cm

C.

1.14×10^{-7} cm

D.

2.28×10^{-7} cm

Answer: A

Solution:

$n_1 \rightarrow$ lower energy level

$n_2 \rightarrow$ higher energy level

$$n_1 + n_2 = 4, \quad n_2 = 3$$
$$n_2 - n_1 = 2, \quad n_1 = 1$$

Rydberg's formula :

$$\frac{1}{\lambda} = R_H Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$
$$\frac{1}{\lambda} = R_H (3)^2 \left[\frac{1}{1^2} - \frac{1}{3^2}\right]$$
$$\frac{1}{\lambda} = 8R_H$$
$$\lambda = \frac{1}{8R_H}$$
$$\lambda = \frac{1}{8 \times 1.1 \times 10^5}$$
$$\lambda = \frac{1000}{8.8} \times 10^{-8} \text{ cm}$$
$$\lambda = 113.63 \times 10^{-8} \text{ cm}$$
$$\lambda \simeq 1.14 \times 10^{-6} \text{ cm}$$

Question10

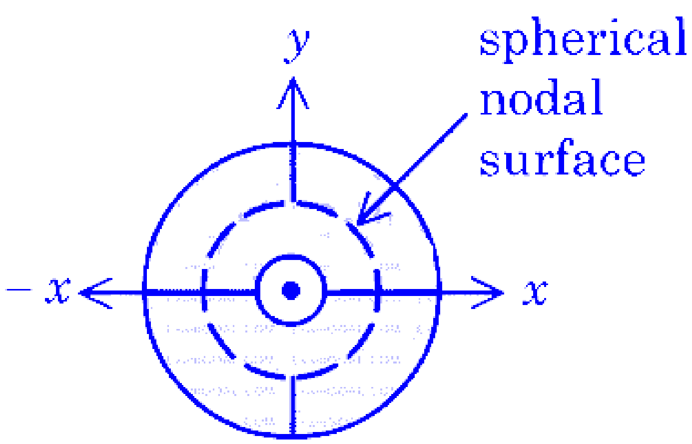


Figure 1. electron probability density for 2 s orbital

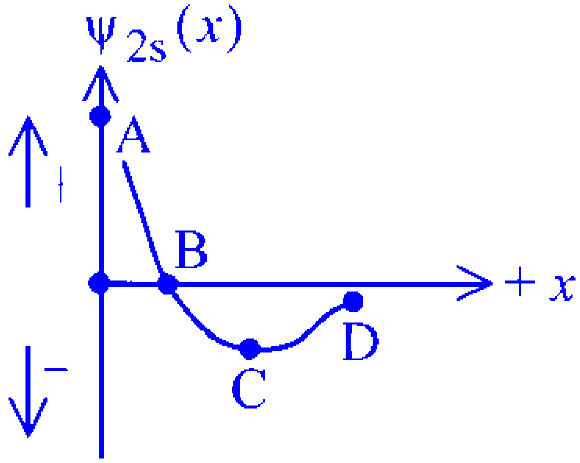


Figure 2. wave function for 2s orbital

Which of the following point in Figure 2 most accurately represents the nodal surface as shown in Figure 1?

JEE Main 2026 (Online) 28th January Morning Shift

Options:

A.

D

B.

B

C.

C

D.

A

Answer: B

Solution:

The nodal surface (as seen in Figure 1) is the place where the probability density becomes zero.

For an s-orbital, this nodal surface is a **spherical node** (a spherical shell around the nucleus).

At a spherical (radial) node, the radial wave function becomes zero.

$\psi_{\mathrm{r}} = 0$

So, in Figure 2, the correct point is where the 2s wave function curve **cuts the horizontal axis** (changes sign). That point represents the spherical node shown in Figure 1.

Question11

The wave numbers of three spectral lines of H atom are considered. Identify the set of spectral lines belonging to Balmer series.

($\mathrm{R}=$ Rydberg constant)

JEE Main 2026 (Online) 28th January Morning Shift

Options:

A.

$\frac{5\mathrm{R}}{36}, \frac{3\mathrm{R}}{16}, \frac{21\mathrm{R}}{100}$

B.

$\frac{7\mathrm{R}}{144}, \frac{3\mathrm{R}}{16}, \frac{16\mathrm{R}}{255}$

C.

$\frac{3\mathrm{R}}{4}, \frac{3\mathrm{R}}{16}, \frac{7\mathrm{R}}{144}$

D.

$\frac{5\mathrm{R}}{36}, \frac{8\mathrm{R}}{9}, \frac{15\mathrm{R}}{16}$

Answer: A

Solution:

Balmer series corresponds to transitions **ending at** $n_1 = 2$.

For hydrogen, the wave number is given by

$\bar{\nu} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right), \quad n_2 > n_1$

For Balmer series: $n_1 = 2$, so

$\bar{\nu} = R\left(\frac{1}{4} - \frac{1}{n_2^2}\right), \quad n_2 = 3, 4, 5, \ldots$

Now compute the first few Balmer lines:

1. For $n_2 = 3$:

$\bar{\nu} = R\left(\frac{1}{4} - \frac{1}{9}\right) = R\left(\frac{9-4}{36}\right) = \frac{5R}{36}$

2. For $n_2 = 4$:

$\bar{\nu} = R\left(\frac{1}{4} - \frac{1}{16}\right) = R\left(\frac{4-1}{16}\right) = \frac{3R}{16}$

3. For $n_2 = 5$:

$\bar{\nu} = R\left(\frac{1}{4} - \frac{1}{25}\right) = R\left(\frac{25-4}{100}\right) = \frac{21R}{100}$

These three values $\left(\frac{5R}{36}, \frac{3R}{16}, \frac{21R}{100}\right)$ match **Option A**.

Correct option: A

Question12

The wavelength of photon 'A' is 400 nm. The frequency of photon 'B' is $10^{16}\ \mathrm{s}^{-1}$. The wave number of photon 'C' is $10^4\ \mathrm{cm}^{-1}$. The correct order of energy of these photons is :

JEE Main 2026 (Online) 28th January Evening Shift

Options:

A.

A > C > B

B.

C > B > A

C.

B > A > C

D.

A > B > C

Answer: C

Solution:

(1) Wavelength of $\mathrm{A}=400 \mathrm{~nm}$.

(2) For photon $\mathrm{B}, \lambda=\frac{\mathrm{C}}{v}=\frac{3 \times 10^8}{10^{16}}$

$=3 \times 10^{-8} \mathrm{~m}=30 \times 10^{-9} \mathrm{~m}=30 \mathrm{~nm}$.

(3) **For photon** C**, wave number is given, and** $\bar{v}=\frac{1}{\lambda}$**. So**
$\mathrm{C}(\lambda)=\frac{1}{\bar{v}}=\frac{1}{10^4}=10^{-4} \mathrm{~cm}=10^{-6} \mathrm{~m}=1000 \mathrm{~nm}$

Now compare the wavelengths: $\lambda_{\mathrm{C}}>\lambda_{\mathrm{A}}>\lambda_{\mathrm{B}}$

Energy of a photon is inversely proportional to wavelength: $\text{Energy}(\mathrm{E}) \propto \frac{1}{\lambda}$

So, smaller wavelength means higher energy. Therefore: $\mathrm{E}_{\mathrm{B}}>\mathrm{E}_{\mathrm{A}}>\mathrm{E}_{\mathrm{C}}$

Question13

What is the energy (in J atom^{-1}) required for the following process?

Li^{2+} (g) → Li^{3+} (g) + e^-

(Take the ionization energy for the H atom in the ground state as 2.18×10^{-18} J atom $^{-1}$)

JEE Main 2026 (Online) 2nd April Morning Shift

Options:

A.

8.72×10^{-18}

B.

1.962×10^{-18}

C.

1.962×10^{-17}

D.

6.54×10^{-17}

Answer: C

Solution:

For the process

$\mathrm{Li}^{2+}(g) \rightarrow \mathrm{Li}^{3+}(g) + e^{-}$

we have to remove the only electron present in Li^{2+}.

Li^{2+} is a hydrogen-like species because it has only one electron.

For a hydrogen-like species, ionization energy is given by

$E = Z^2 \times 2.18 \times 10^{-18}\ \mathrm{J\ atom}^{-1}$

where Z is the atomic number.

For lithium, $Z = 3$.

So,

$E = 3^2 \times 2.18 \times 10^{-18}$

$E = 9 \times 2.18 \times 10^{-18}$

$E = 19.62 \times 10^{-18}$

$E = 1.962 \times 10^{-17}\ \mathrm{J\ atom}^{-1}$

Hence, the required energy is

$\boxed{1.962 \times 10^{-17}\ \mathrm{J\ atom}^{-1}}$

So, the correct option is:

$\boxed{\text{Option C}}$

Question14

Which of the following is correct set of 4 quantum numbers of 19^{th} electron in Chromium (Atomic number $= 24$) in accordance with Aufbau principle?

JEE Main 2026 (Online) 4th April Morning Shift

Options:

A.

$n = 3, l = 2, m = +2, s = +\frac{1}{2}$

B.

$n = 3, l = 2, m = -2, s = +\frac{1}{2}$

C.

$n = 4, l = 1, m = 0, s = +\frac{1}{2}$

D.

$n = 4, l = 0, m = 0, s = +\frac{1}{2}$

Answer: D

Solution:

Using the Aufbau principle, we fill electrons in the order:

$1s,\ 2s,\ 2p,\ 3s,\ 3p,\ 4s,\ 3d, \ldots$

Now let us count electrons one by one up to the 19^{th} electron.

- $1s^2 \Rightarrow 2$ electrons
- $2s^2 \Rightarrow 4$ electrons
- $2p^6 \Rightarrow 10$ electrons
- $3s^2 \Rightarrow 12$ electrons
- $3p^6 \Rightarrow 18$ electrons

So, the 19^{th} electron enters the $4s$ orbital.

For a $4s$ electron:

- $n = 4$
- for s-orbital, $l = 0$
- when $l = 0$, only possible value of m is 0

- for the first electron in this orbital, $s = +\dfrac{1}{2}$

Thus, the set of four quantum numbers is:

$n = 4,\ l = 0,\ m = 0,\ s = +\frac{1}{2}$

So, the correct option is:

$\boxed{\text{Option D}}$

Note: Chromium actually has exceptional configuration $[Ar]\, 3d^5 4s^1$, but the question clearly says "in accordance with Aufbau principle", so we must place the 19^{th} electron in $4s$.

Question15

What is the ratio of wave number of first line (lowest energy line) of Balmer series of H atomic spectrum to first line of its Brackett series?

JEE Main 2026 (Online) 4th April Morning Shift

Options:

A.

5 : 1

B.

5 : 0.81

C.

5 : 1.75

D.

5 : 27

Answer: B

Solution:

For hydrogen spectrum, the wave number is given by the Rydberg formula:

$\bar{\nu} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

where:

- n_1 = lower energy level
- n_2 = higher energy level
- $n_2 > n_1$

Now let us find the first line of each series.

1. First line of Balmer series

For Balmer series:

$n_1 = 2$

The first line means transition from $n_2 = 3$ to $n_1 = 2$.

So,

$\bar{\nu}_B = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = R\left(\frac{1}{4} - \frac{1}{9}\right)$

$\bar{\nu}_B = R\left(\frac{9-4}{36}\right) = \frac{5R}{36}$

2. First line of Brackett series

For Brackett series:

$n_1 = 4$

The first line means transition from $n_2 = 5$ to $n_1 = 4$.

So,

$\bar{\nu}_{Br} = R\left(\frac{1}{4^2} - \frac{1}{5^2}\right) = R\left(\frac{1}{16} - \frac{1}{25}\right)$

$\bar{\nu}_{Br} = R\left(\frac{25-16}{400}\right) = \frac{9R}{400}$

3. Required ratio

$\bar{\nu}_B : \bar{\nu}_{Br} = \frac{5R}{36} : \frac{9R}{400}$

Cancel R:

$= \frac{5}{36} : \frac{9}{400}$

$= \frac{5}{36} \times \frac{400}{9}$

So ratio:

$\bar{\nu}_B : \bar{\nu}_{Br} = \frac{2000}{324} = \frac{500}{81}$

Now,

$\frac{500}{81} = \frac{5 \times 100}{81}$

and

$\frac{100}{81} \approx 1.23$

So,

$\bar{\nu}_B : \bar{\nu}_{Br} = 5 : 0.81$

Therefore, the correct answer is:

$\boxed{\text{Option B } 5 : 0.81}$

Question16

The species having identical radii according to the Bohr's theory are :

A. H (first orbit)

B. He^{+}(first orbit)

C. He^{+}(Second orbit)

D. Li^{2+} (first orbit)

E. Be^{3+} (Second orbit)

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 4th April Evening Shift

Options:

A.

A and C Only

B.

A and E Only

C.

B and E Only

D.

C and D Only

Answer: B

Solution:

For a hydrogen-like species, according to Bohr's theory, the radius of the n^{th} orbit is

$r_n = \frac{n^2 a_0}{Z}$

where:

- n = orbit number
- Z = atomic number
- a_0 = Bohr radius

So, for identical radii, the value of $\frac{n^2}{Z}$ must be same.

Now check each species:

A. H (first orbit)

For hydrogen, $Z = 1, n = 1$

$r = \frac{1^2 a_0}{1} = a_0$

B. He^+ (first orbit)

Here, $Z = 2, n = 1$

$r = \frac{1^2 a_0}{2} = \frac{a_0}{2}$

C. He^+ (second orbit)

Here, $Z = 2, n = 2$

$r = \frac{2^2 a_0}{2} = \frac{4a_0}{2} = 2a_0$

D. Li^{2+} (first orbit)

Here, $Z = 3, n = 1$

$r = \frac{1^2 a_0}{3} = \frac{a_0}{3}$

E. Be^{3+} (second orbit)

Here, $Z = 4, n = 2$

$r = \frac{2^2 a_0}{4} = \frac{4a_0}{4} = a_0$

Now compare:

- A: a_0
- B: $\frac{a_0}{2}$
- C: $2a_0$
- D: $\frac{a_0}{3}$
- E: a_0

So, the identical radii are for **A and E**.

Therefore, the correct answer is:

$\boxed{\text{Option B: A and E Only}}$

Question17

Identify the correct statements from the following :

A. Heisenberg uncertainty principle is applicable to electrons.

B. The size of $2p_x$ orbital is less than the size of $3p_x$ orbital.

C. The energy of 2 s orbital of H atom is equal to the energy of 2 s orbital of Li .

D. The electronic configuration of Cr is $[\mathrm{Ar}] 3\ \mathrm{d}^5 4\ \mathrm{s}^1$

Choose the correct answer from the options given below:

JEE Main 2026 (Online) 5th April Morning Shift

Options:

A.

A, B and C Only

B.

A, B and D Only

C.

B, C and D Only

D.

A, C and D Only

Answer: B

Solution:

Statement A

Heisenberg uncertainty principle is applicable to electrons.

This statement is **correct**.

According to Heisenberg uncertainty principle, it is impossible to determine simultaneously the exact position and exact momentum of a microscopic particle like an electron.

So, **A is true**.

Statement B

The size of $2p_x$ orbital is less than the size of $3p_x$ orbital.

This statement is **correct**.

As the principal quantum number n increases, the size of the orbital increases.

Here,

- $2p_x$ has $n = 2$
- $3p_x$ has $n = 3$

Therefore, size of $3p_x$ orbital is greater than that of $2p_x$ orbital.

So, **B is true**.

Statement C

The energy of $2s$ orbital of H atom is equal to the energy of $2s$ orbital of Li.

This statement is **incorrect**.

For hydrogen atom, energy depends only on principal quantum number n. But for multi-electron atoms like lithium, the energy depends on both n and l due to shielding and penetration effects.

So, the energy of $2s$ orbital in H and Li is **not equal**.

Thus, **C is false**.

Statement D

The electronic configuration of Cr is $[\mathrm{Ar}]\,3d^5\,4s^1$

This statement is **correct**.

Chromium is an exception to the expected configuration. Due to extra stability of half-filled $3d$ subshell, its configuration is:

$[\mathrm{Ar}]\,3d^5\,4s^1$

So, **D is true**.

Final Answer

Correct statements are **A, B and D only**.

Therefore, the correct option is:

$\boxed{\text{Option B}}$

Question18

Arrange the following atomic orbitals of multi electron atoms in order of increasing energy.

A. $n = 3, l = 2, m = +1$

B. $n = 4, l = 0, m = 0$

C. $n = 6, l = 1, m = 0$

D. $n = 5, l = 1, m = +1$

E. $n = 2, l = 1, m = +1$

Choose the correct answer from the options given below:

JEE Main 2026 (Online) 5th April Morning Shift

Options:

A.

$\mathrm{C} < \mathrm{D} < \mathrm{B} < \mathrm{A} < \mathrm{E}$

B.

$\mathrm{B} < \mathrm{A} < \mathrm{E} < \mathrm{C} < \mathrm{D}$

C.

$\mathrm{E} < \mathrm{C} < \mathrm{D} < \mathrm{B} < \mathrm{A}$

D.

$\mathrm{E} < \mathrm{B} < \mathrm{A} < \mathrm{D} < \mathrm{C}$

Answer: D

Solution:

For multi-electron atoms, the energy of an orbital is decided by the $n + l$ rule.

- The orbital with smaller value of $n + l$ has lower energy.
- If two orbitals have the same value of $n + l$, then the orbital with smaller n has lower energy.

Also, energy does not depend on m.

Now let us identify each orbital:

- A: $n = 3,\ l = 2$

  This is $3d$

  $n + l = 3 + 2 = 5$

- B: $n = 4,\ l = 0$

  This is $4s$

  $n + l = 4 + 0 = 4$

- C: $n = 6,\ l = 1$

  This is $6p$

  $n + l = 6 + 1 = 7$

- D: $n = 5,\ l = 1$

  This is $5p$

  $n + l = 5 + 1 = 6$

- E: $n = 2,\ l = 1$

This is $2p$

$n + l = 2 + 1 = 3$

Now arrange in increasing energy:

$2p < 4s < 3d < 5p < 6p$

So,

E < B < A < D < C

Hence, the correct answer is:

$\boxed{\text{Option D}}$

Question19

Which of the following statement(s) is/are true ?

A. If two orbitals have the same value of ($\mathrm{n} + l$), the orbital with lower value of n will have lower energy.

B. Energies of the orbitals in the same subshell increase with increase in atomic number.

C. The size of $2\mathrm{p}_x$ orbital is less than the size of $3\mathrm{p}_x$ orbital.

D. Among 5f, 6s, 4d, 5p and 5d orbitals, none of the orbitals have 2 radial nodes.

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 5th April Evening Shift

Options:

A.

A, B and C only

B.

A and C only

C.

C and D only

D.

A only

Answer: B

Solution:

Statement A: If two orbitals have the same value of $(n+l)$, the orbital with lower value of n will have lower energy.

According to the Bohr-Bury rule (or the $(n+l)$ rule) for the filling of electrons in multi-electron atoms:

- Orbitals with a lower value of $(n+l)$ have lower energy.
- If two orbitals have the same $(n+l)$ value, the orbital with the lower principal quantum number (n) will have the lower energy.

For example, for the 3d orbital, $(n+l)=3+2=5$. For the 4p orbital, $(n+l)=4+1=5$. Since 3d has a lower value of n $(n=3)$ compared to 4p $(n=4)$, the 3d orbital has lower energy.

Therefore, **Statement A is true**.

Statement B: Energies of the orbitals in the same subshell increase with increase in atomic number.

As the atomic number (Z) increases, the nuclear charge increases. This leads to a stronger electrostatic pull on the electrons by the nucleus, which increases the effective nuclear charge (Z_{eff}). Due to this stronger attraction, the electrons are held more tightly, and the energy of the orbitals becomes more negative (it decreases, not increases). For example, the energy of a 2s orbital follows the order:
$E_{2\text{s}}(\text{H}) > E_{2\text{s}}(\text{Li}) > E_{2\text{s}}(\text{Na})$.

Therefore, **Statement B is false**.

Statement C: The size of 2p_x orbital is less than the size of 3p_x orbital.

The size of an orbital is primarily determined by its principal quantum number (n). As the value of n increases, the distance of the electron from the nucleus increases, making the orbital larger. Since $n=3$ for the 3p_x orbital and $n=2$ for the 2p_x orbital, the 3p_x orbital is larger than the 2p_x orbital.

Therefore, **Statement C is true**.

Statement D: Among 5f, 6s, 4d, 5p and 5d orbitals, none of the orbitals have 2 radial nodes.

The number of radial nodes for any orbital is calculated using the formula: $(n-l-1)$.

Let us calculate the radial nodes for the given orbitals:

- For 5f $(n=5, l=3)$: Radial nodes $=5-3-1=1$
- For 6s $(n=6, l=0)$: Radial nodes $=6-0-1=5$
- For 4d $(n=4, l=2)$: Radial nodes $=4-2-1=1$

- For 5p ($n = 5, l = 1$): Radial nodes $= 5 - 1 - 1 = 3$
- For 5d ($n = 5, l = 2$): Radial nodes $= 5 - 2 - 1 = 2$

As we can see, the 5d orbital indeed has exactly 2 radial nodes. Thus, the statement that none of them has 2 radial nodes is incorrect.

Therefore, **Statement D is false**.

Conclusion:

Only statements A and C are true.

The correct answer is **Option B**.

Question20

Match the LIST-I with LIST-II

List-I Orbital		List-II Radial nodes and nodal plane	
A.	2s	I.	1 Radial node + two nodal planes
B.	3s	II.	1 Radial node + one nodal plane
C.	3p	III.	2 Radial nodes + No nodal plane
D.	4d	IV.	1 Radial node + No nodal plane

Choose the correct answer from the options given below:

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A.

A-IV, B-I, C-III, D-II

B.

A-IV, B-II, C-III, D-I

C.

A-III, B-I, C-IV, D-II

D.

A-IV, B-III, C-II, D-I

Answer: D

Solution:

A node is a region in space where the probability of finding an electron is zero. There are two types of nodes:

1. **Radial Nodes (or Spherical Nodes):** These are spherical surfaces where the probability of finding an electron is zero.
2. **Angular Nodes (or Nodal Planes):** These are planes (or cones for d-orbitals) passing through the nucleus where the probability of finding an electron is zero.

To find the number of these nodes for any given orbital, we use two simple formulas based on the principal quantum number (n) and the azimuthal quantum number (l).

- Number of radial nodes = $n - l - 1$
- Number of angular nodes (nodal planes) = l

Remember, the value of l depends on the subshell:

- For an 's' subshell, $l = 0$
- For a 'p' subshell, $l = 1$
- For a 'd' subshell, $l = 2$
- For an 'f' subshell, $l = 3$

Now, let's analyze each orbital from List-I.

A. 2s orbital

- Here, the principal quantum number $n = 2$.
- Since it is an 's' orbital, the azimuthal quantum number $l = 0$.
- Number of radial nodes = $n - l - 1 = 2 - 0 - 1 = 1$.
- Number of nodal planes = $l = 0$.
- So, the 2s orbital has **1 Radial node + No nodal plane**. This matches with **IV** in List-II.

B. 3s orbital

- Here, $n = 3$.
- For an 's' orbital, $l = 0$.
- Number of radial nodes = $n - l - 1 = 3 - 0 - 1 = 2$.
- Number of nodal planes = $l = 0$.
- So, the 3s orbital has **2 Radial nodes + No nodal plane**. This matches with **III** in List-II.

C. 3p orbital

- Here, $n = 3$.
- For a 'p' orbital, $l = 1$.
- Number of radial nodes = $n - l - 1 = 3 - 1 - 1 = 1$.
- Number of nodal planes = $l = 1$.
- So, the 3p orbital has **1 Radial node + one nodal plane**. This matches with **II** in List-II.

D. 4d orbital

- Here, $n = 4$.
- For a 'd' orbital, $l = 2$.
- Number of radial nodes = $n - l - 1 = 4 - 2 - 1 = 1$.
- Number of nodal planes = $l = 2$.
- So, the 4d orbital has **1 Radial node + two nodal planes**. This matches with **I** in List-II.

Let's summarize our findings:

- A matches with IV
- B matches with III
- C matches with II
- D matches with I

Comparing this with the given options, we find that the correct combination is **A-IV, B-III, C-II, D-I**.

Therefore, the correct answer is **Option D**.

Question21

If shortest wavelength of hydrogen atom in Lyman series is x, then longest wavelength in Balmer series of $\mathrm{He}^{+}$is:

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A.

$\frac{9x}{5}$

B.

$\frac{36x}{5}$

C.

$\frac{x}{4}$

D.

$\frac{5x}{9}$

Answer: A

Solution:

For any hydrogen-like atom, the wavelength of emitted radiation is given by the Rydberg formula:

$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

where

- R is the Rydberg constant,
- Z is the atomic number,
- n_1 and n_2 are integers with $n_2 > n_1$.

For hydrogen atom ($Z = 1$) — Lyman series

For the **Lyman series**, $n_1 = 1$ and $n_2 = 2, 3, 4, \ldots$

- The **shortest wavelength** corresponds to **maximum energy difference**, i.e. $n_2 \to \infty$.

$\frac{1}{x} = R(1)^2 \left(\frac{1}{1^2} - 0 \right) = R$

So,

$x = \frac{1}{R}$

For helium ion, He^+ ($Z = 2$) — Balmer series

For the **Balmer series**, $n_1 = 2$ and $n_2 = 3, 4, 5, \ldots$

- The **longest wavelength** corresponds to **minimum energy difference**, i.e. smallest n_2, which is $n_2 = 3$.

$\frac{1}{\lambda} = R(2)^2 \left(\frac{1}{2^2} - \frac{1}{3^2} \right)$

Simplify:

$\frac{1}{\lambda} = 4R \left(\frac{1}{4} - \frac{1}{9} \right) = 4R \left(\frac{9-4}{36} \right) = 4R \times \frac{5}{36} = \frac{5R}{9}$

Thus,

$\lambda = \frac{9}{5R}$

Express in terms of x

We have $x = \frac{1}{R}$.

So,

$\lambda = \frac{9}{5R} = \frac{9x}{5}$

✅ **Final Answer**

$$\boxed{\frac{9x}{5}}$$

That is, **Option A** is correct.

Question22

The Bohr radius of a hydrogen like species is 70.53 pm . The species and the stationary state (n) are respectively

(Given : Hydrogen atom Bohr radius is 52.9 pm)

JEE Main 2026 (Online) 6th April Evening Shift

Options:

A.

$\mathrm{Li}^{2+}, 3$

B.

$\mathrm{He}^{+}, 3$

C.

$\mathrm{He}^{+}, 2$

D.

$\mathrm{Li}^{2+}, 2$

Answer: D

Solution:

We know that the radius of any hydrogen-like species is given by the formula:

$r_{\text{species}} \quad = a_0 \frac{n^2}{Z}$

Here, a_0 is the Bohr radius of hydrogen (given as 52.9 pm), n is the principal quantum number, and Z is the atomic number of the nucleus.

Substituting the given values:

$70.53 = 52.9\left(\frac{n^2}{Z}\right)$

Now, divide both sides by 52.9:

$\frac{n^2}{Z} = \frac{70.53}{52.9} = 1.33 \approx \frac{4}{3}$

To make the fraction $\frac{n^2}{Z} = \frac{4}{3}$, we can take $n = 2$ (so $n^2 = 4$) and $Z = 3$.

Thus, the species is Li^{2+} and the electron is in the $n = 2$ energy level.

$\mathrm{Li}^{2+},\ n = 2 \Rightarrow \frac{4}{3} = 1.33$

Question23

The hydrogen spectrum consists of several spectral lines in Lyman series $(L_1, L_2, \mathrm{L}_3 \ldots; \mathrm{L}_1$ has lowest energy among Lyman series). Similarly it consists of several spectral lines in Balmer series $(\mathrm{B}_1,\ \mathrm{B}_2,\ \mathrm{B}_3 \ldots; \mathrm{B}_1$ has lowest energy among Balmer lines). The energy of L_1 is x times the energy of B_1. The value of x is ____ $\times 10^{-1}$

. (Nearest integer)

JEE Main 2026 (Online) 24th January Morning Shift

Answer: 54

Solution:

For hydrogen, the energy difference for a spectral line is given by

$\Delta E=13.6\times Z^2\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right)$

Step 1: Identify the transitions

In the Lyman series, electrons fall to $n_1=1$. The first line L_1 is the transition $2\rightarrow 1$.

In the Balmer series, electrons fall to $n_1=2$. The first line B_1 is the transition $3\rightarrow 2$.

Step 2: Find energy of L_1

$$\Delta \mathrm{E}\left(\mathrm{L}_1\right)=13.6\times \mathrm{Z}^2\left(\frac{1}{1^2}-\frac{1}{2^2}\right)=13.6\mathrm{Z}^2\times\frac{3}{4}$$

Step 3: Find energy of B_1

$$\Delta \mathrm{E}\left(\mathrm{B}_1\right)=13.6\times \mathrm{Z}^2\times\left(\frac{1}{2^2}-\frac{1}{3^2}\right)=13.6\times \mathrm{Z}^2\times\frac{5}{4\times 9}$$

Step 4: Take the ratio to get x

The factor $13.6Z^2$ cancels in the ratio.

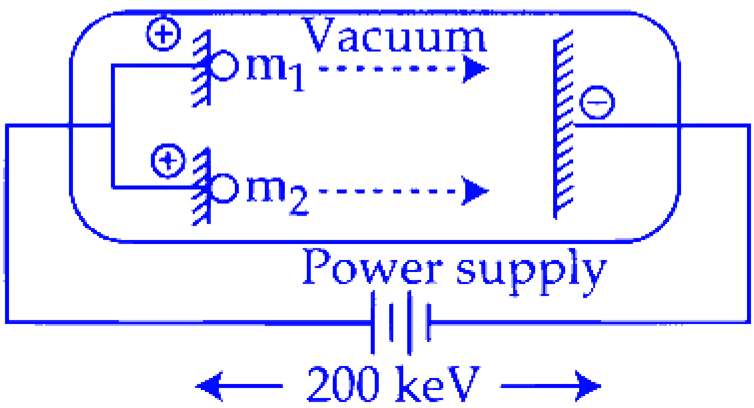
$$\frac{\Delta \mathrm{E}\left(\mathrm{L}_1\right)}{\Delta \mathrm{E}\left(\mathrm{B}_1\right)}=\frac{3}{5}\times 9=\frac{27}{5}=\mathrm{x}$$

Step 5: Write in the asked form

$$\mathrm{x}=\left(\frac{27}{5}\times 10\right)\times 10^{-1}=54\times 10^{-1}$$

Question24

Two positively charged particles m_1 and m_2 have been accelerated across the same potential difference of 200 keV as shown below.



[Given mass of $m_1=1$ amu and $m_2=4$ amu]

The deBroglie wavelength of m_1 will be x times of m_2. The value of x is __________ (nearest integer)

JEE Main 2026 (Online) 28th January Evening Shift

Answer: 2

Solution:

$\lambda_d = \frac{h}{\sqrt{2mK.E.}}$

Here, both particles are accelerated through the same potential difference of 200 keV, so they gain the same kinetic energy (K.E.).

So, in the formula, h and K.E. are the same for both particles. Only the mass m is different.

Therefore, $\lambda_d \propto \frac{1}{\sqrt{m}}$

$$\text{Now, } \frac{(\lambda_d)_{m_1}}{(\lambda_d)_{m_2}} = \sqrt{\frac{m_2}{m_1}} = \sqrt{4} = 2$$

$$(\lambda_d)_{m_1} = 2(\lambda_d)_{m_2}$$

So $x = 2$.

Question25

The surface of sodium metal is irradiated with radiation of wavelength x nm. The kinetic energy of ejected electrons is 2.8×10^{-20} J. The work function of sodium is 2.3 eV. The value of x is ________ $\times 10^2$ nm. (Nearest integer)

(Given: $h = 6.6 \times 10^{-34}$ J s ; $1 \text{ eV} = 1.6 \times 10^{-19}$ J ; $c = 3.0 \times 10^8 \text{ m s}^{-1}$)

JEE Main 2026 (Online) 2nd April Evening Shift

Answer: 5

Solution:

Using Einstein's photoelectric equation,

$h\nu = \phi + K_{\max}$

Also,

$h\nu = \frac{hc}{\lambda}$

So,

$\frac{hc}{\lambda} = \phi + K_{\max}$

Now convert the work function into joule:

$\phi = 2.3 \text{ eV} = 2.3 \times 1.6 \times 10^{-19} \text{ J}$

$\phi = 3.68 \times 10^{-19} \text{ J}$

Given kinetic energy,

$K_{\max} = 2.8 \times 10^{-20} \text{ J}$

Hence total photon energy is

$E = \phi + K_{\max}$

$E = 3.68 \times 10^{-19} + 2.8 \times 10^{-20}$

$E = 3.96 \times 10^{-19} \text{ J}$

Now,

$\lambda = \frac{hc}{E}$

Substitute the values:

$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^{8}}{3.96 \times 10^{-19}}$

$\lambda = \frac{19.8 \times 10^{-26}}{3.96 \times 10^{-19}}$

$\lambda = 5 \times 10^{-7} \text{ m}$

Convert into nm:

$1 \text{ m} = 10^{9} \text{ nm}$

$\lambda = 5 \times 10^{-7} \times 10^{9}$

$\lambda = 5 \times 10^{2} \text{ nm}$

So,

$x = 5$

Therefore, the required nearest integer is

$\boxed{5}$

Question26

Consider two radiations of wavelengths

1. $\lambda_1 = 2000\overset{\circ}{A}$
2. $\lambda_2 = 6000\overset{\circ}{A}$

The ratio of the energies of these two radiations $\left(\frac{E_1}{E_2}\right)$ is _____ (Nearest integer).

JEE Main 2026 (Online) 8th April Evening Shift

Answer: 3

Solution:

We know that the energy of a photon is given by the formula

$E = \frac{hc}{\lambda}$

where

- h = Planck's constant
- c = velocity of light
- λ = wavelength of the radiation.

For the two radiations,

$\frac{E_1}{E_2} = \frac{\frac{hc}{\lambda_1}}{\frac{hc}{\lambda_2}} = \frac{\lambda_2}{\lambda_1}$

$\lambda_1 = 2000\,\text{Å}, \quad \lambda_2 = 6000\,\text{Å}$

$\frac{E_1}{E_2} = \frac{6000}{2000}$

$\frac{E_1}{E_2} = 3$

Final Answer

$$\boxed{\frac{E_1}{E_2} = 3}$$

Hence, the ratio of energies of the two radiations is **3 (nearest integer).**

Question27

The energy of an electron in the first Bohr orbit of the H-atom is -13.6 eV.

The magnitude of energy value of an electron in the first excited state of Be^{3+} is ________ eV (nearest integer value).

JEE Main 2025 (Online) 8th April Evening Shift

Answer: 54

Solution:

To find the energy of an electron in the first excited state of a Be^{3+} ion, we can use the formula for the energy of an electron in a hydrogen-like atom:

$E_T = -13.6 \frac{Z^2}{n^2}$ eV

Given:

Energy of the first orbit (ground state) of the hydrogen atom: $E_1 = -13.6$ eV (where $Z = 1$ and $n = 1$).

For Be^{3+}:

Atomic number $Z = 4$.

First excited state corresponds to $n = 2$.

Calculation for Be^{3+}:

The energy ratio between the hydrogen atom and the Be^{3+} ion can be written as:

$\frac{E_H}{E_{Be^{+3}}} = \frac{Z_1^2}{n_1^2} \times \frac{n_2^2}{Z_2^2}$

Substitute the known values:

$\frac{-13.6}{E_{Be^{+3}}} = \frac{1^2}{1^2} \times \frac{2^2}{4^2}$

$\frac{-13.6}{E_{Be^{+3}}} = \frac{1}{1} \times \frac{4}{16}$

Solving this gives:

$E_{Be^{+3}} = -13.6 \times 4 = -54.4 \text{ eV}$

The magnitude of the energy of the electron in the first excited state of Be^{3+} is $|-54.4| = 54.4$ eV, which is approximately 54 eV when rounded to the nearest integer.

Question28

Radius of the first excited state of Helium ion is given as : $a_0 \rightarrow$ radius of first stationary state of hydrogen atom.

JEE Main 2025 (Online) 22nd January Morning Shift

Options:

A. $r = \frac{a_0}{4}$

B. $r = 2a_0$

C. $r = 4a_0$

D. $r = \frac{a_0}{2}$

Answer: B

Solution:

$r_n = \frac{n^2 a_0}{Z}$

For the helium ion, which is hydrogen-like with a nuclear charge of $Z = 2$, the first excited state corresponds to $n = 2$. Substituting these values:

$r_2 = \frac{(2)^2 a_0}{2} = \frac{4 a_0}{2} = 2 a_0$

Thus, the radius of the first excited state of the helium ion is $2 a_0$.

Question29

Given below are two statements :

Statement (I) : A spectral line will be observed for a $2p_x \rightarrow 2p_y$ transition.

Statement (II) : $2\mathrm{p}_x$ and $2\mathrm{p}_y$ are degenerate orbitals.

In the light of the above statements, choose the correct answer from the options given below :

JEE Main 2025 (Online) 22nd January Evening Shift

Options:

A. Statement I is false but Statement II is true

B. Both Statement I and Statement II are false

C. Statement I is true but Statement II is false

D. Both Statement I and Statement II are true

Answer: A

Solution:

Let us analyze the two statements in the context of atomic orbitals and electronic transitions:

Statement (I)

> "A spectral line will be observed for a $2p_x \rightarrow 2p_y$ transition."

For an emission (or absorption) line to be observed, there must be a difference in energy between the initial and final states.

In a typical hydrogen-like or many-electron atom (without additional external fields or splitting effects), the three $2p$ orbitals ($2p_x$, $2p_y$, $2p_z$) are *degenerate*—i.e., they all have the same energy.

Consequently, a transition from $2p_x$ to $2p_y$ (both having the same energy) would involve **no energy change**. Hence, **no photon** is emitted or absorbed for this "transition." So you **would not** observe a spectral line for such a transition.

Therefore, **Statement (I)** is **false**.

Statement (II)

> "$2p_x$ and $2p_y$ are degenerate orbitals."

Orbitals within the same subshell (e.g., $2p$ subshell) are typically degenerate (same energy) in an isolated atom (especially a hydrogen-like atom).

Thus, $2p_x$ and $2p_y$ (and $2p_z$) do indeed have the same energy.

Therefore, **Statement (II)** is **true**.

Conclusion

Statement (I): **False**

Statement (II): **True**

Hence, the correct choice is:

> **Option A: Statement I is false but Statement II is true.**

Question30

Heat treatment of muscular pain involves radiation of wavelength of about 900 nm . Which spectral line of H atom is suitable for this?

Given : Rydberg constant $\mathrm{R_H = 10^5\ cm^{-1},\ h = 6.6 \times 10^{-34}\ J\ s, c = 3 \times 10^8\ m/s}\Big)$

JEE Main 2025 (Online) 23rd January Morning Shift

Options:

A. Lyman series, $\infty \rightarrow 1$

B. Balmer series, $\infty \rightarrow 2$

C. Paschen series, $5 \rightarrow 3$

D. Paschen series, $\infty \rightarrow 3$

Answer: D

Solution:

To determine which spectral line of the hydrogen atom suits the wavelength for heat treatment of muscular pain (900 nm), we begin by applying the Rydberg equation:

$\frac{1}{\lambda} = \mathrm{R_H Z^2}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

Given:

$\lambda = 900 \text{ nm} = 9 \times 10^{-5} \text{ cm}$

$\mathrm{R_H} = 10^5 \text{ cm}^{-1}$

Hydrogen atom (Z = 1)

We proceed by solving the equation:

$\frac{1}{\lambda \times \mathrm{R_H}} = \frac{1}{n_1^2} - \frac{1}{n_2^2}$

Substituting the known values:

$\frac{1}{9 \times 10^{-5} \text{ cm } \times 10^5 \text{ cm }^{-1}} = \frac{1}{n_1^2} - \frac{1}{n_2^2}$

This simplifies to:

$\frac{1}{9} = \frac{1}{n_1^2} - \frac{1}{n_2^2}$

This condition is satisfied when $n_1 = 3$ and $n_2 = \infty$.

Therefore, the appropriate spectral transition is from $n_2 = \infty$ to $n_1 = 3$, which corresponds to the Paschen series transition $\infty \rightarrow 3$.

Question31

Given below are two statements about X-ray spectra of elements :

Statement (I) : A plot of $\sqrt{v}$ ($v =$ frequency of X-rays emitted) vs atomic mass is a straight line.

Statement (II) : A plot of $v(\nu =$ frequency of X-rays emitted) vs atomic number is a straight line. In the light of the above statements,

choose the correct answer from the options given below :

JEE Main 2025 (Online) 23rd January Evening Shift

Options:

A. Statement I is false but Statement II is true

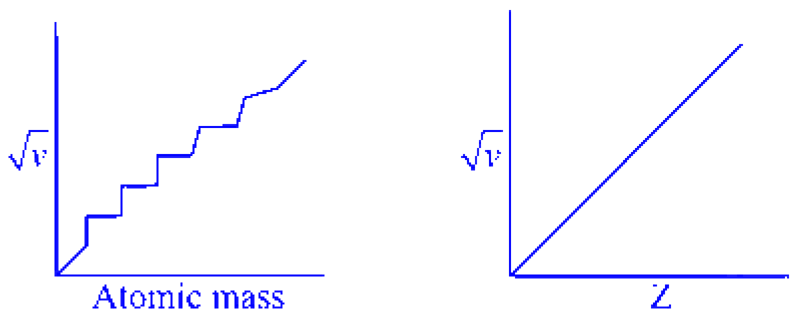
B. Both Statement I and Statement II are true

C. Statement I is true but Statement II is false

D. Both Statement I and Statement II are false

Answer: D

Solution:



Question32

Given below are two statements :

Statement (I) : For a given shell, the total number of allowed orbitals is given by n^2.

Statement (II) : For any subshell, the spatial orientation of the orbitals is given by $-l$ to $+l$ values including zero.

In the light of the above statements, choose the correct answer from the options given below :

JEE Main 2025 (Online) 23rd January Evening Shift

Options:

A. Both Statement I and Statement II are true

B. Both Statement I and Statement II are false

C. Statement I is false but Statement II is true

D. Statement I is true but Statement II is false

Answer: A

Solution:

For a shell total number of orbitals $= \mathrm{n}^2$

Magnetic quantum number have values $(-\ell$ to $+\ell)$ including 0.

Question33

For hydrogen atom, the orbital/s with lowest energy is/are :

(A) $4\mathrm{s}$

(B) $3\mathrm{p}_x$

(C) $3\ \mathrm{d}_{x^2-y^2}$

(D) $3\ \mathrm{d}_{z^2}$

(E) $4\mathrm{p}_z$

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 24th January Evening Shift

Options:

A. (A) only

B. (B) only

C. (B), (C) and (D) only

D. (A) and (E) only

Answer: C

Solution:

In hydrogen atom the orbitals in a shell are degenerate means energy depends only on ' n '

$\therefore \mathrm{E}_{3 \mathrm{p}_{\mathrm{x}}}=\mathrm{E}_{3 \mathrm{~d}_{x^2-y^2}}=\mathrm{E}_{3 \mathrm{~d}_{z^2}}$

Question34

In a multielectron atom, which of the following orbitals described by three quantum numbers will have same energy in absence of electric and magnetic fields?

A. $\mathrm{n}=1, \mathrm{l}=0, \mathrm{~m}_1=0$

B. $\mathrm{n}=2, \mathrm{l}=0, \mathrm{~m}_1=0$

C. $\mathrm{n}=2, \mathrm{I}=1, \mathrm{~m}_1=1$

D. $\mathrm{n}=3, \mathrm{l}=2, \mathrm{~m}_1=1$

E. $\mathrm{n}=3, \mathrm{l}=2, \mathrm{~m}_1=0$

Choose the correct answer from the options given below:

JEE Main 2025 (Online) 28th January Morning Shift

Options:

A. B and C Only

B. C and D Only

C. A and B Only

D. D and E Only

Answer: D

Solution:

	orbital
$A : n = 1, \ell = 0,\ m_\ell = 0$	1s
$B : n = 2, \ell = 0,\ m_\ell = 0$	2s
$C : n = 3, \ell = 1,\ m_\ell = 1$	3p
$D : n = 3, \ell = 2,\ m_\ell = 1$	3d
$E : n = 3, \ell = 2,\ m_\ell = 0$	3d

In absence of electric and magnetic fields, all orbitals of 3d are degenerate

----

Question35

Which of the following is/are not correct with respect to energy of atomic orbitals of hydrogen atom?

(A) 1s < 2p < 3d < 4s

(B) 1s < 2s = 2p < 3s = 3p

(C) 1s < 2s < 2p < 3s < 3p

(D) 1s < 2s < 4s < 3d

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 28th January Evening Shift

Options:

A.

(A) and (B) only

B.

(A) and (C) only

C.

(B) and (D) only

D.

(C) and (D) only

Answer: D

Solution:

First, recall the key fact that **for a hydrogen atom (and hydrogen-like ions)**, all orbitals having the same principal quantum number n have the **same energy**. In other words,

$E_{n\ell}$ depends only on n, and not on ℓ.

Hence, the energy ordering for hydrogen-like atoms follows

$1s \ < \ 2s = 2p \ < \ 3s = 3p = 3d \ < \ 4s = 4p = 4d = 4f \ < \ \ldots$

Analyzing Each Statement

We check each statement against the **true** ordering for the hydrogen atom.

(A) $1s < 2p < 3d < 4s$

Because $2p$ is an $n = 2$ orbital, $3d$ is $n = 3$, and $4s$ is $n = 4$, this partial ordering

$E(1s) \ < \ E(2p) \ < \ E(3d) \ < \ E(4s)$

is consistent with the fact that $n = 1 < n = 2 < n = 3 < n = 4$.

Even though hydrogen also has 2s degenerate with 2p, and 3s and 3p degenerate with 3d, **nothing in (A) contradicts** the correct order $E_1 < E_2 < E_3 < E_4$.

So (A) is **correct** for hydrogen's energy levels as stated.

(B) $1s < 2s = 2p < 3s = 3p$

For hydrogen, indeed $2s = 2p$ in energy, and also $3s = 3p = 3d$.

The statement does *not* mention 3d, but it does correctly say $3s = 3p$. That part is still true for hydrogen.

So for the orbitals it listed, (B) is also **correct**.

(C) $1s < 2s < 2p < 3s < 3p$

This is the kind of ordering that appears in many-electron atoms (where subshell splitting occurs).

But for a hydrogen atom, $2s$ and $2p$ have exactly the **same** energy, so $2s < 2p$ is **not** correct.

Hence (C) is **not** correct for hydrogen.

(D) $1s < 2s < 4s < 3d$

In multi-electron atoms (e.g. the usual "(n+ℓ) rule"), often $4s$ is lower than $3d$.

But for hydrogen, *all* orbitals with $n = 3$ lie below *all* orbitals with $n = 4$. That is, $3d$ is lower in energy than $4s$.

So (D) is **not** correct for hydrogen.

Conclusion

(C) and (D) are **not correct** for the hydrogen atom.

(A) and (B) are correct.

Hence the final answer matches the choice stating:

> **(C) and (D) only** are not correct.

Looking at the given options, that is:

$\boxed{\text{Option (D): (C) and (D) only.}}$

Question36

If a_0 is denoted as the Bohr radius of hydrogen atom, then what is the de-Broglie wavelength (λ) of the electron present in the second orbit of hydrogen atom? [n : any integer]

JEE Main 2025 (Online) 29th January Morning Shift

Options:

A.

$\frac{8\pi a_0}{n}$

B.

$\frac{2a_0}{n\pi}$

C.

$\frac{na_0}{4\pi}$

D.

$\frac{4\pi a_0}{n}$

Answer: A

Solution:

Bohr radius of hychogon atom $\rightarrow a_0$

According to Bohr, the equation used to calculate the angular momentum of an election in a hydrogen atom is

$\mathrm{mvr} = \frac{\mathrm{nh}}{2\pi}$ ..... (1)

$m \rightarrow$ mass of electron

$v \rightarrow$ velocity of electron

$r \rightarrow$ radius of the orbit

$n \rightarrow$ orbit. number. in. which electron is present.

Given that; election is present in second orbit, $n = 2$

The radius of the second orbit $r_2 = a_0 \times 2^2 = 4a_0$

General formula for radius of n^{th} orbit,

$r_n = a_0 \times n^2$

From (1)

$$mvr = n\frac{h}{2\pi}$$
$$2\pi r = n\frac{h}{mv}$$

$\frac{h}{mv} = \lambda$ (de Broglie relation ship, $\lambda \rightarrow$ de Broglie Wavelength

So, $2\pi r = n\lambda$

For the electron in the second orbit, $2\pi r_2 = n\lambda$

Substitute for r_2,

$2\pi \times 4a_0 = n\lambda$
$8\pi a_0 = n\lambda$
$\therefore \lambda = \dfrac{8\pi a_0}{n}$

Correct answer: Option 1) $\frac{8\pi a_0}{n}$

Question37

Given below are two statements :

Statement (I): It is impossible to specify simultaneously with arbitrary precision, both the linear momentum and the position of a particle.

Statement (II) : If the uncertainty in the measurement of position and uncertainty in measurement of momentum are equal for an electron, then the uncertainty in the measurement of velocity is $\geqslant \sqrt{\frac{h}{\pi}} \times \frac{1}{2m}$.

In the light of the above statements, choose the correct answer from the options given below :

JEE Main 2025 (Online) 29th January Evening Shift

Options:

A.

Statement I is false but **Statement II** is true

B.

Both **Statement I** and **Statement II** are true

C.

Both **Statement I** and **Statement II** are false

D.

Statement I is true but **Statement II** is false

Answer: B

Solution:

Statement I : Correct

This statement is describing the uncertainty principle.

It states that it is impossible to measure both the position and the momentum of an object.

Statement II : Correct.

If the uncertainty in the measurement of position (Δx) and the uncertainty in the measurement of momentum (Δp) are equal for an electron, then the uncertainty in the measurement of velocity (Δv) is greater than or equal to $\sqrt{\frac{h}{\pi}}\frac{1}{2m}$

This is derived from the Heisenberg uncertainty principle, $\Delta x\,.\,\Delta p_x \geq \frac{h}{4\pi}$

Given, $\Delta x = \Delta p$

Substitute this in $\Delta x\,.\,\Delta p \geq \frac{h}{4\pi}$ as

$\Delta p\,.\,\Delta p \geq \frac{h}{4\pi}$

$\Delta p^2 \geq \frac{h}{4\pi}$

$\Delta p^2 \geq \frac{h}{4\pi}$

$$\begin{cases} The\, formula\, for\, momentum\,(P)\, is\, P = mV \\ For\, uncertainty\, \Delta P = m\Delta v \end{cases}$$

So, $m\Delta v \geq \sqrt{\frac{h}{4\pi}}$

$\Delta v \geq \sqrt{\frac{h}{4\pi}} \times \frac{1}{m}$

$\Delta v \geq \frac{1}{2} > \sqrt{\frac{h}{\pi}} > \frac{1}{m}$

$\Delta v \geq \sqrt{\frac{h}{\pi}}\frac{1}{2m}$

Both the statements are true (correct). So answer is (2) Both statement I and statement II are true.

Question38

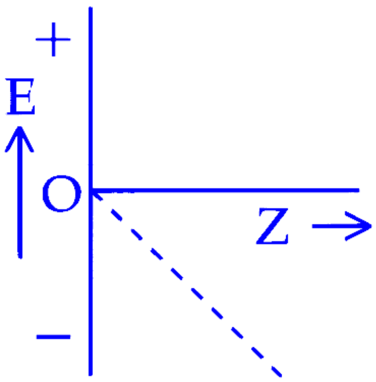
For hydrogen like species, which of the following graphs provides the most appropriate representation of E vs Z plot for a constant n?

[E : Energy of the stationary state, Z : atomic number, n = principal quantum number]

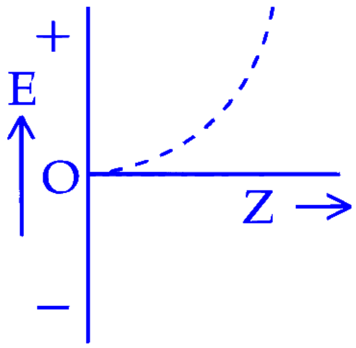
JEE Main 2025 (Online) 29th January Evening Shift

Options:

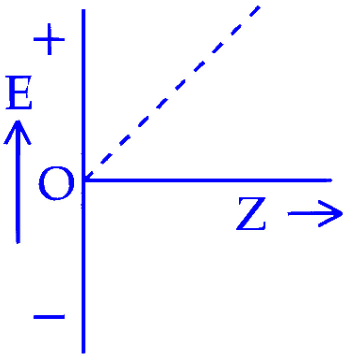
A.



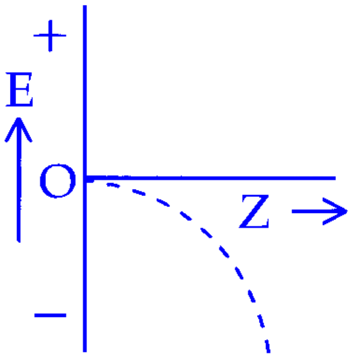
B.



C.



D.



Answer: D

Solution:

$$E\,vs\,Z\,plot\,(cons\tan t\,n)\begin{cases} E - Energy\,of\,the\,stationary\,state \\ Z - atomic\,number \\ n - \Pr incipal\,quantum\,number \end{cases}$$

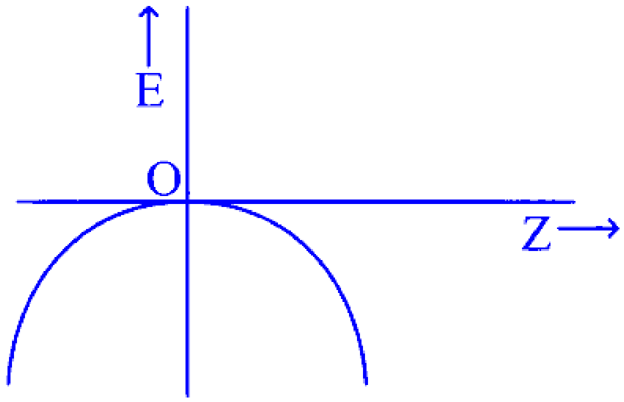
The formula of energy of hydrogen like species is

$E = -13.6\frac{z^2}{n^2}eV$

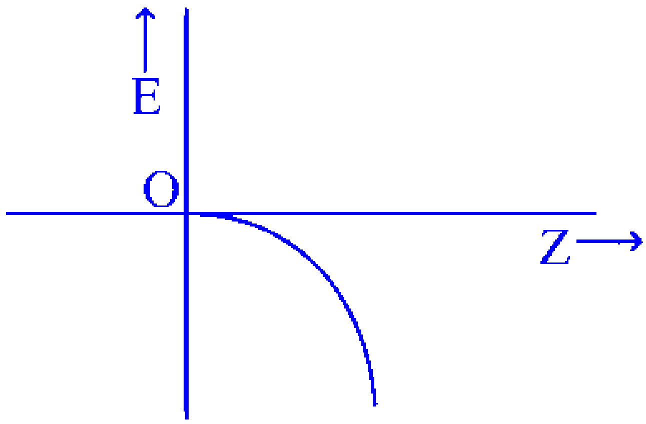
$E \propto -\frac{z^2}{n^2}$

For constant n, $E \propto -Z^2$

Here, E and Z has a quadratic relationship. Since the equation (proportionality) includes a negative sign, the graph of E vs Z will be a downward - opening parabola.



The graph is like an upside down U. In the given options, graph 4 is like half upside down parabola.



The shape is characteristic of negative quadratic function, where the curve opens downward, with the vertex of the parabola at the origin.

So, the answer is option (4).

Question39

According to Bohr's model of hydrogen atom, which of the following statement is incorrect?

JEE Main 2025 (Online) 2nd April Morning Shift

Options:

A. Radius of 4^{th} orbit is four times larger than that of 2^{nd} orbit

B. Radius of 8^{th} orbit is four times larger than that of 4^{th} orbit

C. Radius of 6^{th} orbit is three times larger than that of 4^{th} orbit

D. Radius of 3^{rd} orbit is nine times larger than that of 1^{st} orbit

Answer: C

Solution:

In Bohr's model the radius of the nth orbit is

$r_n = n^2 a_0$

where a_0 is the Bohr radius. Hence for any two orbits m and n:

$\frac{r_m}{r_n} = \frac{m^2}{n^2}$.

Let's check each option:

A: $\frac{r_4}{r_2} = \frac{4^2}{2^2} = \frac{16}{4} = 4 \rightarrow$ correct

B: $\frac{r_8}{r_4} = \frac{8^2}{4^2} = \frac{64}{16} = 4 \rightarrow$ correct

C: $\frac{r_6}{r_4} = \frac{6^2}{4^2} = \frac{36}{16} = 2.25 \; (\neq 3) \rightarrow$ incorrect

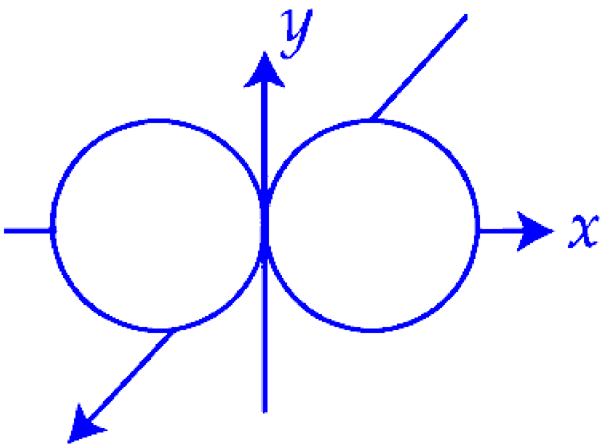
D: $\frac{r_3}{r_1} = \frac{3^2}{1^2} = 9 \rightarrow$ correct

Therefore the incorrect statement is Option C.

Question40

Which of the following statements are true?

(A) The subsidiary quantum number l describes the shape of the orbital occupied by the electron.

(B) 

is the boundary surface diagram of the $2\mathrm{p}_x$ orbital.

(C) The + and - signs in the wave function of the $2p_x$ orbital refer to charge.

(D) The wave function of $2\mathrm{p}_x$ orbital is zero everywhere in the xy plane.

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 2nd April Evening Shift

Options:

A. (A) and (B) only

B. (B) and (D) only

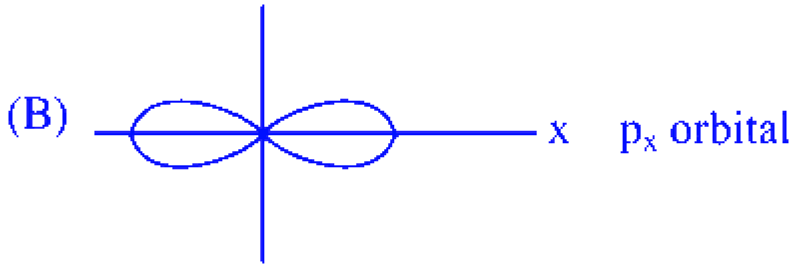
C. (C) and (D) only

D. (A), (B) and (C) only

Answer: A

Solution:

(A) Azimuthal quantum number (ℓ) indicates the shape of orbital occupied by the electron



(C) The + and - sign in the wave function of $2p_x$ orbital refer to the sign (Phase) of the wave function, not the charge

(D) The wave function of $2p_x$ orbital will be zero in yz plane (Nodal plane).

Question41

Which of the following postulate of Bohr's model of hydrogen atom is not in agreement with quantum mechanical model of an atom?

JEE Main 2025 (Online) 3rd April Morning Shift

Options:

A. The electron in a H atom's stationary state moves in a circle around the nucleus.

B. An atom can take only certain distinct energies $\mathrm{E}_1, \mathrm{E}_2, \mathrm{E}_3$, etc. These allowed states of constant energy are called the stationary states of atom.

C. An atom in a stationary state does not emit electromagnetic radiation as long as it stays in the same state.

D. When an electron makes a transition from a higher energy stationary state to a lower energy stationary state, then it emits a photon of light.

Answer: A

Solution:

The postulate that contradicts quantum mechanics is

Option A

“The electron in a H atom’s stationary state moves in a circle around the nucleus.”

In quantum mechanics an electron in a stationary state is described by a delocalized wave-function (an orbital), not by a definite circular orbit.

Question42

For electrons in ' 2 s ' and ' 2 p ' orbitals, the orbital angular momentum values, respectively are :

JEE Main 2025 (Online) 3rd April Evening Shift

Options:

A. 0 and $\sqrt{6} \frac{h}{2 \pi}$

B. 0 and $\sqrt{2} \frac{h}{2 \pi}$

C. $\sqrt{2} \frac{h}{2 \pi}$ and 0

D. $\frac{h}{2 \pi}$ and $\sqrt{2} \frac{h}{2 \pi}$

Answer: B

Solution:

The orbital angular momentum for electrons in orbitals is given by the formula:

Orbital angular momentum $=\sqrt{\ell(\ell+1)} \frac{\mathrm{h}}{2 \pi}$

For the 2s orbital, the quantum number ℓ is 0:

Orbital angular momentum = $\sqrt{0(0+1)} \frac{\mathrm{h}}{2 \pi}=0$

For the 2p orbital, the quantum number ℓ is 1:

Orbital angular momentum = $\sqrt{1(1+1)} \frac{\mathrm{h}}{2 \pi}=\sqrt{2} \frac{\mathrm{h}}{2 \pi}$

Question43

Which one of the following about an electron occupying the 1 s orbital in a hydrogen atom is incorrect?

(Bohr's radius is represented by a_0)

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. The probability density of finding the electron is maximum at the nucleus

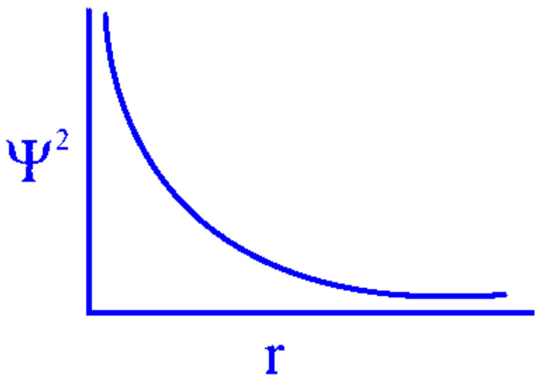
B. The total energy of the electron is maximum when it is at a distance a_0 from the nucleus

C. The electron can be found at a distance $2a_0$ from the nucleus

D. The 1 s orbital is spherically symmetrical

Answer: B

Solution:



1. Ψ^2 = Probability density is maximum at nucleus.

2. Electron can exist upto infinity from nucleus.

3. True

4. Energy of electron is maximum at infinite distance from nucleus.

Question44

Consider the ground state of chromium atom $(Z = 24)$. How many electrons are with Azimuthal quantum number $l = 1$ and $l = 2$ respectively ?

JEE Main 2025 (Online) 4th April Evening Shift

Options:

A. 16 and 5

B. 12 and 5

C. 12 and 4

D. 16 and 4

Answer: B

Solution:

The ground state electron configuration of chromium (Cr, $Z = 24$) is:

Cr: $[\mathrm{Ar}]\, 3d^5\, 4s^1$

Let's break down the configuration:

The noble gas core $[\mathrm{Ar}]$ represents:

$1s^2$ (with $l = 0$)

$2s^2$ (with $l = 0$)

$2p^6$ (with $l = 1$)

$3s^2$ (with $l = 0$)

$3p^6$ (with $l = 1$)

Electrons with azimuthal quantum number $l = 1$ are found in p orbitals:

$2p^6$: 6 electrons

$3p^6$: 6 electrons

Total p-electrons: $6 + 6 = 12$.

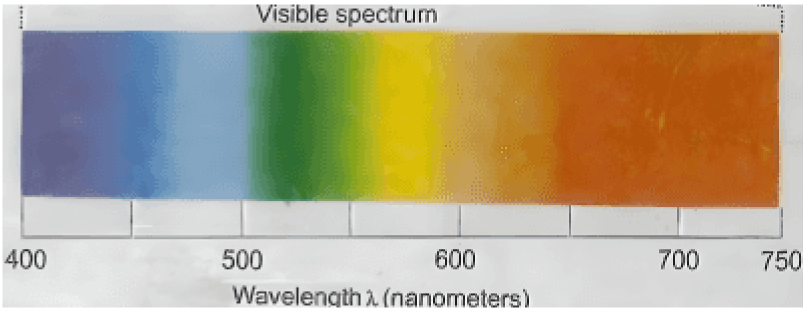
Electrons with azimuthal quantum number $l = 2$ are found in d orbitals:

$3d^5$: 5 electrons

Thus, there are 12 electrons with $l = 1$ and 5 electrons with $l = 2$.

The correct option is Option B.

Question45



Which of the following statements are correct, if the threshold frequency of caesium is 5.16×10^{14} Hz ?

A. When Cs is placed inside a vacuum chamber with an ammeter connected to it and yellow light is focused on Cs , the ammeter shows the presence of current.

B. When the brightness of the yellow light is dimmed, the value of the current in the ammeter is reduced.

C. When a red light is used instead of the yellow light, the current produced is higher with respect to the yellow light.

D. When a blue light is used, the ammeter shows the formation of current.

E. When a white light is used. the ammeter shows formation of current.

Choose the correct answer from the options given below:

JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. B, C and D Only

B. A, C, D and E Only

C. A, B, D and E Only

D. A, D and E Only

Answer: C

Solution:

To determine the correct statements regarding the threshold frequency of cesium, first note that the threshold frequency given is 5.16×10^{14} Hz. Using this frequency, we can calculate the corresponding wavelength, λ, using the equation:

$\lambda = \frac{c}{v} = \frac{3 \times 10^8 \text{ m/s}}{5.16 \times 10^{14} \text{ Hz}}$

This calculation gives $\lambda \approx 581.39\,\text{nm}$.

Key points:

The wavelength calculated is near the yellow light region. Therefore, yellow light has enough energy to cause a photoelectric effect.

When the intensity of light (brightness) decreases, the photocurrent will also decrease, as fewer photons are available to eject electrons.

Red light, with its larger wavelength and lower frequency, does not have enough energy to cause the photoelectric effect in cesium.

Blue light has a higher frequency than yellow light ($v_{\text{blue}} > v_{\text{yellow}}$), meaning it can produce a photoelectric current.

White light includes all visible frequencies, including those above the threshold, so it will also emit a current.

Thus, the correct statements involve using yellow, blue, and white light, without producing an effect with red light:

A: Yellow light produces a current.

B: Dimming yellow light reduces the current.

D: Blue light produces a current.

E: White light produces a current.

Therefore, the correct statements are A, B, D, and E.

Question46

The extra stability of half-filled subshell is due to :

(A) Symmetrical distribution of electrons

(B) Smaller coulombic repulsion energy

(C) The presence of electrons with the same spin in non-degenerate orbitals

(D) Larger exchange energy

(E) Relatively smaller shielding of electrons by one another

Indentify the correct statements :

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A. (B), (D) and (E) only

B. (A), (B), (D) and (E) only

C. (B), (C) and (D) only

D. (A), (B) and (D) only

Answer: B

Solution:

The enhanced stability of a half-filled subshell is attributed to the following factors:

Symmetrical Distribution of Electrons: Electrons are evenly distributed across the available orbitals, leading to a balanced and stable arrangement.

Large Exchange Energy: The energy due to the exchange of electrons with parallel spins among degenerate orbitals is maximized, contributing to increased stability.

Smaller Coulombic Repulsion: The repulsion between electrons is minimized in a half-filled configuration, reducing the overall energy of the subshell.

Smaller Shielding of Electrons: In a half-filled subshell, electrons are more exposed to the nuclear charge, as there is less shielding effect from other electrons, enhancing stability.

Question47

Correct statements for an element with atomic number 9 are:

A. There can be 5 electrons for which $m_s = +\frac{1}{2}$ and 4 electrons for which $m_s = -\frac{1}{2}$.

B. There is only one electron in p_z orbital.

C. The last electron goes to orbital with $n = 2$ and $l = 1$.

D. The sum of angular nodes of all the atomic orbitals is 1.

Choose the correct answer from the options given below:

JEE Main 2025 (Online) 8th April Evening Shift

Options:

A.

A and B Only

B.

A, C and D Only

C.

A and C Only

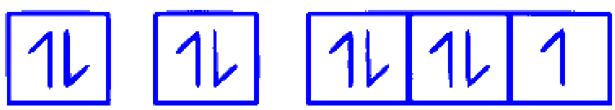
D.

C and D Only

Answer: C

Solution:

Element with atomic number 9 is Fluorine
$F(9) = 1s^2 2s^2 2p^5$



(A) 5 electrons can be up-spin $\left[\mathrm{m_s} = +\frac{1}{2}\right]$ and 4 electrons can be down spin $\left[\mathrm{m_s} = -\frac{1}{2}\right]$

(B) Unpaired electron can be in anyone of p_x, p_y or p_z orbital

(C) Last electron is in 2 p subshell with $\mathrm{n}=2, \ell=1$

(D) Angular node for s-orbital $=0$ while of each p -orbital $=1$

Sum of all angular node $=3$

Question48

Which of the following electronic configuration would be associated with the highest magnetic moment?

[27-Jan-2024 Shift 1]

Options:

A.

$[Ar]3d^7$

B.

$[Ar]3d^8$

C.

$[Ar]3d^3$

D.

$[Ar]3d^6$

Answer: D

Solution:

	$3d^7$	$3d^8$	$3d^3$	$3d^6$
No.of. unpaired e^-	3	2	3	4
Spin only Magnetic moment	$\sqrt{15}$ BM	$\sqrt{8}$ BM	$\sqrt{15}$ BM	$\sqrt{24}$ BM

Question49

Consider the following complex ions

$P = [FeF_6]^{3-}$

$Q = [V(H_2O)_6]^{2+}$

$R = [Fe(H_2O)_6]^{2+}$

The correct order of the complex ions, according to their spin only magnetic moment values (in B.M.) is :

[27-Jan-2024 Shift 1]

Options:

A.

$R < Q < P$

B.

$R < P < Q$

C.

$Q < R < P$

D.

$Q < P < R$

Answer: C

Solution:

$[FeF_6]^{3-}$: Fe^{+3} : $[Ar]3d^5$

1	1	1	1	1

F : Weak field Ligand

No. of unpaired electron's $= 5$

$\mu = \sqrt{5(5+2)}$

$\mu = \sqrt{35}$ BM

$[V(H_2O)_6]^{+2}$: V^{+2} : $3d^3$

1	1	1		

No. of unpaired electron's $= 3$

$\mu = \sqrt{3(3+2)}$

$\mu = \sqrt{15}$ BM

$[Fe(H_2O)_6]^{+2}$: Fe^{+2} : $3d^6$

1⇂	1	1	1	1

H_2O : Weak field Ligand

No. of unpaired electron's $= 4$

No. of unpaired electron's $= 4$

$\mu = \sqrt{4(4+2)}$

$\mu = \sqrt{24}$ BM

Question50

The electronic configuration for Neodymium is: [Atomic Number for Neodymium 60]

[27-Jan-2024 Shift 1]

Options:

A.

$[Xe]4f^4 6s^2$

B.

$[Xe]5f^4 7s^2$

C.

$[Xe]4f^6 6s^2$

D.

$[Xe]4f^1 5d^1 6s^2$

Answer: A

Solution:

Electronic configuration of Nd(Z = 60) is;

$[Xe]4f^4 6s^2$

Question51

The number of electrons present in all the completely filled subshells having n = 4 and s = + 1/2 is_____

(Where n = principal quantum number and s= spin quantum number)

[27-Jan-2024 Shift 1]

Answer: 16

Solution:

$n = 4$ can have,

	4s	4p	4d	4f
Total e^-	2	6	10	14
Total e^- with $S = +\frac{1}{2}$	1	3	5	7

So, Ans. 16

Question52

Total number of ions from the following with noble gas configuration is______

$Sr^{2+}(Z = 38)$, $Cs^{+}(Z = 55)$, $La^{2+}(Z = 57)Pb^{2+}$
$(Z = 82)$, $Yb^{2+}(Z = 70)$ and $Fe^{2+}(Z = 26)$

[27-Jan-2024 Shift 2]

Answer: 2

Solution:

Noble gas configuration $= ns^2np^6$

$[Sr^{2+}] = [Kr]$

$[Cs^{+}] = [Xe]$

$[Yb^{2+}] = [Xe]4f^{14}$

$[La^{2+}] = [Xe]5d^{1}$

$[Pb^{2+}] = [Xe]4f^{14}5d^{10}6s^{2}$

$[Fe^{2+}] = [Ar]3d^{6}$

Question53

The correct set of four quantum numbers for the valence electron of rubidium atom (Z = 37) is:

[29-Jan-2024 Shift 1]

Options:

A.

5, 0, 0, + 1/2

B.

5, 0, 1, + 1/2

C.

5, 1, 0, + 1/2

D.

5, 1, 1, + 1/2

Answer: A

Solution:

$\mathrm{Rb} = [\mathrm{Kr}]5s^1$

$\mathrm{n} = 5$

$l = 0$

$\mathrm{m} = 0$

$s = +1/2 \text{ or } -1/2$

Question54

Choose the correct answer from the options given below :-

	List I (Spectral Series for Hydrogen)		List II (Spectral Region Higher
A.	Lyman	I.	Infrared region
B.	Balmer	II.	UV region
C.	Paschen	III.	Infrared region
D.	Pfund	IV.	Visible region

[29-Jan-2024 Shift 2]

Options:

A.

A-II, B-III, C-I, D-IV

B.

A-I, B-III, C-II, D-IV

C.

A-II, B-IV, C-III, D-I

D.

A-I, B-II, C-III, D-IV

Answer: C

Solution:

A - II, B - IV, C - III, D - I

Fact based.

Question55

Given below are two statements:

Statement-I: The orbitals having same energy are called as degenerate orbitals.

Statement-II: In hydrogen atom, 3p and 3d orbitals are not degenerate orbitals.

In the light of the above statements, choose the most appropriate answer from the options given

[30-Jan-2024 Shift 1]

Options:

A.

Statement-I is true but Statement-II is false

B.

Both Statement-I and Statement-II are true.

C.

Both Statement-I and Statement-II are false

D.

Statement-I is false but Statement-II is true

Answer: A

Solution:

For single electron species the energy depends upon principal quantum number ' n ' only. So, statement II is false.

Statement I is correct definition of degenerate orbitals.

Question56

Number of spectral lines obtained in He+spectra, when an electron makes transition from fifth excited state to first excited state will be___

[30-Jan-2024 Shift 2]

Answer: 10

Solution:

5^{th} excited state $\Rightarrow n_1 = 6$

1^{st} excited state $\Rightarrow n_2 = 2$

$\Delta n = n_1 - n_2 = 6 - 2 = 4$

Maximum number of spectral lines

$$= \frac{\Delta n(\Delta n + 1)}{2} = \frac{4(4+1)}{2} = 10$$

Question57

The ionization energy of sodium in $kJmol^{-1}$. If electromagnetic radiation of wavelength 242 nm is just sufficient to ionize sodium atom is___

[31-Jan-2024 Shift 1]

Answer: 494

Solution:

$$E = \frac{1240}{\lambda(nm)} eV$$

$$= \frac{1240}{242} eV$$

$$= 5.12 eV$$

$$= 5.12 \times 1.6 \times 10^{-19}$$

$$= 8.198 \times 10^{-19} J/ \text{ atom}$$

$$= 494 \, kJ/mol$$

Question58

The four quantum numbers for the electron in the outer most orbital of potassium (atomic no. 19) are

[31-Jan-2024 Shift 2]

Options:

A.

n = 4, l = 2, m = −1, s = + 1/2

B.

n = 4, l = 0, m = 0, s = + 1/2

C.

n = 3, l = 0, m = 1, s = + 1/2

D.

n = 2, l = 0, m = 0, s = + 1/2

Answer: B

Solution:

$_{19}K\ \ 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1.$

Outermost orbital of potassium is 4 s orbital

$n = 4, 1 = 0, m_1 = 0,\ s = \pm\frac{1}{2}.$

Question59

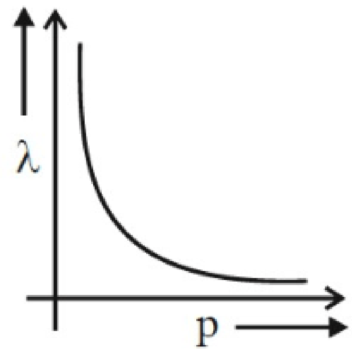
According to the wave-particle duality of matter by de-Broglie, which of the following graph plot presents most appropriate

relationship between wavelength of electron (λ) and momentum of electron (p) ?

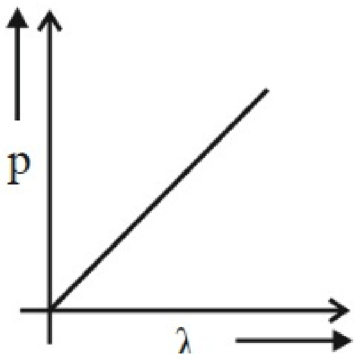
[1-Feb-2024 Shift 1]

Options:

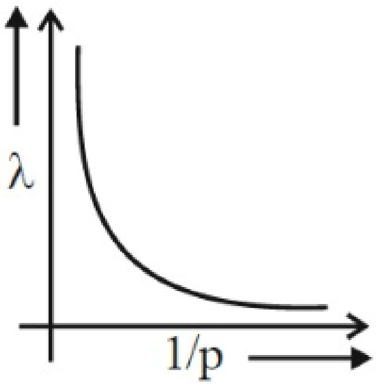
A.



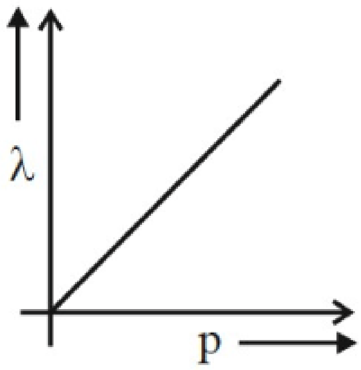
B.



C.



D.



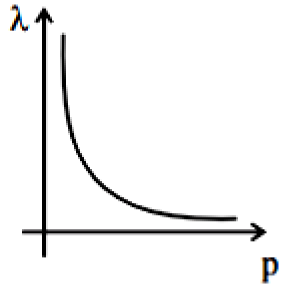
Answer: A

Solution:

$\lambda = \frac{h}{p}\left[\lambda \propto \frac{1}{p}\right]$

$\Rightarrow \lambda p = h(\text{ constant })$

So, the plot is a rectangular hyperbola



Question60

The number of radial node/s for 3p orbital is:

[1-Feb-2024 Shift 2]

Options:

A.

1

B.

4

C.

2

D.

3

Answer: A

Solution:

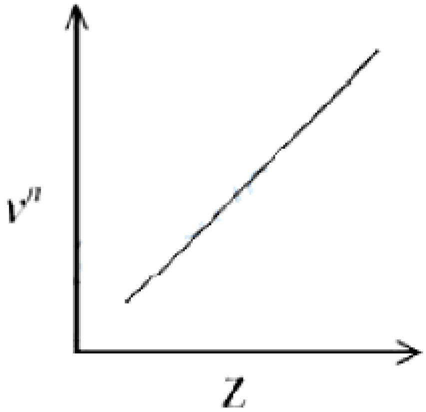
For 3p : $n = 3, \ell = 1$

Number of radial node $= n - \ell - 1$

$= 3 - 1 - 1 = 1$

Question61

It is observed that characteristic X-ray spectra of elements show regularity. When frequency to the power 'n' i.e. v^n of X-rays emitted is plotted against atomic number ' Z ', following graph is obtained.



The value of ' n ' is
[24-Jan-2023 Shift 1]

Options:

A. 1

B. 2

C. $\frac{1}{2}$

D. 3

Answer: C

Solution:

According to Henry Moseley $\sqrt{v} \alpha z - b$

So $n = \frac{1}{2}$

Question62

If wavelength of the first line of the Paschen series of hydrogen atom is 720 nm, then the wavelength of the second line of this series is _____nm
(Nearest integer)
[24-Jan-2023 Shift 1]

Answer: 492

Solution:

$$\frac{1}{(\lambda_1)_P} = R_H Z^2\left(\frac{1}{9} - \frac{1}{16}\right)$$

$$\frac{1}{(\lambda_2)_P} = R_H Z^2\left(\frac{1}{9} - \frac{1}{25}\right)$$

$$\frac{(\lambda_2)_P}{(\lambda_1)_P} = \frac{\frac{7}{16 \times 9}}{\frac{16}{25 \times 9}} = \frac{25 \times 7}{16 \times 16}$$

$$(\lambda_2)_P = \frac{25 \times 7}{16 \times 16} \times 720$$

$$(\lambda_2)_P = 492 \text{ nm}$$

Question63

The number of s-electrons present in an ion with 55 protons in its unipositive state is
[24-Jan-2023 Shift 2]

Options:

A. 8

B. 9

C. 12

D. 10

Answer: D

Solution:

$Z = 55[Cs] \Rightarrow [Xe]6s^1$
$[Cs^+] \Rightarrow [Xe]$ i.e. upto $5s$ count e^- of s-subshell
i.e. $1s, 2s, 3s, 4s, 5s \Rightarrow 10$ electrons

Question64

The radius of the 2 nd orbit of Li^{2+} is x. The expected radius of the 3 rd orbit of Be^{3+} is
[25-Jan-2023 Shift 1]

Options:

A. $\frac{9}{4}x$

B. $\frac{4}{9}x$

C. $\frac{27}{16}x$

D. $\frac{16}{27}x$

Answer: C

Solution:

Li^{2+}

$r_2 = x = k \times \frac{2^2}{3} = \frac{4k}{2}$

$\frac{y}{x} = \frac{9}{4} \times \frac{3}{4} = \frac{27}{16}$

$y = \frac{27}{16}x$

Be^{3+}

$r_3 = y = k \times \frac{3^2}{4}$

Question65

How many of the following metal ions have similar value of spin only magnetic moment in gaseous state?________
(Given: Atomic number: V, 23; Cr, 24; Fe, 26; Ni, 28)
$V^{3+} \cdot Cr^{3+}, Fe^{2+}, Ni^{3+}$
[25-Jan-2023 Shift 1]

Answer: 2

Solution:

$\mu_s = \sqrt{n(n+2)}BM$ ($n =$ no. of unpaired electrons)

V^{3+} : $[Ar]3d^2 4s^0$ 2

Cr^{3+} : $[Ar]3d^3 4s^0$ 3

Fe^{2+} : $[Ar]3d^6 4s^0$ 4

Ni^{3+} : $[Ar]3d^7 4s^0$ 3

Cr^{3+}& Ni^{3+} have same value of μ_s

Question66

The shortest wavelength of hydrogen atom in Lyman series is λ. The longest wavelength in Balmer series of He^{+} is
[29-Jan-2023 Shift 1]

Options:

A. $\frac{5}{9\lambda}$

B. $\frac{9\lambda}{5}$

C. $\frac{36\lambda}{5}$

D. $\frac{5\lambda}{9}$

Answer: B

Solution:

For H : $\frac{1}{\lambda} = R_H \times 1^2 \left(\frac{1}{1^2} - \frac{1}{\infty^2} \right) \ldots (1)$

$\frac{1}{\lambda_{He^+}} = R_H \times 2^2 \times \left(\frac{1}{4} - \frac{1}{9} \right) \ldots (2)$

From (1) & (2) $\frac{\lambda_{He^+}}{\lambda} = \frac{9}{5}$

$\lambda_{He^+} = \lambda \times \frac{9}{5}$

$\lambda_{He^+} = \frac{9\lambda}{5}$

Question67

Assume that the radius of the first Bohr orbit of hydrogen atom is 0.6Å. The radius of the third Bohr orbit of He^{+} is ______ picometer. (Nearest Integer)
[29-Jan-2023 Shift 2]

Answer: 270

Solution:

$$r \propto \frac{n^2}{Z}$$
$$r_{He^+} = r_H \times \frac{n^2}{Z}$$
$$r_{He^+} = 0.6 \times \frac{(3)^2}{2}$$
$$= 2.7 Å$$
$$r_{He^+} = 270 \text{ pm}$$

Question68

The energy of one mole of photons of radiation of frequency 2×10^{12} Hz in $Jmol^{-1}$ is _______.
(Nearest integer)
(Given: $h = 6.626 \times 10^{-34}$ Js
$N_A = 6.022 \times 10^{23} mol^{-1}$)
[30-Jan-2023 Shift 1]

Answer: 798

Solution:

For one photon $E = hv$
For one mole photon,
$$E = 6.023 \times 10^{23} \times 6.626 \times 10^{-34} \times 2 \times 10^{12}$$
$$= 798.16 J$$
$$\approx 798 J$$

Question69

Maximum number of electrons that can be accommodated in shell with n = 4 are:
[30-Jan-2023 Shift 2]

Options:

A. 16

B. 32

C. 50

D. 72

Answer: B

Solution:

The number of electrons in the orbitals of sub-shell of $n = 4$ are

4s	2
4p	6
4d	10
4f	14
(Total)	32

Question70

The wave function (Ψ) of 2 s is given by

$$\Psi_{2\,s} = \frac{1}{2\sqrt{2\pi}}\left(\frac{1}{a_0}\right)^{1/2}\left(2 - \frac{r}{a_0}\right)e^{-r/2a_0}$$

At $r = r_0$, radial node is formed. Thus, r_0 in terms of a_0
[30-Jan-2023 Shift 2]

Options:

A. $r_0 = a_0$

B. $r_0 = 4a_0$

C. $r_0 = \frac{a_0}{2}$

D. $r_0 = 2a_0$

Answer: D

Solution:

At node $\Psi_{2s} = 0$

$\therefore 2 - \frac{r_0}{a_0} = 0$

$\therefore r_0 = 2a_0$

Question71

Which transition in the hydrogen spectrum would have the same wavelength as the Balmer type transition from n = 4 to n = 2 of He^+spectrum
[31-Jan-2023 Shift 1]

Options:

A. n = 2 to n = 1

B. n = 1 to n = 3

C. n = 1 to n = 2

D. n = 3 to n = 4

Answer: A

Solution:

He^+ ion:

$$\frac{1}{\lambda(H)} = R(1)^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$\frac{1}{\lambda(He^+)} = R(2)^2\left[\frac{1}{2^2} - \frac{1}{4^2}\right]$$

Given $\lambda(H) = \lambda(He^+)$

$$R(1)^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] = R(4)\left[\frac{1}{2^2} - \frac{1}{4^2}\right]$$

$$\frac{1}{n_1^2} - \frac{1}{n_2^2} = \frac{1}{1^2} - \frac{1}{2^2}$$

On comparing $n_1 = 1 \& n_2 = 2$

Ans. 1

Question72

Arrange the following orbitals in decreasing order of energy ?
(A) n = 3, 1 = 0, m = 0
(B) n = 4, 1 = 0, m = 0
(C) n = 3, 1 = 1, m = 0
(D) n = 3, l = 2, m = 1
The correct option for the order is :
[31-Jan-2023 Shift 2]

Options:

A. B > D > C > A

B. D > B > C > A

C. A > C > B > D

D. D > B > A > C

Answer: B

Solution:

(A) $n = 3;\ l = 0;\ m = 0;\ 3\ s$ orbital
(B) $n = 4;\ 1 = 0;\ m = 0;\ 4\ s$ orbital
(C) $n = 3;\ 1 = 1;\ m = 0;\ 3p$ orbital
(D) $n = 3;\ 1 = 2;\ m = 0;\ 3d$ orbital

As per Hund's rule energy is given by $(n+1)$ value.
If value of $(n+1)$ remains same then energy is given by n only.

Question73

Electrons in a cathode ray tube have been emitted with a velocity of $1000 ms^{-1}$. The number of following statements which is/are true about the emitted radiation is _______.
Given : $h = 6 \times 10^{-34}$ Js, $m_e = 9 \times 10^{-31}$ kg.
(A) The deBroglie wavelength of the electron emitted is 666.67 nm.
(B) The characteristic of electrons emitted depend upon the material of the electrodes of the cathode ray tube.
(C) The cathode rays start from cathode and move towards anode.
(D) The nature of the emitted electrons depends on the nature of the gas present in cathode ray tube.
[1-Feb-2023 Shift 1]

Answer: 2

Solution:

(A) $V_e = 1000 m/s; h = 6 \times 10^{-34} Js;$
$m_e = 9 \times 10^{-31} kg$
$$\lambda = \frac{h}{mv} = \frac{6 \times 10^{-34}}{9 \times 10^{-31} \times 1000} = 666.67 \times 10^{-9} m$$
$= 666.67$ nm
(B) The characteristic of electrons emitted is independent of the material of the electrodes of the cathode ray tube.
(C) The cathode rays start from cathode and move towards anode.
(D) The nature of the emitted electrons is independent on the nature of the gas present in cathode ray tube.

Question74

Which one of the following sets of ions represents a collection of isoelectronic species?
(Given : Atomic Number: F : 9, Cl : 17, Na = 11,
Mg = 12, Al = 13, K = 19, Ca = 20, Sc = 21)
[1-Feb-2023 Shift 2]

Options:

A. (Li^{+}, Na^{+}, Mg^{2+}, Ca^{2+}.

B. Ba^{2+}, Sr^{2+}, K+, Ca^{2+}

C. CN^{3-}, O^{2-}, F^{-}, S^{2-}

D. (K^{+}, Cl^{-}, Ca^{2+}, Sc^{3+}.

Answer: D

Solution:

K^{+} Cl^{1} Ca^{2+} Sc^{3+}
18 18 18 18

Question75

The wavelength of an electron of kinetic energy 4.50×10^{-29} J is ...... $\times 10^{-5}$ m. (Nearest integer)
Given : mass of electron is 9×10^{-31} kg, h = 6.6×10^{-34} J s
[6-Apr-2023 shift 1]

Answer: 7

Solution:

$$\lambda_d = \frac{h}{mv} = \frac{h}{\sqrt{2\,mKE}} = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9 \times 10^{-31} \times 4.5 \times 10^{-29}}}$$

$$= \frac{6.6 \times 10^{-34}}{\sqrt{9^2 \times 10^{-60}}}$$

$$= \frac{6.6 \times 10^{-34}}{9 \times 10^{-30}} = \frac{6.6}{9} \times 10^{-4}$$

$= 7.3 \times 10^{-5}$m

Therefore Ans $= 7$

Question76

If the radius of the first orbit of hydrogen atom a_0, then de Broglie's wavelength of electron in 3^{rd} orbit is [6-Apr-2023 shift 2]

Options:

A. $\frac{\pi a_0}{6}$

B. $\frac{\pi a_0}{3}$

C. $6\pi a_0$

D. $3\pi a_0$

Answer: C

Solution:

$$(r_3)_H = \frac{a_0 n^2}{Z} = a_0 \times 3^2 = 9a_0$$

$2\pi r = n\lambda$

$\Rightarrow 2\pi \times 9a_0 = 3\lambda$

$\Rightarrow \lambda = 6\pi a_0$

Question77

The number of following statements which is/are incorrect is __________

[8-Apr-2023 shift 1]

Options:

A. Line emission spectra are used to study the electronic structure

B. The emission spectra of atoms in the gas phase show a continuous spread of wavelength from red to violet

C. An absorption spectrum is like the photographic negative of an emission spectrum

D. The element helium was discovered in the sun by spectroscopic method

Answer: A

Solution:

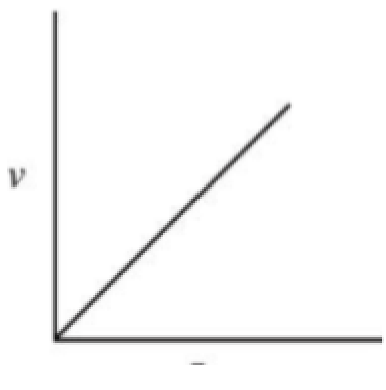
Fact

Question78

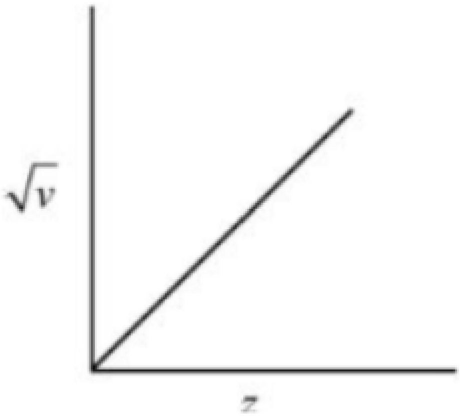
Henry Moseley studied characteristic X-ray spectra of elements. The graph which represents his observation correctly is
Given v = frequency of X-ray emitted
Z = atomic number
[8-Apr-2023 shift 2]

Options:

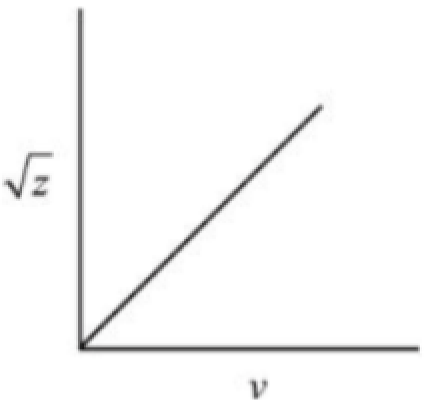
A.



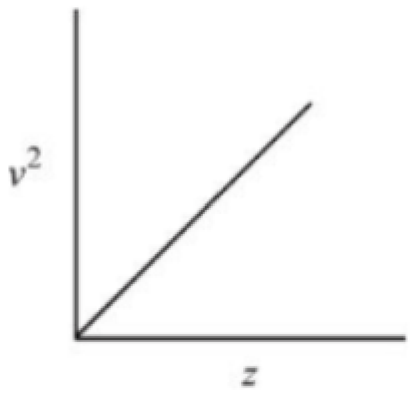
B.



C.



D.



Answer: B

Solution:

$\sqrt{v} \alpha Z$

Question79

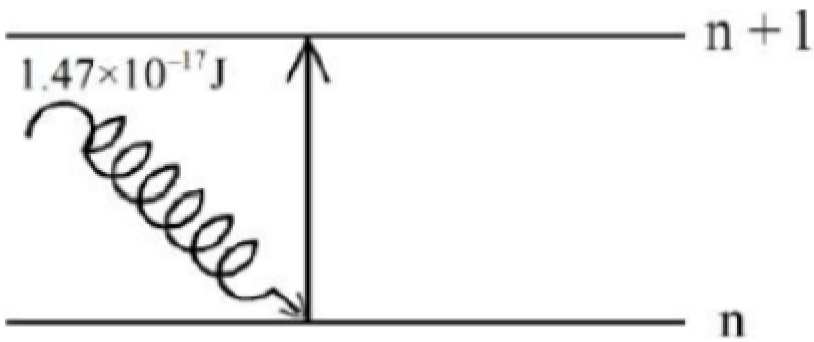
The number of atomic orbitals from the following having 5 radial nodes is _______ 7 s, 7p, 6 s, 8p, 8d
[8-Apr-2023 shift 2]

Answer: 3

Solution:

No. of radial node
$= n - \ell - 1$
For $6S \rightarrow 6 - 0 - 1 = 5$
$7P \rightarrow 7 - 1 - 1 = 5$
$8d \rightarrow 8 - 2 - 1 = 5$

Question80



The electron in the nth orbit of Li^{2+} is excited to (n + 1) orbit using the radiation of energy 1.47×10^{-17}J(as shown in the diagram). The value of n is ________
Given: $R_H = 2.18 \times 10^{-18}$J
[10-Apr-2023 shift 2]

Answer: 1

Solution:

$$\Delta E = R_H Z^2 \left(\frac{1}{n_1^{\ 2}} - \frac{1}{n_2^{\ 2}} \right)$$
$$1.47 \times 10^{-17} = 2.18 \times 10^{-18} \times 9 \left(\frac{1}{n^2} - \frac{1}{(n+1)^2} \right)$$
$$\frac{1.47}{1.96} = \frac{3}{4} = \frac{1}{n^2} - \frac{1}{(n+1)^2}$$
So, $n = 1$

Question81

For a metal ion, the calculated magnetic moment is 4.90 BM. This metal ion has ________ number of unpaired electrons.
[10-Apr-2023 shift 2]

Answer: 4

Solution:

$\mu = 4.90$ BM.
$\mu = \sqrt{n(n+2)}$
So, $n = 4$

Question82

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A : In the photoelectric effect electrons are ejected from the metal surface as soon as the beam of light of frequency greater than threshold frequency strikes the surface.
Reason R: When the photon of any energy strikes an electron in the atom transfer of energy from the photon to the electron takes place.
In the light of the above statements, choose the most appropriate answer from the options given below:
[11-Apr-2023 shift 1]

Options:

A. A is correct but R is not correct

B. A is not correct but R is correct

C. Both A and R correct and R is the correct explanation of A

D. Both A and R are correct but R is NOT the correct explanation of A

Answer: A

Solution:

Assertion A is correct but Reason is not correct.

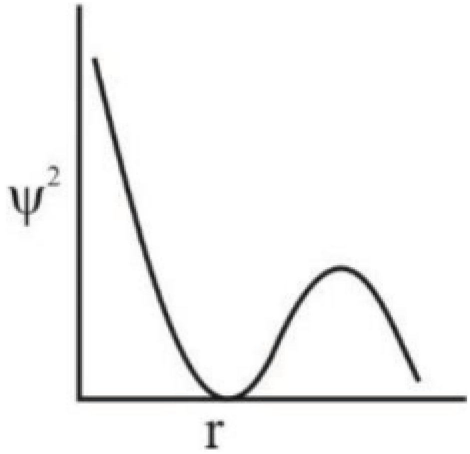
Question83

The number of correct statements from the following is
A. For 1 s orbital, the probability density is maximum at the nucleus
B. For 2 s orbital, the probability density first increases to maximum and then decreases sharply to zero.
C. Boundary surface diagrams of the orbitals encloses a region of 100% probability of finding the electron.
D. p and d-orbitals have 1 and 2 angular nodes respectively
E. probability density of p-orbital is zero at the nucleus
[11-Apr-2023 shift 2]

Answer: 3

Solution:

A, D and E statements are correct.



For 2 s orbital, the probability density first decreases and then increases.
At any distance from nucleus the probability density of finding electron is never zero and it always have some finite value.

Question84

Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R
Assertion A: 5f electrons can participate in bonding to a far greater extent than 4f electrons
Reason R: 5f orbitals are not as buried as 4f orbitals In the light of the above statements,
choose the correct answer from the options given below
[12-Apr-2023 shift 1]

Options:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is NOT the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A

Solution:

Due to this reason actinoids participate in more bonding.

Question85

Values of work function (W_0) for a few metals are gives below

Metal	Li	Na	K	Mg	Cu	Ag
W_0/eV	2.42	2.3	2.25	3.7	4.8	4.3

The number of metals which will show photoelectric effect when light of wavelength 400 nm falls on it is
Given: $h = 6.6 \times 10^{-34}$ J s
$c = 3 \times 10^{8} ms^{-1}$
$e = 1.6 \times 10^{-19}C$
[12-Apr-2023 shift 1]

Answer: 3

Solution:

$E(ev) = \frac{1240}{400} = 3.1$ ev
Mg, Cu, Ag

Question86

The energy of an electron in the first Bohr orbit of hydrogen atom is -2.18×10^{-18}J. Its energy in the third Bohr orbit is _______.
[13-Apr-2023 shift 1]

Options:

A. $\frac{1}{27}$ of this value

B. $\frac{1}{9}$ th of this value

C. One third of this value

D. Three times of this value

Answer: B

Solution:

$$E_{1,1} = -2.18 \times 10^{-18} J$$
$$E_{3,1} = E_{1,1} \times \frac{1^2}{3^2}$$
$$E_{3,1} = \frac{1}{9} \times E_{1,1}$$

Question87

The orbital angular momentum of an electron in 3 s orbital is $\frac{xh}{2\pi}$. The value of x is _______ (nearest integer)
[13-Apr-2023 shift 2]

Answer: 0

Solution:

Orbital angular momentum $= \sqrt{1(1+1)} \frac{h}{2\pi}$

Value of 1 for $s = 0$

Question88

The total number of isoelectronic species from the given set is ________.
O^{2-}, F^{-}, Al, Mg^{2+}, Na^{+}, O^{+}, Mg, Al^{3+}, F
[15-Apr-2023 shift 1]

Answer: 5

Solution:

O^{-2},	F^{-}	mg^{+2}	Na^{+1}	Al^{+3}	Isoelectronic species
$10e^{-}$	$10e^{-}$	$10e^{-}$	$10e^{-}$	$10e^{-}$	

Question89

Consider the following pairs of electrons
(A)
(a)n = 3, l = 1, $m_1 = 1$, $m_s = +\frac{1}{2}$
(b) n = 3, l = 2, $m_1 = 1$, $m_s = +\frac{1}{2}$
(B)
(a) n = 3, l = 2, $m_1 = -2$, $m_s = -\frac{1}{2}$
(b) n = 3, l = 2, $m_1 = -1$, $m_s = -\frac{1}{2}$
(C)
(a)n = 4, l = 2, $m_1 = 2$, $m_s = +\frac{1}{2}$
(b) n = 3, l = 2, $m_1 = 2$, $m_s = +\frac{1}{2}$
The pairs of electrons present in degenerate orbitals is/are :
[24-Jun-2022-Shift-1]

Options:

A. Only (A)

B. Only (B)

C. Only (C)

D. (B) and (C)

Answer: B

Solution:

For degenerate orbitals, only the value of m must be different.
The value of $(n+1)$ must be the same.Hence, the pair of electrons with quantum numbers given in (B) are degenerate.

Question90

The energy of one mole of photons of radiation of wavelength 300nm is (Given : h $= 6.63 \times 10^{-34}$J s, $N_A = 6.02 \times 10^{23}$mol^{-1}, c $= 3 \times 10^{8}$ms^{-1}) [24-Jun-2022-Shift-2]

Options:

A. 235kJ mol^{-1}

B. 325kJ mol^{-1}

C. 399kJ mol^{-1}

D. 435kJ mol^{-1}

Answer: C

Solution:

Energy of one photon

$$E = \frac{1240}{\lambda(\text{nm})} \text{eV}$$

$$= \frac{1240}{300}$$

$$= 4.1333 \text{ eV}$$

$\therefore$ Energy of one mole of photon

$= 4.1333 \times 6.02 \times 10^{23}$ eV
$= 4.1333 \times 6.02 \times 10^{23} \times 1.6 \times 10^{-19}$J
$= \dfrac{4.1333 \times 6.02 \times 10^{23} \times 1.6 \times 10^{-19}}{1000}$ kJ
$= 399$ kJ/mol

Question91

The pair, in which ions are isoelectronic with Al^{3+} is : [25-Jun-2022-Shift-1]

Options:

A. Br^{-} and Be^{2+}

B. Cl^{-} and Li^{+}

C. S^{2-} and K^{+}

D. O^{2-} and Mg^{2+}

Answer: D

Solution:

O^{2-}, Mg^{2+} and Al^{3+} are isoelectronic. All have 10 electrons.

Question92

The longest wavelength of light that can be used for the ionisation of lithium atom (Li) in its ground state is $x \times 10^{-8}$m. The value of x is___ (Nearest Integer). (Given : Energy of the electron in the first shell of the hydrogen atom is -2.2×10^{-18}J; h = 6.63×10^{-34} Js and c = 3×10^{8}ms^{-1}) [25-Jun-2022-Shift-1]

Answer: 4

Solution:

Bohr model is not valid for lithium atom (Li) as Bohr model is valid for only single electronic species, so it would be valid for Li^{+2} but not Li atom.

Question93

The minimum energy that must be possessed by photons in order to produce the photoelectric effect with platinum metal is :
[Given: The threshold frequency of platinum is $1.3 \times 10^{15} s^{-1}$ and $h = 6.6 \times 10^{-34} Js$.]
[25-Jun-2022-Shift-2]

Options:

A. $3.21 \times 10^{-14} J$

B. $6.24 \times 10^{-16} J$

C. $8.58 \times 10^{-19} J$

D. $9.76 \times 10^{-20} J$

Answer: C

Solution:

The minimum energy possessed by photons will be equal to the work function of the metal.

$\therefore E_{min} = h\nu_0 J$

$= 6.6 \times 10^{-34} \times 1.3 \times 10^{15}$

$= 8.58 \times 10^{-19} J$

Question94

If the radius of the 3^{rd} Bohr's orbit of hydrogen atom is r_3 and the radius of 4^{th} Bohr's orbit is r_4. Then :
[26-Jun-2022-Shift-1]

Options:

A. $r_4 = \frac{9}{16} r_3$

B. $r_4 = \frac{16}{9} r_3$

C. $r_4 = \frac{3}{4} r_3$

D. $r_4 = \frac{4}{3} r_3$

Answer: B

Solution:

We know,
$r = r_0 \times \frac{n^2}{z}$
For hydrogen atom,
$\therefore r_3 = r_0 \times \frac{3^2}{1}$
$\Rightarrow r_0 \times \frac{r_3}{9}$
and $r_4 = r_0 \times \frac{4^2}{1}$
$= \frac{r_3}{9} \times 16$
$= \frac{16}{9} r_3$

Question95

The number of radial and angular nodes in 4d orbital are, respectively
[26-Jun-2022-Shift-2]

Options:

A. 1 and 2

B. 3 and 2

C. 1 and 0

D. 2 and 1

Answer: A

Solution:

We know,
Radial nodes $= n - l - 1$
and Angular nodes $= l$
For 4d orbital,
$n = 4$
$l = 2$
$\therefore$ Radial nodes $= 4 - 2 - 1 = 1$
Angular nodes $= 2$

Question96

If the uncertainty in velocity and position of a minute particle in space are, $2.4 \times 10^{-26} (ms^{-1})$ and $10^{-7} (m)$ respectively. The mass of the particle in g is____ (Nearest integer)
(Given : $h = 6.626 \times 10^{-34}$ Js)
[27-Jun-2022-Shift-1]

Answer: 22

Solution:

We know from hisenberg uncertainty principle
$\Delta x \cdot \Delta p = \frac{h}{4\pi}$
$\Rightarrow \Delta x \cdot m \Delta v = \frac{h}{4\pi}$
Given,

$\Delta x = 10^{-7} m$
$\Delta x = 2.4 \times 10^{-26} m/ s$
$h = 6.626 \times 10^{-34} Js$
$\therefore 10^{-7} \times m \times 2.4 \times 10^{-26} = \dfrac{6.626 \times 10^{-34}}{4\pi}$
$\Rightarrow 2.4m = \dfrac{6.626}{4\pi \times 10}$
$\Rightarrow m = 0.022\ kg$
$\Rightarrow m = 22\ gm$

Question97

Consider the following set of quantum numbers.

	n	1	m_1
A.	3	3	3
B.	3	2	2
C.	2	1	+1
D.	2	2	+2

The number of correct sets of quantum numbers is __________. [27-Jun-2022-Shift-2]

Answer: 2

Solution:

For A,
Given $n = 3$ and $l = 3$
but we know maximum value of $I = n - 1$.
$\therefore$ I can't be equal to n.
So, Set A of quantum numbers is not possible.
For B,
Given $n = 3, l = 2, m = -2$
Here, $l = 2$ which follow the rule $l = n - 1$.
And we know possible value of m is $-I$ to $+I$.

here possible value of $m = -2$ to $+2$
∴ This Set B is valid set of quantum numbers.
For C,
Given $n = 2, I = 1, m = +1$
Here $I = 1$ which follows the rule $I = n - 1$.
For $I = 1$ possible value of $m = -1$ to $+1$
Here $m = +1$. So value of m is valid.
∴ Set C is valid set of quantum numbers.
For D,
Given $n = 2, I = 2, m = +2$
$I = 2$ does not follow the rule $I = n - 1$ rule.
∴ Set D is not valid set of quantum numbers.

Question98

If the work function of a metal is 6.63×10^{-19}J, the maximum wavelength of the photon required to remove a photoelectron from the metal is___ nm. (Nearest integer)
[Given : $h = 6.63 \times 10^{-34}$Js, and $c = 3 \times 10^{8}ms^{-1}$]
[28-Jun-2022-Shift-1]

Answer: 300

Solution:

Given,
$$\text{Work function } = 6.63 \times 10^{-19}\text{J}$$
$$= \frac{6.63 \times 10^{-19}}{1.6 \times 10^{-19}}$$
$$= 4.14\text{ eV}$$
We know,
$$E = \frac{1240}{\lambda(\text{nm})}$$
$$\Rightarrow 4.14 = \frac{1240}{\lambda}$$
$$\lambda = 300\text{ nm}$$

Question99

Consider the following statements :
(A) The principal quantum number ' n ' is a positive integer with values of ' n ' = 1, 2, 3, ...
(B) The azimuthal quantum number 'I' for a given 'n' (principal quantum number) can have values as 'I' = 0, 1, 2,..... n
(C) Magnetic orbital quantum number 'm' for a particular 'l' (azimuthal quantum number) has (2l + 1) values.
(D) ±1/2 are the two possible orientations of electron spin.
(E) For I = 5, there will be a total of 9 orbital
Which of the above statements are correct?
[28-Jun-2022-Shift-2]

Options:

A. (A), (B) and (C)

B. (A), (C), (D) and (E)

C. (A), (C) and (D)

D. (A), (B), (C) and (D)

Answer: C

Solution:

(A) Principle quantum number n is a positive integer and it's possible values are $\mathrm{n}=1,2,3 \ldots \ldots$
$\therefore \mathrm{A}$ is correct.
(B) Azimuthal quantum number 'I' for a given ' n ' can have values as $\mathrm{I}=0,1,2 \ldots \ldots(\mathrm{n}-1)$
$\therefore$ Statement B is wrong.
(C) Magnetic orbital quantum number m_{l} for particular I has values from $-\mathrm{I}$ to $+1$ including zero means $2 \mathrm{l}+1$ values.
$\therefore$ Statement C is correct.
(D) $\pm \frac{1}{2}$ are the two possible orientation of electron spin.
$\therefore$ Statement D is correct.
(E) For $\mathrm{I}=5$, there will be a total of 11 orbitals.
$\mathrm{l}=0 \Rightarrow 5$ subshell
$\mathrm{l}=1 \Rightarrow \mathrm{p}$ subshell
$\mathrm{l}=2 \Rightarrow \mathrm{d}$ subshell
$\mathrm{l}=3 \Rightarrow \mathrm{f}$ subshell
$\mathrm{l}=4 \Rightarrow \mathrm{g}$ subshell
$\mathrm{l}=5 \Rightarrow \mathrm{h}$ subshell
We know,
Number of orbital in any subshell $=2 \mathrm{l}+1$.
$\therefore$ For h subshell, number of orbitals $=2 \times 5+1=11$

Question100

Which of the following statements are correct?
(A) The electronic configuration of Cr is $[Ar]3d^5 4s^1$.
(B) The magnetic quantum number may have a negative value.
(C) In the ground state of an atom, the orbitals are filled in order of their increasing energies.
(D) The total number of nodes are given by $n - 2$.
Choose the most appropriate answer from the options given below : [29-Jun-2022-Shift-1]

Options:

A. (A), (C) and (D) only

B. (A) and (B) only

C. (A) and (C) only

D. (A), (B) and (C) only

Answer: D

Solution:

(A) $Cr(24) = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$
$= [Ar]3d^5 4s^1$
(B) Magnetic quantum number (m) values ranging from $-I$ to $+I$ including zero.
$\therefore$ It can have negative value.
(C) According to Aufbau rule, electrons are filled first in these orbitals which have low energy.
$\therefore$ Statement C is correct.
(D) We know,
Number of Radial nodes $= n - I - 1$
and number of Angular nodes $= 1$
$\therefore$ Total nodes $= n - I - 1 + 1 = n - 1$

Question101

The electronic configuration of Pt (atomic number 78) is: [29-Jun-2022-Shift-1]

Options:

A. $[Xe]4f^{14}5d^{9}6s^{1}$

B. $[Kr]4f^{14}5d^{10}$

C. $[Xe]4f^{14}5d^{10}$

D. $[Xe]4f^{14}5d^{8}6s^{2}$

Answer: A

Solution:

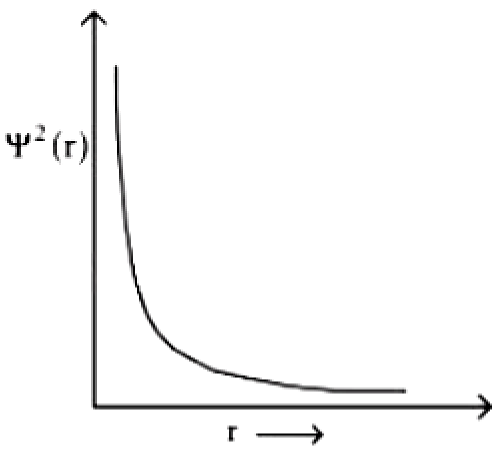
Atomic number of Pt is 78
Electronic configuration is $-{}_{78}Pt \rightarrow [Xe]4f^{14}5d^{9}6\ s^{1}$ (Exceptional electronic configuration)

Question102

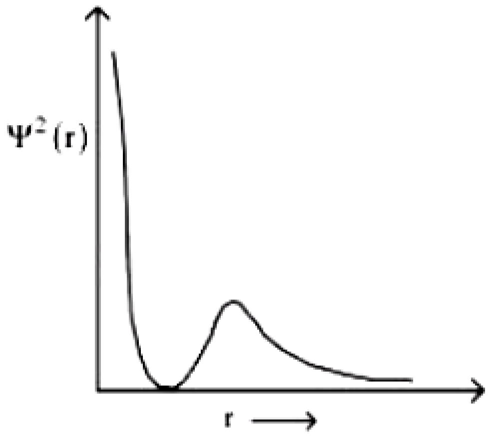
Which of the following is the correct plot for the probability density $\psi^{2}(r)$ as a function of distance ' r ' of the electron from the nucleus for 2s orbital?
[29-Jun-2022-Shift-2]

Options:

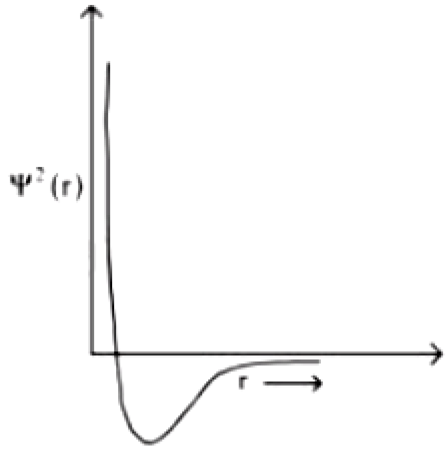
A.



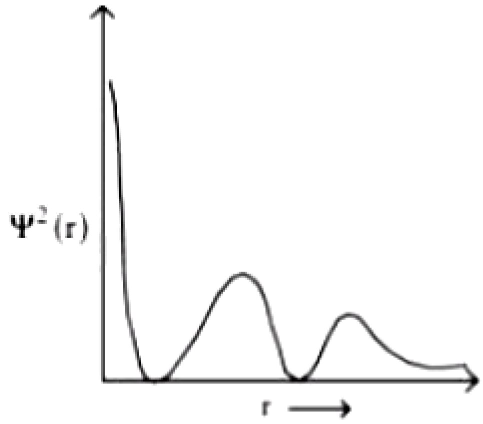
B.



C.



D.



Answer: B

Solution:

Formula for number of radial nodes in n^{th} orbital $= n - l - 1$
For $2s$, number of radial nodes $= 2 - 0 - 1 = 1$ and value of ψ^2 is always positive.

Question103

Which of the following sets of quantum numbers is not allowed? [25-Jul-2022-Shift-1]

Options:

A. $n = 3, 1 = 2, m_{l} = 0, s = +\frac{1}{2}$

B. $n = 3, l = 2, m_{l} = -2, s = +\frac{1}{2}$

C. $n = 3, l = 3, m_{l} = -3, s = -\frac{1}{2}$

D. $n = 3, l = 0, m_{l} = 0, s = -\frac{1}{2}$

Answer: C

Solution:

If $n = 3$, then possible values of $l = 0, 1, 2$
But in option (C), the value of 1 is given ' 3 ', this is not possible.

Question104

When the excited electron of a H atom from n = 5 drops to the ground state, the maximum number of emission lines observed are______ [25-Jul-2022-Shift-2]

Answer: 10

Solution:

Maximum number of emission lines

$$= \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

$n_2 = 5$

$n_1 = 1$

$$\Rightarrow \frac{(5-1)(5-1+1)}{2} = 10$$

Hence maximum number of emission lines observed are 10 .

Question105

The wavelength of an electron and a neutron will become equal when the velocity of the electron is x times the velocity of neutron. The value of x is ________.(Nearest Integer)

(Mass of electron is 9.1×10^{-31} kg and mass of neutron is 1.6×10^{-27} kg)

[26-Jul-2022-Shift-1]

Answer: 1758

Solution:

$$\lambda_e = \frac{h}{m_e \times V_e}, \quad \lambda_N = \frac{h}{m_N \times V_N}$$

$\lambda_e = \lambda_N$ When $V_e = xV_N$

$$\frac{1}{m_e V_e} = \frac{1}{m_N \times V_N}$$

$$\frac{m_N}{m_e} = \frac{V_e}{V_N} = x$$

$$x = \frac{1.6 \times 10^{-27}}{9.1 \times 10^{-31}}$$

$$= 0.17582 \times 10^4$$

$$\simeq 1758$$

Question106

Consider an imaginary ion ${}_{22}^{48}X^{3-}$. The nucleus contains ' a ' % more neutrons than the number of electrons in the ion. The value of ' a ' is ________ . [nearest integer]
[26-Jul-2022-Shift-2]

Answer: 4

Solution:

Number of electrons in ${}_{22}^{48}X^{3-}$ is 25 .
Number of neutrons $= 48 - 22 = 26.$
% increase in the number of neutrons over electrons
$= \left(\frac{26-25}{25} \right) 100 = 4\%$
$\therefore a = 4$

Question107

Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: Energy of 2s orbital of hydrogen atom is greater than that of 2s orbital of lithium.
Reason R: Energies of the orbitals in the same subshell decrease with increase in the atomic number.
In the light of the above statements, choose the correct answer from the options given below.
[27-Jul-2022-Shift-1]

Options:

A. Both A and R are true and R is the correct explanation of A.

B. Both A and R are true but R is N OT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Answer: A

Solution:

As the atomic number increases then the potential energy of electrons present in same shell becomes more and more negative. And therefore total energy also becomes more negative.

$E_{total} = -13.6 \frac{Z^2}{n^2} eV$

∴ Energies of the orbitals in the same subshell decreases with increase in atomic number.

Question108

The correct decreasing order of energy for the orbitals having, following set of quantum numbers :
(A) n = 3, I = 0, m = 0
(B) n = 4, I = 0, m = 0
(C) n = 3, I = 1, m = 0
(D) n = 3, l = 2, m = 1
is :
[27-Jul-2022-Shift-2]

Options:

A. (D) >(B) > (C) > (A)

B. (B) >(D) > (C) > (A)

C. (C) > (B) > (D) > (A)

D. (B) > (C) > (D) > (A)

Answer: A

Solution:

(A) $n + \ell = 3 + 0 = 3$
(B) $n + \ell = 4 + 0 = 4$
(C) $n + \ell = 3 + 1 = 4$
(D) $n + \ell = 3 + 2 = 5$
Higher $n + \ell$ value, higher the energy \& if same $n + \ell$ value, then higher n value, higher the energy.
Thus : $D > B > C > A$.

Question109

Identify the incorrect statement from the following.
[28-Jul-2022-Shift-1]

Options:

A. A circular path around the nucleus in which an electron moves is proposed as Bohr's orbit.

B. An orbital is the one electron wave function (ψ) in an atom.

C. The existence of Bohr's orbits is supported by hydrogen spectrum.

D. Atomic orbital is characterised by the quantum numbers n and l only.

Answer: D

Solution:

Atomic orbital is characterised by the quantum numbers n, l and m.
Hence option D is incorrect.

Question110

If the wavelength for an electron emitted from H-atom is 3.3×10^{-10}m, then energy absorbed by the electron in its ground state compared to minimum energy required for its escape from the atom, is times. (Nearest integer)
[Given : h = 6.626×10^{-34}J s]
Mass of electron $= 9.1 \times 10^{-31}$ kg
[28-Jul-2022-Shift-2]

Answer: 2

Solution:

$$\lambda = \frac{h}{mv}$$

$$\Rightarrow mv = \frac{h}{\lambda} = \frac{6.626 \times 10^{-34} \text{kg} \frac{m^2}{\sec^2} \times \sec}{3.3 \times 10^{-10} m}$$

$$mv = \frac{6.626 \times 10^{-24}}{3.3} = 2 \times 10^{-24} \text{kg m sec}^{-1}$$

Kinetic energy $= \frac{1}{2} mv^2$

$$= \frac{(mv)^2}{2m}$$

$$= \frac{(2 \times 10^{-24})^2}{2 \times 9.1 \times 10^{-31} \text{kg}}$$

$$= 2.18 \times 10^{-18} J$$

$$= 21.8 \times 10^{-19} J$$

Total energy absorbed = Ionization energy + Kinetic energy

$$= (21.76 + 21.8) \times 10^{-19}$$

$$= 43.56 \times 10^{-19} J$$

$\approx$2 times of $21.76 \times 10^{-19} J$

Question111

The minimum uncertainty in the speed of an electron in an one dimensional region of length $2a_o$ (Where a_o = Bohr radius 52.9 pm) is ________ $km\ s^{-1}$

(Given : Mass of electron $= 9.1 \times 10^{-31}$ kg, Planck's constant $h = 6.63 \times 10^{-34}$ Js)

[29-Jul-2022-Shift-1]

Answer: 548

Solution:

Heisenberg's uncertainty principle

$\Delta x \times \Delta P_x \geq \frac{h}{4\pi}$

$\Rightarrow 2a_0 \times m \, \Delta v_x = \frac{h}{4\pi}(\text{ minimum })$

$\Rightarrow \Delta v_x = \frac{h}{4\pi} \times \frac{1}{2a_0} \times \frac{1}{m}$

$= 6.63 \times 10^{-34}$

$4 \times 3.14 \times 2 \times 52.9 \times 10^{-12} \times 9.1 \times 10^{-31}$

$= 548273 ms^{-1}$

$= 548.273 kms^{-1}$

$= 548 kms^{-1}$

Question112

Given below are the quantum numbers for 4 electrons.
A. $n = 3, 1 = 2, m_1 = 1, m_s = +1/2$
B. $n = 4, 1 = 1, m_1 = 0, m_s = +1/2$
C. $n = 4, 1 = 2, m_1 = -2, m_s = -1/2$
D. $n = 3, 1 = 1, m_1 = -1, m_s = +1/2$
The correct order of increasing energy is
[29-Jul-2022-Shift-2]

Options:

A. D < B < A < C

B. D < A < B < C

C. B < D < A < C

D. B < D < C < A

Answer: B

Solution:

Energy of the sub-shell is given by, $(n+1)$ rule.

$A \Rightarrow 3d \Rightarrow n + 1 = 5$

$B \Rightarrow 4p \Rightarrow n + \lambda = 5$

$C \Rightarrow 4d \Rightarrow n + \ell \Rightarrow 6$

$D \Rightarrow 3s \Rightarrow (n+\ell) = 4$
$D < A < B < C$

Question113

Electromagnetic radiation of wavelength 663nm is just sufficient to ionise the atom of metal A. The ionisation energy of metal A in $kJ\ mol^{-1}$ is ....(Rounded off to the nearest integer)
$[h = 6.63 \times 10^{-34} J - s,\ c = 3.00 \times 10^{8} ms^{-1},$
$N_A = 6.02 \times 10^{23} mol^{-1}]$
[25 Feb 2021 Shift 2]

Answer: 180

Solution:

Energy of EMR = IE of the metal (A)

$$= h\nu = \frac{hc}{\lambda} atom^{-1} = -\frac{hc}{\lambda} \times N_A mol^{-1}$$

$$= \frac{(6.63 \times 10^{-34}) \times (3 \times 10^{8}) \times (6.02 \times 10^{23})}{(663 \times 10^{-9})} J\ mol^{-1}$$

$[\because \lambda = 663nm = 663 \times 10^{-9} m]$

$$= 180600 J\ mol^{-1} = 180.6 kJ\ mol^{-1} \sim eq 180 kJ\ mol^{-1}$$

Question114

According to Bohr's atomic theory,
I. kinetic energy of electron is $\propto \frac{Z^2}{n^2}$
II. the product of velocity (v) of electron and principal quantum number (n), $'vn' \propto Z^2$.
III. frequency of revolution of electron in an orbit is $\propto \frac{Z^3}{n^3}$
IV. coulombic force of attraction on the electron is $\propto \frac{Z^3}{n^4}$

Choose the most appropriate answer from the options given below. [24 Feb 2021 Shift 2]

Options:

A. Only III

B. Only I

C. I, III and IV

D. I and IV

Answer: D

Solution:

According to Bohr's theory,

I. $KE \propto \frac{Z^2}{n^2}$ or $13.6 \propto \frac{Z^2}{n^2} \frac{(eV)}{(\text{ atom })}$ (∴ Correct)

II. Speed of electron $\propto \frac{Z}{n}$

(Here, Z = atomic number, $n =$ number of shells)

∴ $v \times n \propto Z$ (∴ Incorrect)

III. Frequency of revolution of electron $= \frac{v}{2\pi r}$

Frequency $\propto \frac{Z^2}{n^3}$ $\left(\because v \propto \frac{z}{n}, r \propto \frac{n^2}{z}\right)$ (∴ Incorrect)

IV. $F = \frac{kq_1q_2}{r^2} = \frac{kZe^2}{r^2}$

$F = \frac{Z}{\left(\frac{n^2}{Z}\right)^2}$

$F \propto \frac{Z^3}{n^4}$ (∴ Correct)

Hence, only I, and IV are correct.

Question115

A ball weighing 10g is moving with a velocity of $90ms^{-1}$. If the uncertainty in its velocity is 5%, then the uncertainty in its position is ……… $\times 10^{-33}$m (Rounded off to the nearest integer). [Given,

$h = 6.63 \times 10^{-34} J - s$]
[26 Feb 2021 Shift 2]

Answer: 1

Solution:

According to Heisenberg's uncertainty equation,
$\Delta x \times m \Delta v \geq \frac{h}{4\pi}$ $(\therefore m = 10g = 10 \times 10^{-3} kg, v = 90 ms^{-1})$
Uncertainty in position,

$$\Delta x = \frac{h}{4\pi} \times \frac{1}{m \times \Delta v(\text{ with uncertainty })}$$

$$= \frac{6.63 \times 10^{-34}}{4 \times 3.14} \times \frac{1}{(10 \times 10^{-3}) \times \left(90 \times \frac{5}{100}\right)} m$$

$$= 1.17 \times 10^{-33} m$$

$$= 1 \times 10^{-33} m$$

Question116

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) In TlI_3, isomorphous to CsI_3, the metal is present in +1 oxidation state.
Reason (R) Tl metal has fourteen f -electrons in the electronic configuration.
In the light of the above statements, choose the most appropriate answer from the options given below.
[26 Feb 2021 Shift 2]

Options:

A. A is correct but R is not correct.

B. Both A and R are correct and R is the correct explanation of A.

C. A is not correct but R is correct

D. Both A and R are correct but R is not correct explanation of A.

Answer: B

Solution:

Assertion (A) is correct. TII_3 and CsI_3 are the triiodide, I_3^- (polyhalide) compounds of TI^+ and Cs^+ ions. Due to inert pair effect of group -13 elements, $T1^+$ is more stable than TH^{3+}. Cs and Tl belong to the same period (6th).
Sizes of Cs^+ and $T1^+$ are nearly same.
So, geometrical network and lattice pattern of both CsI_3 and TII_3 are same (bcc lattice). CsI_3 and TII_3 are also able to form mixed crystals when crystallisation is carried out from a mixture of saturated solutions of CsI_3 and TII_3. These properties enable TII_3 and CsI_3 to be isomorphous crystals.
Reason (R) is also correct. Electronic configuration of Cs $(Z = 55)$ and $TI(Z = 81)$ as follows
$_{55}Cs : [_{54}Xe]6s^1$
$_{81}Tl : [_{54}Xe]4f^{14}{}_{5d}{}^{10}6s^2 6p^1$
So, Tl has fourteen f-electrons in the anti-penultimate (4th) shell. A and R are individually correct and R is the correct explanation of A.

Question117

The orbital having two radial as well as two angular nodes is [26 Feb 2021 Shift 1]

Options:

A. 3p

B. 4f

C. 4d

D. 5d

Answer: D

Solution:

Number of radial nodes $= (n - I - 1)$
[$n =$ principal quantum number, $I =$ az imuthal quantum number] Number of angular nodes $= 1$
(a) $3p(n = 3, I = 1) \Rightarrow 1 \quad 1$

(b) $4f\,(n=4, I=3) \Rightarrow 0 \;\; 3$
(c) $4d\,(n=4, I=2) \Rightarrow 1 \;\; 2$
(d) $5d\,(n=5, I=2) \Rightarrow 2 \;\; 2$
So, $5d$-orbital has two radial as well as two angular nodes (option-d). Note $I=0$ for s-orbital, $I=1$ for p-orbital, $I=2$ for d-orbital and $I=3$ for $f-$orbital.

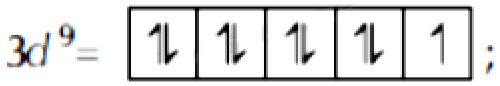
Question118

The spin only magnetic moment of a divaler ion in aqueous solution (atomic number $=29$) is .... BM.
[25 Feb 2021 Shift 2]

Answer: 2

Solution:

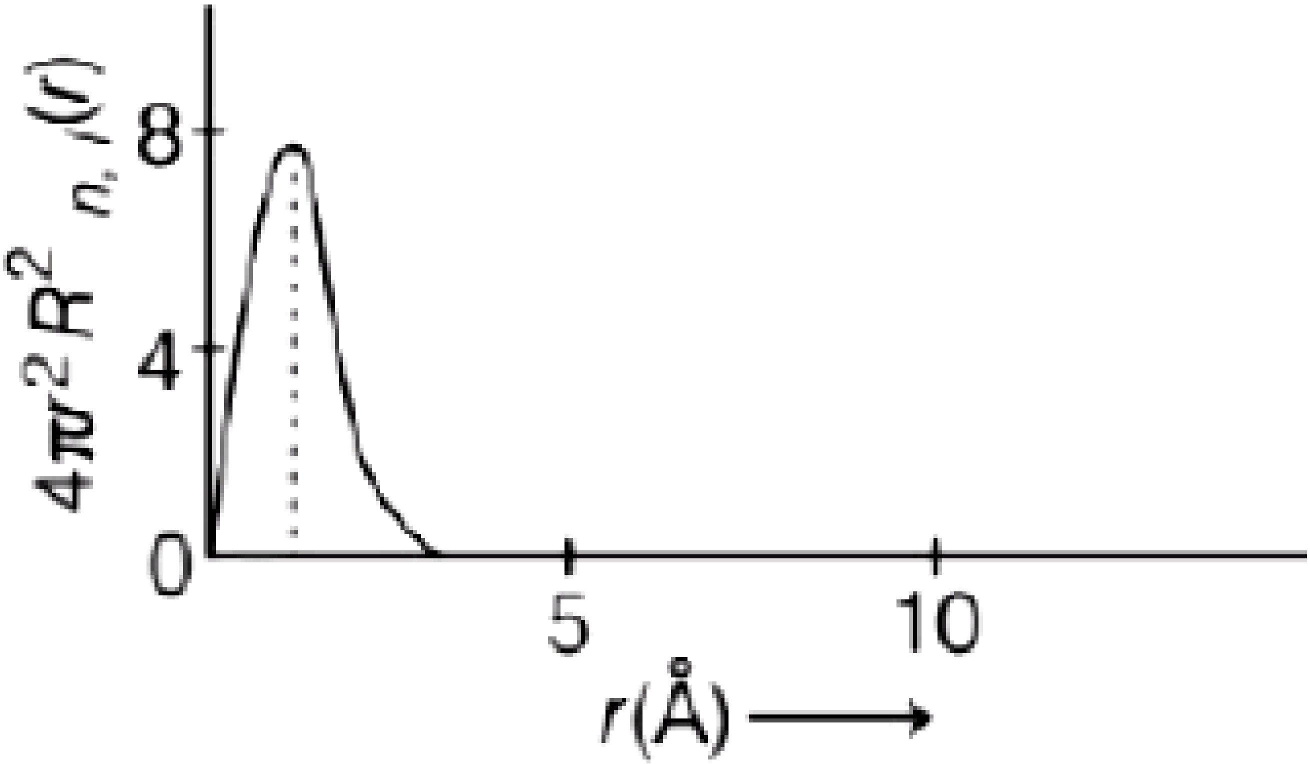
$Z=29[Cu] \xrightarrow{-2e^-} Cu^{2+}=[Ar]3d^9$

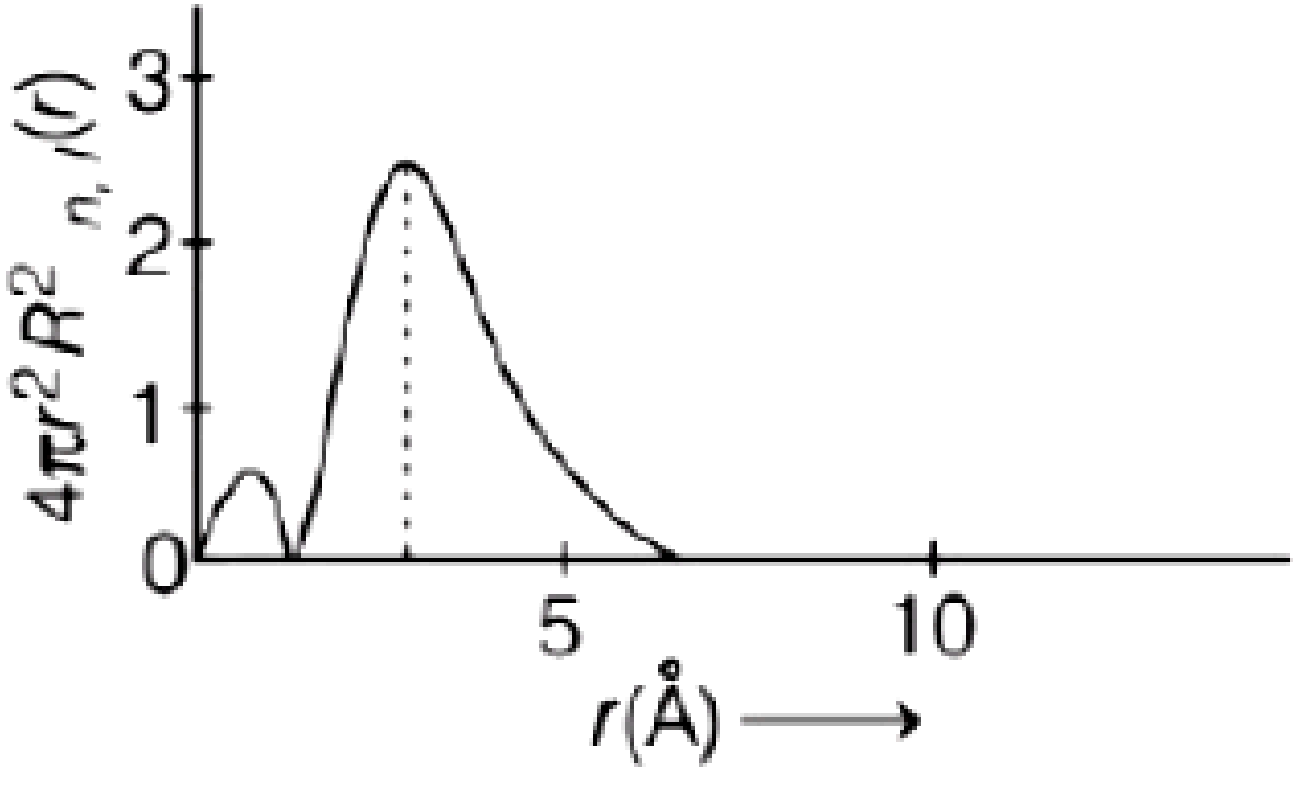


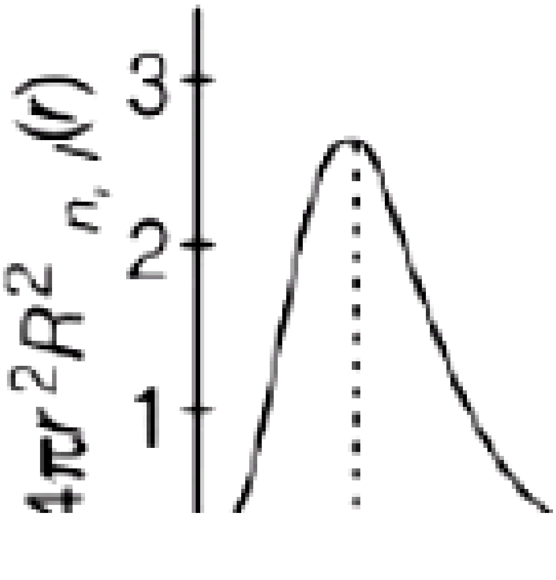
Number of unpaired electron, $n=1$
∴ Spin only magnetic moment,
$\mu=\sqrt{n(n+2)}BM=\sqrt{1(1+2)}BM=\sqrt{3}BM$
$=1.73BM=2BM$

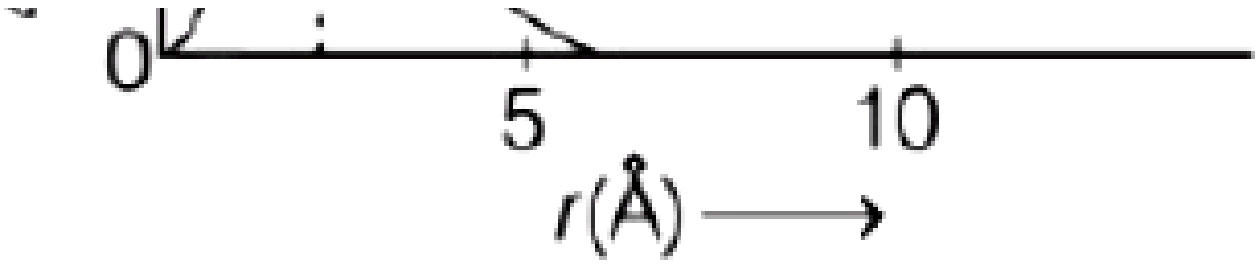
Question119

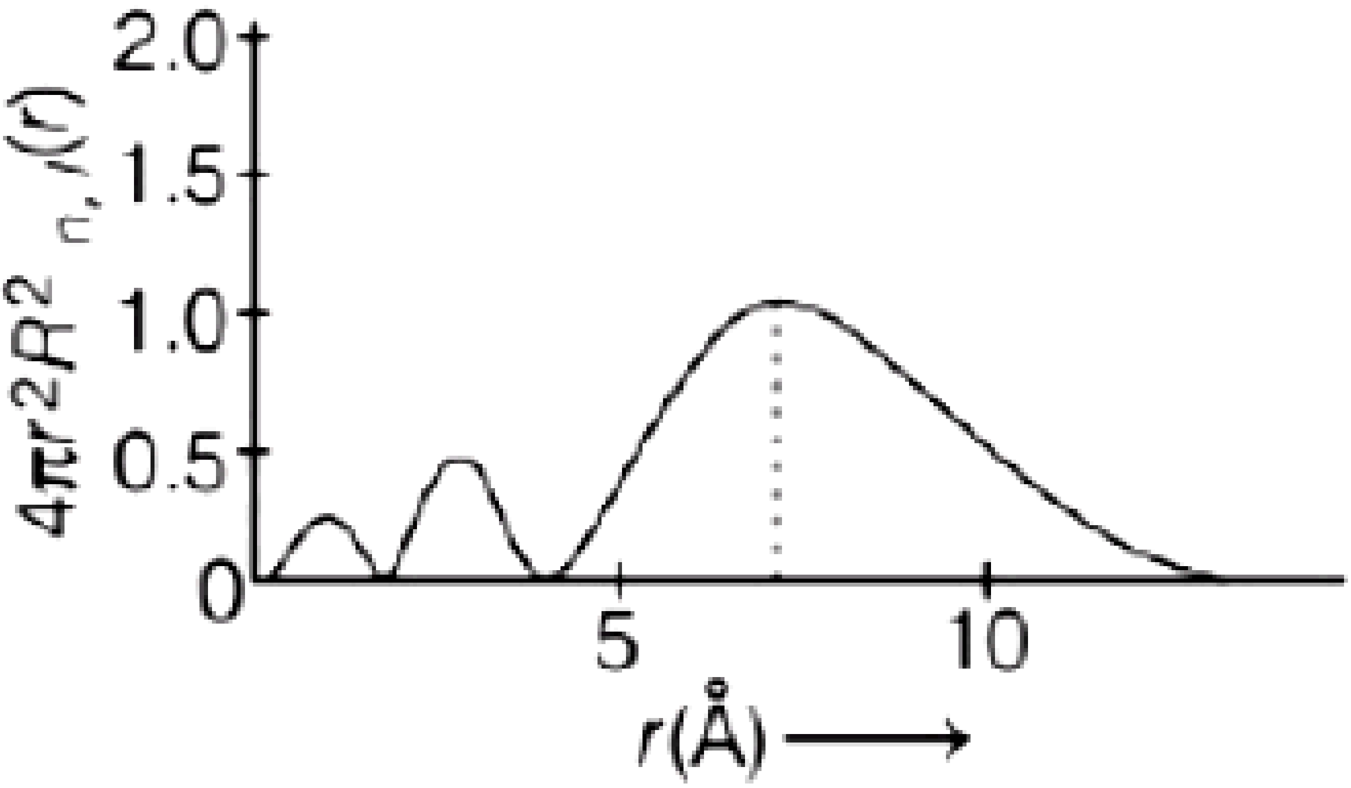
The plots of radial distribution functions for various orbitals of hydrogen atom against ' r ' are given below.


$4\pi r^2 R^2{}_{n,l}(r)$
8
4
0
5
10
r(Å)


$4\pi r^2 R^2{}_{n,l}(r)$
3
2
1
0
5
10
r(Å)


$4\pi r^2 R^2{}_{n,l}(r)$
3
2
1





The correct plot for 3s-orbital is
[25 Feb 2021 Shift 1]

Options:

A. (A)

B. (B)

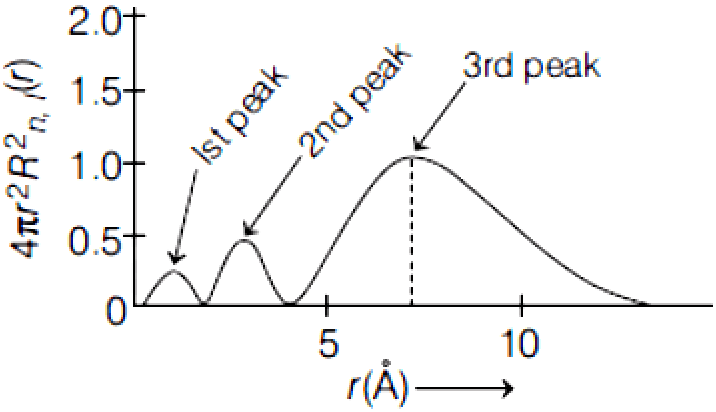
C. (C)

D. (D)

Answer: D

Solution:

The correct plot for 3s-orbital is



For 3s,
value of I = 0
value of n = 3
Number of peak = n − I = 3 − 0 = 3
In graph D, three peaks are present, so this is the correct plot for 3s-orbital.

Question120

A proton and a Li^{3+} nucleus are accelerated by the same potential. If λ_{Li} and λ_P denote the de Broglie wavelengths of Li^{3+} and proton respectively, then the value of $\frac{\lambda_{Li}}{\lambda_P}$ is $x \times 10^{-1}$. The value of x is _____(Rounded off to the nearest integer) (Mass of Li^{3+} = 8.3 mass of proton)
[24feb2021shift1]

Answer: 2

Solution:

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

$$\frac{\lambda_{Li}}{\lambda_P} = \sqrt{\frac{m_P(e)V}{m_{Li}(3e)(V)}} \quad m_{Li} = 8.3m_P$$

$$\frac{\lambda_{Li}}{\lambda_P} = \sqrt{\frac{1}{8.3 \times 3}} = \frac{1}{5} = 0.2 = 2 \times 10^{-1}$$

Question121

Given below are two statements.
Statement I Bohr's theory accounts for the stability and line spectrum of Li^{+}ion.
Statement II Bohr's theory was unable to explain the splitting of spectral lines in the presence of a magnetic field.
In the light of the above statements, choose the most appropriate answer from the options given below:
[18 Mar 2021 Shift 2]

Options:

A. Both statements I and II are true.

B. Statement I is false but statement II is true.

C. Both statements I and II are false.

D. Statement I is true but statement II is false.

Answer: B

Solution:

Statement I is false, because Bohr's theory accounts for the stability and spectrum of single electronic species (e.g. He^{+}, Li^{2+} etc.) but Li^{+}has two electrons.
Bohr's theory fails to explain splitting of spectral lines in presence of magnetic field i.e. Zeeman effect.
∴ Statement II is true.

Question122

When light of wavelength 248nm falls on a metal of threshold energy 3.0eV , the de-Broglie wavelength of emitted electrons is ...... Å.
[Round off to the nearest integer]
[Use: $\sqrt{3} = 1.73$, $h = 6.63 \times 10^{-34}$J s
$m_e = 9.1 \times 10^{-31}$kg, $c = 3.0 \times 10^{8}ms^{-1}$

$1eV = 1.6 \times 10^{-19} J$]
[16 Mar 2021 Shift 1]

Answer: 9

Solution:

Given wavelength, if incident light $(\lambda) = 248nm = 2480Å$
Threshold energy $= 3.0eV$
We know, for electron de-Broglie wavelength is given by
$\lambda = \frac{h}{mv}$
$\lambda = \frac{h}{\sqrt{2m(KE)}}$
$KE = q \times V$ (where, $q =$ charge of particle)
$V =$ voltage applied
$\lambda = \frac{h}{\sqrt{2m(qV)}}$
Putting mass of electron, $m_e = 9.1 \times 10^{-31} kg$
Charge of electron, $q = 1.6 \times 10^{-19} C$
$\lambda = \sqrt{\frac{150}{V}} Å$....(i)
We know, $E_{applied} = \varphi + K_{max}$....(ii)
where, $\varphi =$ threshold energy
$\varphi = 3eV$
$E_{applied} = \frac{12400}{\lambda_{Å}} = \frac{12400}{2480} = 5eV$
Putting value in Eq. (ii),
$5 = 3 + K_{max}$
$\Rightarrow KE_{max} = 2eV$
Voltage $= 2$
$\lambda_e = \sqrt{\frac{150}{2}} = 8.6Å$ [Eq. (i)]
Nearest integer $= 9$

Question123

A certain orbital has no angular nodes and two radial nodes. The orbital is
[18 Mar 2021 Shift 1]

Options:

A. 2s

B. 3s

C. 3p

D. 2p

Answer: B

Solution:

As, we know,
For 2s orbital, $n=2, I=0$
Angular node $(1)=0$
Radial nodes $=n-1-1$ Rightarrow $2-0-1=1$
For 2p orbital, $n=2$ and $I=1$
Angular node $(1)=1$
Therefore, number of radial nodes $=2-1-1=0$
It has one angular and zero radial node.

For 3p orbital, $n=3, I=1$
Angular node $(1)=1$
It has one angular and one radial node.
For 3s orbital, $n=3, I=0$
No. of angular node $(1)=0$
No. of radial node Rightarrow $n-I-1$
$=3-0-1=2=2$
Hence, for 3 s-orbital has no angular node and 2 radial nodes.

Question124

A certain orbital has n = 4 and $m_l = -3$. The number of radial nodes in this orbital is (Round off to the nearest integer). [17 Mar 2021 Shift 1]

Answer: 0

Solution:

Given, $n = 4, m_l = -3$, so $l = 3$
Possible value of $l = 0, 1, 2, 3$
As $m = -3$ is possible only for $l = 3$
For $l = 3, m_l = -3, -2, -1, 0, 1, 2, 3$
So, this is $4f$-orbital.
Number of radial nodes $= n - l - 1$
$= 4 - 3 - 1$
$= 0$
So, there is no radial node in $4f$ orbital.

Question125

The number of orbitals with n = 5, m_1 = +2 is (Round off to the nearest integer).
[16 Mar 2021 Shift 2]

Answer: 3

Solution:

Given, $n = 5, m_l = +2$
For $n = 5$, possible value of $l = 0, 1, 2, 3, 4$
For $l = 0, m_l = 0$
$l = 1, m_l = -1, 0, 1$
$l = 2, m_l = -2, -1, 0, 1, 2$
$l = 3, m_l = -3, -2, -1, 0, 1, 2, 3$
$l = 4, m_l = -4, -3, -2, -1, 0, 1, 2, 3, 4$
Possible value of m_l for a given value of l
$= 0, \pm 1, \pm 2, \pm 3 \ldots \pm l$
So, number of orbitals having $n = 5$ and $m_l = \pm 2$ are 3 .

Question126

If the Thompson model of the atom was correct, then the result of Rutherford's gold foil experiment would have been:

[27 Jul 2021 Shift 2]

Options:

A. All of the α-particles pass through the gold foil without decrease in speed.

B. α-Particles are deflected over a wide range of angles.

C. All α-particles get bounced back by 180°

D. α-Particles pass through the gold foil deflected by small angles and with reduced speed.

Answer: D

Solution:

As in Thomson model, protons are diffused (charge is not centred) α - particles deviate by small angles and due to repulsion from protons, their speed decreases.

Question127

Given below are two statements :
Statement I : Rutherford's gold foil experiment cannot explain the line spectrum of hydrogen atom.
Statement II : Bohr's model of hydrogen atom contradicts Heisenberg's uncertainty principle.
In the light of the above statements, choose the most appropriate answer from the options given below :
[27 Jul 2021 Shift 1]

Options:

A. Statement I is false but statement II is true.

B. Statement I is true but statement II is false.

C. Both statement I and statement II are false.

D. Both statement I and statement II are true.

Answer: D

Solution:

Rutherford's gold foil experiment only proved that electrons are held towards nucleus by electrostatic forces of attraction and move in circular orbits with very high speeds.
Bohr's model gave exact formula for simultaneous calculation of speed & distance of electron from the nucleus, something which was deemed impossible according to Heisenberg.

Question128

An accelerated electron has a speed of $5 \times 10^6 ms^{-1}$ with an uncertainty of 0.02%. The uncertainty in finding its location while in motion is $x \times 10^{-9} m$.
The value of x is __________ . (Nearest integer)
[Use mass of electron $= 9.1 \times 10^{-31} kg, h = 6.63 \times 10^{-34} J\ s, \pi = 3.14$]
[25 Jul 2021 Shift 2]

Answer: 58

Solution:

$\Delta v = \frac{0.02}{100} \times 5 \times 10^6 = 10^3 m/s$

$\Delta x \,.\, \Delta v = \frac{h}{4\pi m}$

$x \times 10^{-9} \times 10^3 = \frac{6.63 \times 10^{-34}}{4 \times 3.14 \times 9.1 \times 10^{-31}} x \times 10^{-9} \times 10^3 = 0.058 \times 10^{-3}$

$x = \frac{0.058 \times 10^{-6}}{10^{-9}} = 58$

Question129

A source of monochromatic radiation of wavelength 400nm provides 1000J of energy in 10 seconds. When this radiation falls on the surface of sodium, $x \times 10^{20}$ electrons are ejected per second. Assume

that wavelength 400nm is sufficient for ejection of electron from the surface of sodium metal. The value of x is _______.
(Nearest integer)
($h = 6.626 \times 10^{-34}$ J s)
[25 Jul 2021 Shift 1]

Answer: 2

Solution:

Total energy provided by
Source per second $= \frac{1000}{10} = 100\text{J}$
Energy required to eject electron $= \frac{hc}{\lambda}$
$= \frac{6.626 \times 10^{-34}}{400 \times 10^{-9}} \times 3 \times 10^{8}$
Number of electrons ejected $= \frac{100}{\frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{400 \times 10^{-9}}}$
$= \frac{400 \times 10^{-7} \times 10^{26}}{6.626 \times 3}$
$= \frac{40 \times 10^{-20}}{6.626 \times 3}$
$= 2.01 \times 10^{20}$

Question130

The wavelength of electrons accelerated from rest through a potential difference of 40kV is $x \times 10^{-12}$m. The value of x is ________ . (Nearest integer)
Given : Mass of electron $= 9.1 \times 10^{-31}$kg
Charge on an electron $= 1.6 \times 10^{-19}$C
Planck's constant $= 6.63 \times 10^{-34}$J S
[20 Jul 2021 Shift 2]

Answer: 6

Solution:

De-broglie-wave length of electron:

$\lambda_e = \frac{h}{\sqrt{2m(KE)}}$ { $\because e^-$ is accelerated from rest $\Rightarrow KE = q \times V$

$\lambda = \frac{h}{\sqrt{2mqv}}$

$= \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 1.6 \times 10^{-19} \times 9.1 \times 10^{-31} \times 40 \times 10^3}}$

$= 0.614 \times 10^{-11} m$

$= 6.14 \times 10^{-12} m$

Nearest integer $= 6$

Question131

The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is equal to $\frac{h^2}{xma_0^2}$. The value of 10x is .... (a_0 is radius of Bohr's orbit) (Nearest integer)
[Given, $\pi = 3.14$]
[27 Aug 2021 Shift 1]

Answer: 3155

Solution:

$\because$ Angular moment is given as,

$mvr = \frac{nh}{2\pi}$

$\therefore$ Kinetic energy $= \frac{n^2h^2}{8\pi^2 mr^2} = \frac{4h^2}{8\pi^2 m(4a_0)^2}$

$= \left(\frac{4}{8\pi^2 \times 16}\right)\frac{h^2}{ma_0^2} = \frac{h^2}{xma_0^2}$

$x = \frac{4}{8\pi^2 \times 16}$
$\Rightarrow x = 315.507$
$10x = 3155.07$

Question132

Given below are two statements.
Statement I According to Bohr's model of an atom, qualitatively the magnitude of velocity of electron increases with decrease in positive charges on the nucleus as there is no strong hold on the electron by the nucleus.
Statement II According to Bohr's model of an atom, qualitatively the magnitude of velocity of electron increases with decrease in principal quantum number.In the light of the above statements, choose the most appropriate answer from the options given below. [26 Aug 2021 Shift 1]

Options:

A. Both statement I and statement II are false.

B. Both statement I and statement II are true.

C. Statement I is false but statement II is true.

D. Statement I is true but statement II is false.

Answer: C

Solution:

According to Bohr's atom, velocity of electron is given by,
$v \propto \frac{Z}{n}$ ...(i)
where, v = velocity of electron
where, v = velocity of electron
n = principal quantum number
From Eq. (i), velocity of electron is directly proportional to atomicnumber of atom, corresponding to positive charge. So, as Z increases velocity also increases.
Hence, statement I is false. Also, velocity of electron is inversely proportional to n i.e. as ‘n’ decreases velocity of electron increases.
So, statement II is true.

Question133

The number of photons emitted by a monochromatic (single frequency) infrared range finder of power 1 mW and wavelength of 1000 nm, in 0.1 second is $x \times 10^{13}$. The value of x is .... (Nearest integer) ($h = 6.63 \times 10^{-34}$ Js, $c = 3.00 \times 10^{8} ms^{-1}$)
[27 Aug 2021 Shift 2]

Answer: 50

Solution:

Power of the source $= 1\text{ mW} = 10^{-3}\text{W}$
Energy of one photon, $E = hv = \frac{hc}{\lambda}$
where, $h = 6.6 \times 10^{-34}$ Js
$c = 3 \times 10^{8} m/s$ and
λ = 1000 nm
$= 1000 \times 10^{-9}m\ (1\text{ nm} = 10^{-9}m)$
$= 10^{-6}m$
$E = \frac{6.6 \times 10^{-34} \times 3 \times 10^{8}}{10^{-6}} \cong 20 \times 10^{-20}J$
Power = number of photons emitted in one second × energy of 1 photon.
$10^{-3} = n \times 20 \times 10^{-20}$
$\Rightarrow n = \frac{10^{-3}}{20 \times 10^{-20}} = 0.5 \times 10^{16}$
Number of photons emitted in $1s = 0.5 \times 10^{16}$
Number of photons emitted in $0.1s = 0.5 \times 10^{16} \times 0.1 = 50 \times 10^{13}$
$x \times 10^{13} = 50 \times 10^{13}$
x = 50

Question134

A metal surface is exposed to 500 nm radiation. The threshold frequency of the metal for photoelectric current is 4.3×10^{14} Hz. The velocity of ejected electron is .... $\times 10^{5} ms^{-1}$. (Nearest integer)

[Use h = 6.63×10^{-34} Js, $m_e = 9.0 \times 10^{-31}$ kg]
[26 Aug 2021 Shift 2]

Answer: 5

Solution:

According to photoelectric effect $= \frac{hc}{\lambda} = hv_0 + \frac{1}{2}mv^2$
(where, h = Planck's constant, λ = wavelength
$v_0 =$ threshold frequency, v = velocity of electron
m = mass of electron, c = speed of light)

$$\Rightarrow \frac{6.63 \times 10^{-34} \times 310^8}{500 \times 10^{-9}}$$

$$= 6.63 \times 10^{-34} \times 4.3 \times 10^{14} + \frac{1}{2}mv^2$$

$$\frac{6.63 \times 30 \times 10^{-19}}{50} = 6.63 \times 4.3 \times 10^{-20} + \frac{1}{2}mv^2$$

$$11.271 \times 10^{-20}J = \frac{1}{2} \times 9 \times 10^{-31} \times v^2$$

$$v = 5 \times 10^5 m/s$$

∴ x = 5

Question135

The value of magnetic quantum number of the outermost electron of Zn^+ ion is
[31 Aug 2021 Shift 2]

Answer: 0

Solution:

The configuration of $Zn^{+} = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$
The outermost electron is in s-orbital.
For s-orbital, azimuthal quantum number (l) = 0
Magnetic quantum number, m = − l to + l
For l = 0, m = 0

Question136

Ge(Z = 32) in its ground state electronic configuration has x completely filled orbitals with $m_1 = 0$. The value of x is

[31 Aug 2021 Shift 1]

Answer: 7

Solution:

Ge (Z = 32) electronic configuration is

Ge (32)	$1s^2$	$2s^2$	$2p^6$	$3s^2$	$3p^6$	$4s^2$	$3d^{10}$	$4p^2$
Orbital	↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑ ↑
Value of m_l (magnetic quantum number)	0	0	−1 0 1	0	−1 0 1	0	−2 −1 0 1 2	−1 0 +1

For s-orbitals (1s, 2s, 3s and 4s) $m_1 = 0\{2 \times 0\}$
For p-orbitals (2p, 3p)$m_1 = 0\{(-1) \times 2 + 0 \times 2 + (+1) \times 2 = 0\}$

For d-orbitals (3d)$m_1 = 0 \left\{ \begin{array}{c} (-2) \times 2 + (-1) \times 2 + 0 \times 2 \\ (+1) \times 2 + (+2) \times 2 = 0 \end{array} \right\}$

∴ Completely filled orbitals with m_1 = 0 are seven (7).

Question137

A 50 watt bulb emits monochromatic red light of wavelength of 795 nm. The number of photons emitted per second by the bulb is $x \times 10^{20}$. The value of x is ......

[Given, $h = 6.63 \times 10^{-34}$ Js and $c = 3.0 \times 10^{8} ms^{-1}$]

[1 Sep 2021 Shift 2]

Answer: 2

Solution:

Energy of photon is given as

$E = \frac{nhc}{\lambda}$ ...(i)

where, E = energy of photon (50 W),

n = number of photon

$h =$ Planck's constant $(6.63 \times 10^{-34}\ \mathrm{Js})$

$c =$ speed of light $(3 \times 10^{8}\mathrm{m/s})$

$\lambda =$ wavelength of light $(795 \times 10^{-9}\mathrm{m})$

E = 50 W = 50 J = energy of photon

$$50\mathrm{J} = \frac{n \times 6.63 \times 10^{-34}\mathrm{Js} \times 3 \times 10^{8}\mathrm{m/s}}{795 \times 10^{-9}\mathrm{m}}$$

$$\Rightarrow n = \frac{50 \times 795 \times 10^{-9}}{6.63 \times 10^{-34} \times 3 \times 10^{8}} = 1998.49 \times 10^{17} = 1.998 \times 10^{20} = 2 \times 10^{20}$$

$\therefore x = 2$

∴ Answer is 2.

Question138

For the Balmer series in the spectrum of H atom, $\bar{v} = R_H \left\{ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right\}$, the correct statements among (I) to (IV) are:

(I) As wavelength decreases, the lines in the series converge

(II) The integer n_1 is equal to 2

(III) The lines of longest wavelength corresponds to $n_2 = 3$

(IV) The ionization energy of hydrogen can be calculated from wave number of these lines

[Jan. 08,2020 (I)]

Options:

A. (I), (III), (IV)

B. (I), (II), (III)

C. (I), (II), (IV)

D. (II), (III), (IV)

Answer: B

Solution:

In the Balmer series of H-atom the transition takes place from the higher oribtal to $n = 2$. Therefore the longest wave length corresponds to $n_1 = 2$ and $n_2 = 3$. As the wave length decreases, the lines in the series converges. Hence, statement I, II, III are the correct statements among the given options.

Question139

The radius of the second Bohr orbit, in terms of the Bohr radius, a_0, in Li^{2+} is:
[Jan. 08, 2020 (II)]

Options:

A. $\frac{2a_0}{3}$

B. $\frac{4a_0}{9}$

C. $\frac{4a_0}{3}$

D. $\frac{2a_0}{9}$

Answer: C

Solution:

$r = \frac{a_0 n^2}{Z}$

For Li^{2+}, $r = \frac{a_0(2)^2}{3} = \frac{4a_0}{3}$

Question140

The de Broglie wavelength of an electron in the 4^{th} Bohr orbit is: [Jan. 09, 2020 (I)]

Options:

A. $2\pi a_0$

B. $4\pi a_0$

C. $6\pi a_0$

D. $8\pi a_0$

Answer: D

Solution:

$$2\pi r = n\lambda$$
$$r = \frac{n^2 a_0}{Z}$$
$$2\pi \times \frac{4^2}{1} a_0 = 4\lambda$$
$$\lambda = 2\pi \times \frac{4}{1} a_0$$
$$\lambda = 8\pi a_0$$

Question141

The number of orbitals associated with quantum numbers $n = 5, m_s = +\frac{1}{2}$ is: [NV, Jan. 07,2020(I)]

Options:

A. 11

B. 25

C. 50

D. 15

Answer: B

Solution:

The possible number of orbitals in a shell in term of "n ' is n^2
$\therefore n=5; n^2=25$

Question142

The difference between the radii of 3 rd and 4 th orbits of Li^{2+} is ΔR_1. The difference between the radii of 3 rd and 4 th orbits of He^+ is ΔR_2. Ratio $\Delta R_1 : \Delta R_2$ is :
[Sep .05,2020 (I)]

Options:

A. 8 : 3

B. 3 : 8

C. 2 : 3

D. 3 : 2

Answer: C

Solution:

$r=0.529\frac{n^2}{Z}Å$
For Li^{2+},
$(r_{Li^{2+}})_{n=4}-(r_{Li^{2+}})_{n-3}=\frac{0.529}{3}[4^2-3^2]=\Delta R_1$
For He^+,
$(r_{He^+})_{n=4}-(r_{He^+})_{n=3}=\frac{0.529}{2}[4^2-3^2]=\Delta R_2$
$\frac{\Delta R_1}{\Delta R_2}=\frac{2}{3}$

Question143

The region in the electromagnetic spectrum where the Balmer series lines appear is:
[Sep. 04, 2020 (I)]

Options:

A. Visible

B. Microwave

C. Infrared

D. Ultraviolet

Answer: A

Solution:

In hydrogen spectrum maximum lines of Balmer series lies in visible region.

Question144

The shortest wavelength of H atom in the Lyman series is λ_1. The longest wavelength in the Balmer series is He^+ is:
[Sep. 04, 2020 (II)]

Options:

A. $\frac{36\lambda_1}{5}$

B. $\frac{5\lambda_1}{9}$

C. $\frac{9\lambda_1}{5}$

D. $\frac{27\lambda_1}{5}$

Answer: C

Solution:

Shortest wavelength → Max. energy $(\infty \to 1)$
For Lyman series of H atom,

$$\frac{1}{\lambda_1} = R_H (1)^2 \left[\frac{1}{1} - 0 \right]$$

$$\Rightarrow \frac{1}{\lambda_1} = R_H \Rightarrow R_H = \frac{1}{\lambda_1}$$

For Balmer series of He^+,

$$\frac{1}{\lambda} = R_H (2)^2 \left[\frac{1}{2^2} - \frac{1}{3^2} \right] \Rightarrow \frac{1}{\lambda} = R_H (4) \left(\frac{9-4}{36} \right)$$

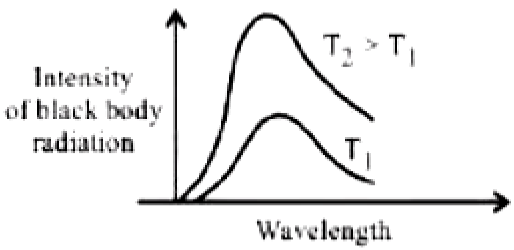
$$\Rightarrow \frac{1}{\lambda} = \frac{5R_H}{9} \Rightarrow \lambda = \frac{9}{5R_H} = \frac{9\lambda_1}{5}$$

Question145

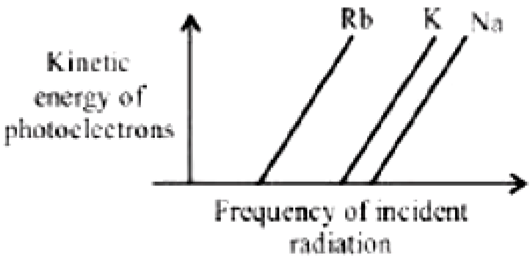
The figure that is not a direct manifestation of the quantum nature of atoms is :
[Sep. 02, 2020 (I)]

Options:

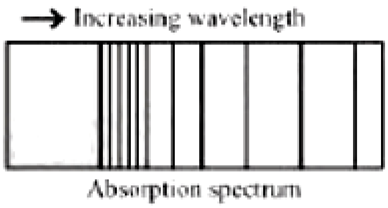
A.



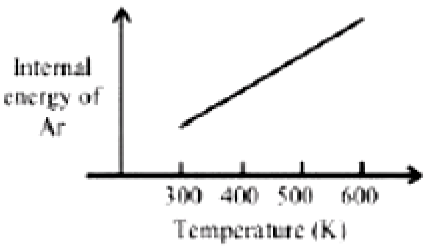
B.



C.



D.



Answer: D

Solution:

(d) (a), (b) and (c) are according to quantum theory but (d) is statement of kinetic theory of gases.

Question146

The work function of sodium metal is 4.41×10^{-19}J . If photons of wavelength 300nm are incident on the metal, the kinetic energy of the ejected clectrons will be
$(h = 6.63 \times 10^{-34}$J s; $c = 3 \times 10^{8}$m/s) $\times 10^{-21}$J
[NV, Sep. 02, 2020(II)]

Answer: 222

Solution:

Metal (Work function = E_0)

$E = E_0 + (KE)_{max}$

$\frac{hc}{\lambda} = 4.41 \times 10^{-19} + KE$

$\frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{300 \times 10^{-9}} = 4.41 \times 10^{-19} + KE$

So, $(KE)_{max} = 6.63 \times 10^{-19} - 4.41 \times 10^{-19}$

$= 2.22 \times 10^{-19} J = 222 \times 10^{-21} J$

Question147

In the sixth period, the orbitals that are filled are: [Sep. 05,2020 (I)]

Options:

A. 6s, 4f , 5d , 6p

B. 6s, 5d , 5f , 6p

C. 6s, 5f , 6d , 6p

D. 6s, 6p, 6d , 6f

Answer: A

Solution:

	$6s$	$4f$	$5d$	$6p$
$n+1$	$6+0$	$4+3$	$5+2$	$6+1$
	⇩	⇩	⇩	⇩
	6	7	7	7

Thus, order of orbitals filled are
$6s < 4f < 5d < 6p$

Question148

The correct statement about probability density (except at infinite distance from nucleus) is:
[Sep. 05, 2020 (II)]

Options:

A. It can be zero for 1s orbital

B. It can be negative for 2p orbital

C. It can be zero for 3p orbital

D. It can never be zero for 2s orbital

Answer: C

Solution:

Radial node $= n - 1 - 1$
$\therefore 1s \Rightarrow 0(\psi^2 \neq 0)$
$2s \Rightarrow 1(\psi^2 = 0)$
$2p \Rightarrow 0(\psi^2 \neq 0)$
$3p \Rightarrow 1(\psi^2 = 0)$
Probability density (ψ^2) can be zero for $3p$ orbital other than infinite distance. It has one radial node.
Thus, statement (c) is correct.

Question149

Consider the hypothetical situation where the azimuthal quantum number, l, takes values 0, 1, 2, ... n + 1, where n is the principal quantum number. Then, the element with atomic number:
[Sep. 03, 2020 (II)]

Options:

A. 9 is the first alkali metal

B. 13 has a half-filled valence subshell

C. 8 is the first noble gas

D. 6 has a 2p -valence subshell

Answer: B

Solution:

Under the given situation for $n = 1, l = 0, 1, 2$
$n = 2, l = 0, 1, 2, 3$
$n = 3, l = 0, 1, 2, 3, 4$
According to $(n + l)$ rule of order of filling of subshells will be:
$1s1p1d\ 2s2p3s2d\ 3f$
Atomic number $1s^2 1p^4$
Atomic number $1s^2 1p^6 1d^1$
Atomic number $1s^2 1p^6$
Atomic number $1s^2 1p^6 1d^5$
Therefore option (b) is correct. Atomic number of first noble gas will be $18(1s^2 1p^6 1d^{10})$.

Question150

The number of subshells associated with n = 4 and m = −2 quantum numbers is:
[Sep. 02,2020 (II)]

Options:

A. 8

B. 2

C. 16

D. 4

Answer: B

Solution:

For $n = 4$ possible values of $l = 0, 1, 2, 3$; only $l = 2$ and $l = 3$ can have $m = -2$. So possible subshells are 2.

Question151

What is the work function of the metal if the light of wavelength 4000Å generates photoelectrons of velocity $6 \times 10^{5} ms^{-1}$ from it ?
(Mass of electron $= 9 \times 10^{-31} kg$
Velocity of light $= 3 \times 10^{8} ms^{-1}$
Planck's constant $= 6.626 \times 10^{-34} J s$
Charge of electron $= 1.6 \times 10^{-19} J eV^{-1}$)
[Jan. 12, 2019 (I)]

Options:

A. 0.9eV

B. 3.1eV

C. 2.1eV

D. 4.0eV

Answer: C

Solution:

$E = hv = \frac{hc}{\lambda}$

$E = \frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{4000 \times 10^{-10}} = 4.97 \times 10^{-19} J$

$= \frac{4.97 \times 10^{-19} J}{1.6 \times 10^{-10} J eV^{-1}} = 3.1 eV$

$K E = \frac{1}{2} mv^{2} = \frac{1}{2} \times 9 \times 10^{-31} kg \times (6 \times 10^{5} ms^{-1})^{2}$

$= 1.62 \times 10^{-19} J\ [1J = kg \cdot m^{2} s^{-2}]$

$= 1 cV$

According to photoelectric effect,

$K.E. = hv - hv_{0}$

$hv_{0} = hv - K.E$

Work function $(W_{0}) = E - K.E$

$= 3.1 - 1 = 2.1 eV$

Question152

Heat treatment of muscular pain involves radiation of wavelength of about 900nm. Which spectral line of H atom is suitable for this purpose?
[$R_H = 1 \times 10^5 cm^{-1} \cdot h = 6.6 \times 10^{-34} J\ s, c = 3 \times 10^8 ms^{-1}$]
[Jan. 11, 2019 (I)]

Options:

A. Paschen, $\infty \rightarrow 3$

B. Paschen, $5 \rightarrow 3$

C. Balmer, $\infty \rightarrow 2$

D. Lyman, $\infty \rightarrow 1$

Answer: A

Solution:

$$\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$n_1 = 3, n_2 = \infty$

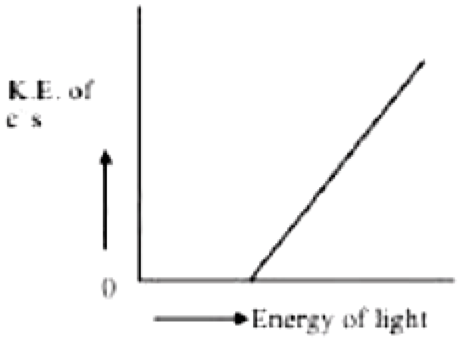
$$\frac{1}{\lambda} = R\left(\frac{1}{9}\right) \Rightarrow \lambda = \frac{9}{R} = \frac{9}{10^5} = 9 \times 10^{-5} cm = 900nm$$

Question153

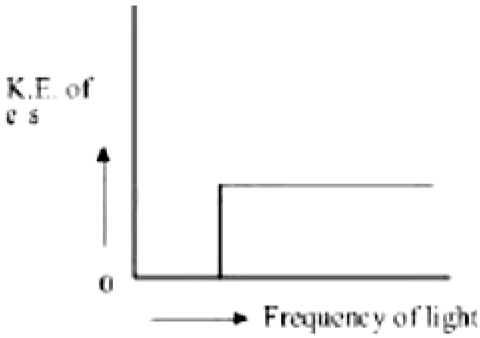
Which of the graphs shown below does not represent the relationship between incident light and the electron cjected from metal surface?
[Jan. 10, 2019 (I)]

Options:

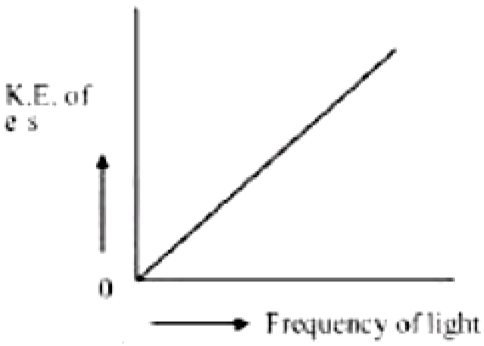
A.



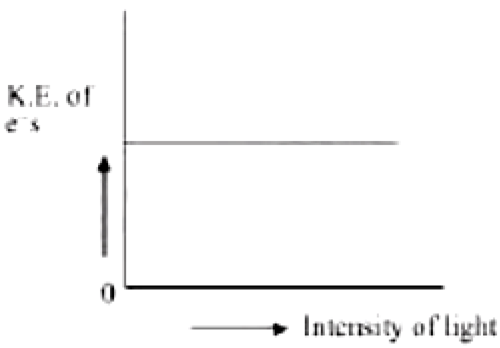
B.



C.



D.



Answer: C

Solution:

$K.E. = hv - hv_0$
where, $v =$ Frequency of incident radiation
$v_0 =$ Threshold frequency
KE is independent of intensity but it depends on frequency of light. Intensity is directly proportional to the no. of electrons emitted.

Question154

The ground state energy of hydrogen atom is $-13.6eV$. The energy of second excited state of He^+ ion in eV is:
[Jan. 10, 2019 (II)]

Options:

A. -54.4

B. -3.4

C. -6.04

D. -27.2

Answer: C

Solution:

According to Bohr's model energy in n^{th} state
$= -13.6 \times \frac{Z^2}{n^2} eV$
For second excited state, of He^+, $n = 3$
$\therefore E_3(He^+) = -13.6 \times \frac{2^2}{3^2} eV = -6.04 eV$

Question155

For emission line of atomic hydrogen from $n_i = 8$ to $n_f = n$ the plot of wave number $(\bar{v})$ against $\left(\frac{1}{n^2}\right)$ will be (The Rydberg constant, R_H is

in wave number unit)
[Jan. 9,2019 (I)]

Options:

A. Linear with intercept $-R_H$

B. Non linear

C. Linear with slope R_H

D. Linear with slope $-R_H$

Answer: D

Solution:

As we know,
$\bar{v} = -R_H \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right) Z^2$ (where, $Z = 1$)
After putting the values, we get
$\bar{v} = -R_H \left(\frac{1}{n^2} - \frac{1}{8^2} \right) \Rightarrow \bar{v} = \frac{R_H}{64} - \frac{R_H}{n^2}$
Comparing to $y = mx + c$, we get
$x = \frac{1}{n^2}$ and $m = -R_H$ (slope)

Question156

If the de Broglie wavelength of the electron in n^{th} Bohr orbit in a hydrogenic atom is equal to $1.5\pi a_0$(a_0 is Bohr radius), then the value of n/z is:
[Jan. 12, 2019 (II)]

Options:

A. 0.40

B. 1.50

C. 1.0

D. 0.75

Answer: D

Solution:

Given $\lambda = 1.5\pi a_0$
$n\lambda = 2\pi r$ . . . (i)
Radii of stationary states (r) is expressed as:
$r = a_0 \frac{n^2}{z}$ . . . (ii)
From eqn (i) and (ii)
$n\lambda = \frac{2\pi a_0 n^2}{z}; \lambda = \frac{2\pi a_0 n}{z}$
$1.5\pi a_0 = 2\pi a_0 \frac{n}{z}$
$\frac{n}{z} = \frac{1.5}{2} = 0.75$

Question157

The de Broglie wavelength (λ) associated with a photoelectron varies with the frequency (v) of the incident radiation as, [v_0 is threshold frequency]:
[Jan. 11, 2019 (II)]

Options:

A. $\lambda \propto \frac{1}{(v - v_0)}$

B. $\lambda \propto \frac{1}{(v - v_0)^{\frac{1}{4}}}$

C. $\lambda \propto \frac{1}{(v - v_0)^{\frac{3}{2}}}$

D. $\lambda \propto \frac{1}{(v - v_0)^{\frac{1}{2}}}$

Answer: D

Solution:

According to de-Broglie wavelength equation,

$\lambda = \frac{h}{mv} \Rightarrow \lambda \propto \frac{1}{v}$

According to photoelectric effect,

$hv - hv_0 = \frac{1}{2}mv^2; v - v_0 = \frac{1}{2} = \frac{mv^2}{h}$

$v - v_0 \propto v^2$

$v \propto (v - v_0)^{1/2}$

$\therefore \lambda \propto \frac{1}{(v - v_0)^{1/2}}$

Question158

Which of the following combination of statements is true regarding the interpretation of the atomic orbitals?
[Jan. 9,2019 (II)|]

Options:

A. An electron in an orbital of high angular momentum stays away from the nucleus than an electron in the orbital of lower angular momentum.

B. For a given value of the principal quantum number, the size of the orbit is inversely proportional to the azimuthal quantum number.

C. According to wave mechanics, the ground state an- gular momentum is equal to $\frac{h}{2\pi}$

D. The plot of ψ vs r for various azimuthal quantum numbers, shows peak shifting towards higher r value.
(a) (a), (d) (b) (a), (b) (c) (a), (c) (d) (b), (c)

Answer: A

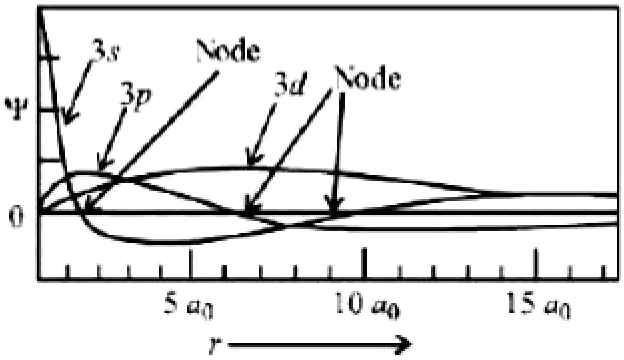
Solution:

(a) Angular momentum $(L) = \frac{nh}{2\pi}$

Therefore, as n increases, L also increases.

(b) $r \propto \frac{n^2}{z}$

(c) For $n=1, L=\frac{h}{2\pi}$
(d) As l increases, the peak of ψ vs r shifts towards higher ' r ' value.



Question159

Among the following, the energy or 2s or orbital is lowest in [April 12, 2019 (II)]

Options:

A. K

B. H

C. Li

D. Na

Answer: A

Solution:

As the value of Z (atomic number) increases, energy of orbitals decreases (becomes more-ve value)
$\therefore$ Order of energy of $2s$ orbital is $H > Li > Na > K$

Question160

The ratio of the shortest wavelength of two spectral series of hydrogen spectrum is found to be about 9. The spectral series are : [April 10, 2019 (II)]

Options:

A. Lyman and Paschen

B. Balmer and Brackett

C. Brackett and Pfund

D. Paschen and Pfund

Answer: A

Solution:

For determined shortest wavelength, $n_2 = \infty$

Lyman series $\bar{v}_L = \frac{1}{\lambda_L} = R\left[\frac{1}{(1)^2} - \frac{1}{\infty^2}\right]$

Paschen series $\bar{v}_P = \frac{1}{\lambda_P} = R\left[\frac{1}{(3)^2} - \frac{1}{\infty^2}\right]$

$\frac{\bar{v}_L}{\bar{v}_P} = \frac{\lambda_P}{\lambda_L} = 9$

Question161

For any given series of spectral lines of atomic hydrogen, let $\Delta\bar{v} = \bar{v}_{max} - \bar{v}_{min}$ be the difference in maximum and minimum frequencies in cm^{-1}. The ratio $\Delta\bar{v}_{Lyman} / \Delta\bar{v}_{Balmer\ i\ is}$: [April 9, 2019 (I)]

Options:

A. 4 : 1

B. 9 : 4

C. 5 : 4

D. 27 : 5

Answer: B

Solution:

$\bar{v} \propto \Delta E$
For H-atom
$\bar{v} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$
For Lyman series,
$\bar{v}(\max) = 13.6\left(1 - \frac{1}{\infty}\right)$

$\bar{v}(\min) = 13.6\left(1 - \frac{1}{4}\right)$

$\therefore \bar{v}_{max} - \bar{v}_{min} = 13.6\left(\frac{1}{4}\right)$
For Balmer series,
$\bar{v}(\max) = 13.6\left(\frac{1}{4} - \frac{1}{\infty}\right)$

$\bar{v}(\min) = 13.6\left(\frac{1}{4} - \frac{1}{9}\right)$

$\therefore\ \bar{v}_{max} - \bar{v}_{min} = 13.6\left(\frac{1}{9}\right)$

So, $\frac{\Delta \bar{v}_{Lyman}}{\Delta \bar{v}_{Balmer}} = \frac{9}{4}$

Question162

Which one of the following about an electron occupying the 1s orbital in a hydrogen atom is incorrect? (The Bohr radius is represented by a_0).

[April 9, 2019 (II)]

Options:

A. The probability density of finding the electron is maximum at the nucleus.

B. The electron can be found at a distance $2a_0$ from the nucleus.

C. The magnitude of the potential energy is double that of its kinetic energy on an average.

D. The total energy of the electron is maximum when it is at a distance a_0 from the nucleus.

Answer: D

Solution:

The total energy of the electron is minimum at a distance of a_0 from the nucleus for $1s$ orbital.

Question163

If p is the momentum of the fastest electron ejected from a metal surface after the irradiation of light having wavelength m, then for 1.5p momentum of the photoelectron, the wavelength of the light should be:
(Assume kinetic energy of cjected photoelectron to be very high in comparison to work function):
[April 8, 2019 (II)]

Options:

A. $\frac{3}{4}\lambda$

B. $\frac{1}{2}\lambda$

C. $\frac{2}{3}\lambda$

D. $\frac{4}{9}\lambda$

Answer: D

Solution:

In photoelectric effect,

$\frac{hc}{\lambda} = w + KE$ of electron

Given that KE of ejected photoelectron is very high in comparison to work function w.

$\frac{hc}{\lambda} = KE$

$\frac{hc}{\lambda} = \frac{1}{2}mv^2\left(\frac{m}{m}\right)$

$\frac{hc}{\lambda} = \frac{1}{2}\cdot\frac{m^2v^2}{m}$

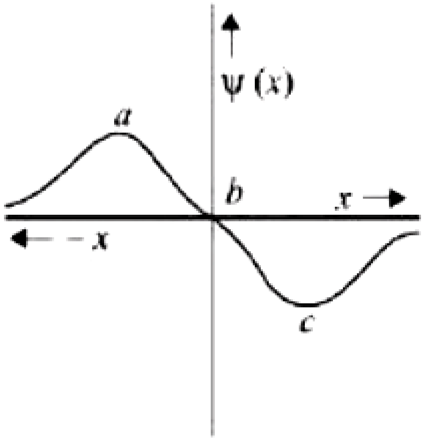
$\frac{hc}{\lambda} = \frac{P^2}{2m}$

New wavelength

$$\frac{hc}{\lambda_1} = \frac{(1.5P)^2}{2m} \Rightarrow \lambda_1 = \frac{4}{9}\lambda$$

Question164

The electrons are more likely to be found:



[April 12, 2019 (I)]

Options:

A. in the region a and c

B. in the region a and b

C. only in the region a

D. only in the region 0

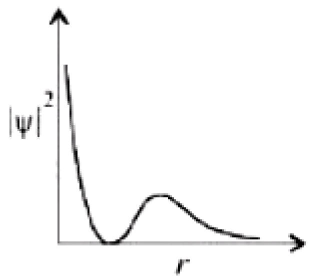
Answer: A

Solution:

Probability of finding an electron will have maximum value at both 'a' and 'c'. There is zero probability of finding an electron at 'b'.

Question165

The graph between $|\psi|^2$ and r (radial distance) is shown below. This represents :



[April 10, 2019 (I)]

Options:

A. 3s orbital

B. 2s orbital

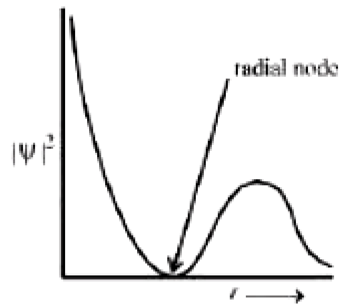
C. 1s orbital

D. 2p orbital

Answer: B

Solution:

The given probability density curve is for 2s orbital due to the presence of only one radial node. 1s and 2p orbital do not have any radial node and 3s orbital has two radial nodes. Hence, option (b) is correct.



Question166

The isoelectronic set of ions is
[April 10, 2019 (I)]

Options:

A. N^{3-}, O^{2-}, F^- and Na^+

B. N^{3-}, Li^{+}, Mg^{2+} and O^{2-}

C. F^{-}, Li^{+}, Na^{+} and Mg^{2+}

D. Li^{+}, Na^{+}, O^{2-} and F^{-}

Answer: A

Solution:

Atomic numbers of N, O, F and Na are $7, 8, 9$ and 11 respectively. Therefore, total number of electrons in each of N^{3-}, O^{2}, F^{-}, and Na^{+} are 10 and hence they are isoelectronic.

Question167

The quantum number of four electrons are given below:

I. $n = 4, l = 2, m_l = -2, m_s = -1/2$
II. $n = 3, l = 2, m_l = 1, m_s = +1/2$
III. $n = 4, l = 1, m_l = 0, m_s = +1/2$
IV. $n = 3, l = 1, m_l = 1, m_s = -1/2$
The correct order of their increasing energies will be:
[April 8, 2019(I)]

Options:

A. IV < III < II < I

B. I < II < III < IV

C. IV < II < III < I

D. I < III < II < IV

Answer: C

Solution:

				$n+l$
(I)	$n=4$	$l+2$	$4d$	6
(II)	$n=3$	$l+2$	$3d$	5
(III)	$n=4$	$l+1$	$4p$	5
(IV)	$n=3$	$l+1$	$3p$	4

The energy of an atomic orbital increases with increasing $n+\ell$. For identical values of $n+\ell$, energy increases with increasing value of n. Therefore the correct order of energy is:

Question168

The size of the iso-electronic species Cl^-, Ar and Ca^{2+} is affected by: [April 8, 2019 (I)]

Options:

A. azimuthal quantum number of valence shell

B. electron-electron interaction in the outer orbitals

C. principal quantum number of valence shell

D. nuclear charge

Answer: D

Solution:

Iso-electronic species differ in size due to different effective nuclear charge.

Question169

Which of the following statements is false?
[Online April 16, 2018]

Options:

A. Splitting of spectral lines in electrical field is called Stark effect

B. Frequency of emitted radiation from a black body goes from a lower wavelength to higher wavelength as the temperature increases

C. Photon has momentum as well as wavelength

D. Rydberg constant has unit of energy

Answer: B

Solution:

When temperature is increased, black body emits high energy radiation from higher wavelength to lower wavelength.

Question170

Ejection of the photoelectron from metal in the photoelectric effect experiment can be stopped by applying 0.5V when the radiation of 250nm is used. The work function of the metal is: [Online April 15, 2018 (I)]

Options:

A. 4eV

B. 5.5eV

C. 4.5eV

D. 5eV

Answer: C

Solution:

$\lambda = 250\text{nm}$

$E = \frac{hc}{\lambda} = \frac{1240\text{eV . nm}}{250\text{nm}} = 4.96\text{eV}$

$KE = \text{stopping potential} = 0.5\text{eV}$

$E = W_0 + K.E$

$4.96 = W_0 + 0.5$

$W_0 = 4.46 \approx 4.5\text{eV}$

Question171

The de-Broglie's wavelength of electron present in first Bohr orbit of 'H' atom is:
[Online April 15, 2018 (II)]

Options:

A. $4 \times 0.529\text{Å}$

B. $2\pi \times 0.529\text{Å}$

C. $\frac{0.529}{2\pi}\text{Å}$

D. 0.529A

Answer: B

Solution:

First Bohr orbit of H atom has radius $r = 0.529\text{Å}$ Also, the angular momentum is quantised.

$mvr = \frac{h}{2\pi}$

$2\pi r = \frac{h}{mV} = \lambda$

$\therefore \lambda = 2\pi \times 0.529\text{Å}$

Question172

The radius of the second Bohr orbit for hydrogen atom is:
(Plank's const. $h = 6.6262 \times 10^{-34}$ J s; mass of electron $= 9.1091 \times 10^{-31}$ kg; charge of electron $e = 1.60210 \times 10^{-19}$ C;

permittivity of vaccum $E_0 = 8.854185 \times 10^{-12} kg^{-1} m^{-3} A^2$)
[2017]

Options:

A. 1.65Å

B. 4.76Å

C. 0.529Å

D. 2.12Å

Answer: D

Solution:

Radius of n^{th} Bohr orbit in H-atom $= 0.53n^2$Å
Radius of II Bohr orbit $= 0.53 \times (2)^2 = 2.12$Å

Question173

If the shortest wavelength in Lyman series of hydrogen atom is A, then the longest wavelength in Paschen series of He^+ is : [Online April 8,2017]

Options:

A. $\frac{5A}{9}$

B. $\frac{9A}{5}$

C. $\frac{36A}{5}$

D. $\frac{36A}{7}$

Answer: D

Solution:

For Lyman series (shortest wavelength)

$n_1 = 1, n_2 = \infty$

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\Rightarrow \frac{1}{A} = 1^2R\left(\frac{1}{1} - \frac{1}{\infty}\right) \Rightarrow \frac{1}{A} = R$$

Longest wavelength $= 1^{st}$ line

$n_1 = 3, n_2 = 4$

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{3^2} - \frac{1}{4^2}\right) \Rightarrow \frac{1}{\lambda} = \frac{R7}{36}$$

$$R = \frac{1}{A}$$

$$\frac{1}{\lambda} = \frac{\frac{1}{A} \times 7}{36} \Rightarrow \frac{1}{\lambda} = \frac{7}{36A} \Rightarrow \lambda = \frac{36A}{7}$$

Question174

The electron in the hydrogen atom undergoes transition from higher orbitals to orbital of radius 211.6pm. This transition is associated with :
[Online April 9, 2017]

Options:

A. Lyman series

B. Balmer series

C. Paschen series

D. Brackett serics

Answer: B

Solution:

$$r = 0.529 \times \frac{n^2}{Z}Å$$

$$r = 211.6pm = 2.11Å$$

$$\Rightarrow 0.529 \times \frac{n^2}{Z} = 2.11Å$$

$n = 2$ (Balmer series)

Question175

A stream of electrons from a heated filaments was passed two charged plates kept at a potential difference V esu. If 'e' and m are charge and mass of an electron, respectively, then the value of h/λ (where λ is wavelength associated with electron wave) is given by: [2016]

Options:

A. $\sqrt{meV}$

B. $\sqrt{2meV}$

C. meV

D. 2meV

Answer: B

Solution:

As electron of charge 'e' is passed through 'V' volt, kinetic energy of electron will be eV

Wavelength of electron wave $(\lambda) = \frac{h}{\sqrt{2m \cdot K.E}}$

$\lambda = \frac{h}{\sqrt{2meV}} \quad \therefore \frac{h}{\lambda} = \sqrt{2meV}$

Question176

The total number of orbitals associated with the principal quantum number 5 is:
[Online April 9,2016]

Options:

A. 20

B. 25

C. 10

D. 5

Answer: B

Solution:

Number of orbitals in a shell $= n^2 = (5)^2 = 25.$

Question177

Which of the following is the energy of a possible excited state of hydrogen? [2015]

Options:

A. -3.4eV

B. $+6.8\text{eV}$

C. $+13.6\text{eV}$

D. -6.8eV

Answer: A

Solution:

Total energy $= -\frac{13.6}{n^2}Z^2\text{eV}$
where $n = 2, 3, 4....$
Putting $n = 2$
$E_T = -\frac{13.6}{4} = -3.4\text{eV}$

Question178

At temperature T , the average kinetic energy of any particle is $\frac{3}{2}$kT . The de Broglie wavelength follows the order :
[Online April 11, 2015]

Options:

A. Visible photon > Thermal neutron > Thermal electron

B. Thermal proton > Thermal electron > Visible photon

C. Thermal proton > Visible photon > Thermal electron

D. Visible photon > Thermal electron > Thermal neutron

Answer: D

Solution:

Kinetic energy of any particle $= \frac{3}{2}kT$

Also $K.E. = \frac{1}{2}mv^2$

$\frac{1}{2}mv^2 = \frac{3}{2}kT \Rightarrow v^2 = \frac{3kT}{m}$

$v = \sqrt{\frac{3kT}{m}}$

de-broglie wavelength $= \lambda = \frac{h}{mv} = \frac{h}{m\sqrt{\frac{3kT}{m}}}$

$\lambda = \frac{h}{\sqrt{3kT\,m}}; \lambda \propto \frac{1}{\sqrt{m}}$

Mass of electron < mass of neutron λ (electron) >λ (neutron)

Question179

If the principal quantum number n = 6, the correct sequence of filling of electrons will be:
[Online April 10,2015]

Options:

A. $ns \rightarrow (n-2)f \rightarrow np \rightarrow (n-1)d$

B. $ns \rightarrow (n-2)f \rightarrow (n-1)d \rightarrow np$

C. $ns \rightarrow np \rightarrow (n-1)d \rightarrow (n-2)f$

D. $ns \rightarrow (n-1)d \rightarrow (n-2)f \rightarrow np$

Answer: B

Solution:

According to Aufbau principle, the sequence of filling electrons in sixth period is
$6s - 4f - 5d - 6p$ i.e., $(ns) \rightarrow (n-2)f \rightarrow (n-1)d \rightarrow np$

Question180

If λ_0 and λ be threshold wavelength and wavelength of incident light, the velocity of photoelectron ejected from the metal surface is: [Online April 11,2014]

Options:

A. $\sqrt{\frac{2h}{m}(\lambda_0 - \lambda)}$

B. $\sqrt{\frac{2hc}{m}(\lambda_0 - \lambda)}$

C. $\sqrt{\frac{2hc}{m}\left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0}\right)}$

D. $\sqrt{\frac{2h}{m}\left(\frac{1}{\lambda_0} - \frac{1}{\lambda}\right)}$

Answer: C

Solution:

The kinetic energy of the ejected electron is given by the equation
$hv = hv_0 + \frac{1}{2}mv^2 \quad \because v = \frac{c}{\lambda}$
or $\frac{hc}{\lambda} = \frac{hc}{\lambda_0} + \frac{1}{2}mv^2$

$$\frac{1}{2}mv^2 = \frac{hc}{\lambda} - \frac{hc}{\lambda_0} = hc\left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0}\right)$$

$$\therefore \; v^2 = \frac{2hc}{m}\left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0}\right)$$

or $v = \sqrt{\frac{2hc}{m}\left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0}\right)}$

Question181

The energy of an electron in first Bohr orbit of H-atom is 13.6eV . The energy value of electron in the excited state of Li^{2+} is: [Online April 9, 2014]

Options:

A. −27.2eV

B. 30.6eV

C. −30.6eV

D. 27.2eV

Answer: C

Solution:

For Li^{2+} ion

$$E = -13.6 \times \frac{Z^2}{n^2} eV = -13.6 \times \frac{(3)^2}{(2)^2}$$

$$= \frac{-13.6 \times 9}{4} = -30.6eV$$

Question182

Based on the equation:

[Online April 11,2014] $\Delta E = -2.0 \times 10^{-18} J \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$ the wavelength of the light that must be absorbed to excite hydrogen electron from level n = 1 to level n = 2 will be:
($h = 6.625 \times 10^{-34}$J s, $C = 3 \times 10^8 ms^{-1}$)
[Online April 11, 2014]

Options:

A. 1.325×10^{-7}m

B. 1.325×10^{-10}m

C. 2.650×10^{-7}m

D. 5.300×10^{-10}m

Answer: A

Solution:

$$\Delta E = -2.0 \times 10^{-18} \times \left(\frac{1}{2^2} - \frac{1}{1^2} \right)$$
$$= -2.0 \times 10^{-18} \times \frac{-3}{4}$$
$$= 1.5 \times 10^{-18} J$$
$$\Delta E = \frac{hc}{\lambda}$$
$$\lambda = \frac{hc}{\Delta E} = \frac{6.625 \times 10^{-34} js \times 3 \times 10^8 ms^{-1}}{1.5 \times 10^{-18} J}$$
$$= 1.325 \times 10^{-7} m$$

Question183

If m and e are the mass and charge of the revolving electron in the orbit of radius r for hydrogen atom, the total energy of the revolving electron will be:
[Online April 12,2014]

Options:

A. $\frac{1}{2} \frac{e^2}{r}$

B. $-\frac{e^2}{r}$

C. $\frac{me^2}{r}$

D. $-\frac{1}{2} \frac{e^2}{r}$

Answer: D

Solution:

Total energy of a revolving electron is the sum of its kinetic and potential energy.
Total energy $= K.E. + P.E$
$= \frac{e^2}{2r} + \left(-\frac{e^2}{r}\right); \ = -\frac{e^2}{2r}$

Question184

Excited hydrogen atom emits light in the ultraviolet region at 2.47×10^{15} Hz. With this frequency, the energy of a single photon is: ($h = 6.63 \times 10^{-34}$ J s)
[Online April 12, 2014]

Options:

A. 8.041×10^{-40} J

B. 2.680×10^{-19} J

C. 1.640×10^{-18} J

D. 6.111×10^{-17} J

Answer: C

Solution:

$$
\begin{aligned}
E &= hv \\
&= 6.63 \times 10^{-34} \times 2.47 \times 10^{15} \\
&= 1.640 \times 10^{-18} J
\end{aligned}
$$

Question185

Ionization cnergy of gaseous N a atoms is 495.5kJ mol $^{-1}$. The lowest possible frequency of light that ionizes a sodium atom is ($h = 6.626 \times 10^{-34}$ J s, $N_A = 6.022 \times 10^{23}$ mol $^{-1}$) [Online April 19,2014]

Options:

A. $7.50 \times 10^{4} s^{-1}$

B. $4.76 \times 10^{14} s^{-1}$

C. $3.15 \times 10^{15} s^{-1}$

D. $1.24 \times 10^{15} s^{-1}$

Answer: D

Solution:

$$
\text{Energy } = N_A hv
$$

$$
495.5 = 6.023 \times 10^{23} \times 6.6 \times 10^{-34} \times v
$$

$$
v = \frac{495.5 \times 10^{3} J}{6.023 \times 10^{23} \times 6.6 \times 10^{-34}} = 12.4 \times 10^{14}
$$

$$
= 1.24 \times 10^{15} s^{-1}
$$

Question186

The de-Broglie wavelength of a particle of mass 6.63g moving with a velocity of 100ms $^{-1}$ is: [Online April 12, 2014]

Options:

A. 10^{-33}m

B. 10^{-35}m

C. 10^{-31}m

D. 10^{-25}m

Answer: A

Solution:

de $(\lambda) = \frac{h}{mv}$

$= \frac{6.63 \times 10^{-34} \text{J s}}{6.63 \times 10^{-3} \text{kg} \times 100 \text{m/s}} = 10^{-33}\text{m}$

Question187

The correct set of four quantum numbers for the valence electrons of rubidium atom (Z = 37) is:
[2014]

Options:

A. $5, 0, 0, +\frac{1}{2}$

B. $5, 1, 0, +\frac{1}{2}$

C. $5, 1, 1, +\frac{1}{2}$

D. $5, 0, 1, +\frac{1}{2}$

Answer: A

Solution:

The electronic configuration of Rubidium $(\text{Rb} = 37)$ is
$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 5s^1$

Since last electron enters in 5s orbital
Hence $n = 5, l = 0, m = 0, s = \pm \frac{1}{2}$

Question188

Energy of an electron is given by $E = -2.178 \times 10^{-18} J \left(\frac{Z^2}{n^2} \right)$

Wavelength of light required to excite an electron in an hydrogen atom from level n = 1 to n = 2 will be:
$(h = 6.62 \times 10^{-34} J\,s$ and $c = 3.0 \times 10^{8} ms^{-1})$
[2013]

Options:

A. $1.214 \times 10^{-7} m$

B. $2.816 \times 10^{-7} m$

C. $6.500 \times 10^{-7} m$

D. $8.500 \times 10^{-7} m$

Answer: A

Solution:

$\Delta E = 2.178 \times 10^{-18} \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = \frac{hc}{\lambda}$
$\Rightarrow 2.178 \times 10^{-18} \times \frac{3}{4} = \frac{hc}{\lambda}$
$= \frac{6.62 \times 10^{-34} \times 3 \times 10^{8}}{\lambda}$
$\lambda = \frac{6.62 \times 10^{-34} \times 3 \times 10^{8} \times 4}{2.178 \times 10^{-18} \times 3}$
$= 1.214 \times 10^{-7} m$

Question189

The wave number of the first emission line in the Balmer series of H-Spectrum is:
(R = Rydberg constant):
[Online April 22, 2013]

Options:

A. $\frac{5}{36}R$

B. $\frac{9}{400}R$

C. $\frac{7}{6}R$

D. $\frac{3}{4}R$

Answer: A

Solution:

$$\bar{v} = RZ^2\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$
$$= R\left(\frac{1}{4} - \frac{1}{9}\right) = \frac{5R}{36}$$

Question190

The de Broglie wavelength of a car of mass 1000kg and velocity 36 km/hr is :
[Online April 23, 2013]

Options:

A. 6.626×10^{-34}m

B. 6.626×10^{-38}m

C. 6.626×10^{-31}m

D. 6.626×10^{-30}m

Answer: B

Solution:

$\lambda = \dfrac{h}{mv}$

$h = 6, 6 \times 10^{-34} J\ s$

$m = 1000 kg$

$v = 36 km/hr = \dfrac{36 \times 10^3}{60 \times 60} m/sec = 10 m/sec$

$\therefore\ \lambda = \dfrac{6.6 \times 10^{-34}}{10^3 \times 10} = 6.6 \times 10^{-38} m$

Question191

In an atom how many orbital(s) will have the quantum numbers; $n = 3, l = 2$ and $m_l = +2$? [Online April 9, 2013]

Options:

A. 5

B. 3

C. 1

D. 7

Answer: C

Solution:

$n = 3, l = 2$ means $3d$ orbital

+2	+1	0	−1	−2

i.e. in an atom only one orbital can have the value $m_l = +2$

Question192

Given
(A) $n = 5, m = +1$
(B) $n = 2, \ell = 1, m_f = -1, m_s = -1/2$
The maximum number of electron(s) in an atom that can have the quantum numbers as given in (A) and (B) are respectively:
[Online April 25, 2013]

Options:

A. 25 and 1

B. 8 and 1

C. 2 and 4

D. 4 and 1

Answer: B

Solution:

(i) $n = 5$ means $l = 0, 1, 2, 3, 4$
since $m = +1$
hence total no. of electrons will be
$= 0(\text{ from } s) + 2(\text{ from } p) + 2(\text{ from } d) + 2(\text{ from } f) + 2(\text{ from } g)$
$= 0 + 2 + 2 + 2 + 2 = 8$
(ii) $n = 2, l = 1, m_l = -1, m_s = -1/2$ represent $2p$ orbital with one electron.

Question193

The limiting line in Balmer series will have a frequency of (Rydberg constant, $R_\infty = 3.29 \times 10^{15}$ cycles/s)
[Online May 7,2012]

Options:

A. $8.22 \times 10^{14} s^{-1}$

B. $3.29 \times 10^{15} s^{-1}$

C. $3.65 \times 10^{14} s^{-1}$

D. $5.26 \times 10^{13} s^{-1}$

Answer: A

Solution:

$$v = R\left(\frac{1}{n_1^{2}} - \frac{1}{n_2^{2}}\right) Hz$$
$$v = 3.29 \times 10^{15}\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right)$$
$$= 8.22 \times 10^{14} s^{-1}$$

Question194

If the radius of first orbit of H atom is a_0, the de-Broglie wavelength of an electron in the third orbit is [Online May 12, 2012]

Options:

A. $4\pi a_0$

B. $8\pi a_0$

C. $6\pi a_0$

D. $2\pi a_0$

Answer: C

Solution:

$$r_n = a_0 n^2$$
$$r = a_0 \times (3)^2 = 9a_0$$
$$mvr = \frac{nh}{2\pi}; \ mv = \frac{nh}{2\pi r} = \frac{3h}{2\pi \times 9a_0} = \frac{h}{6\pi a_0}$$
$$\lambda = \frac{h}{mv} = \frac{n}{h} \times 6\pi a_0 = 6\pi a_0$$

Question195

If the kinetic energy of an electron is increased four times, the wavelength of the de-Broglie wave associated with it would become [Online May 19, 2012]

Options:

A. one fourth

B. half

C. four times

D. two times

Answer: B

Solution:

de − Broglie wavelength is given by:

$\lambda = \frac{h}{mv} \ldots$ (i)

$K.E. = \frac{1}{2}mv^2$

$v^2 = \frac{2K \cdot E}{m}$

$v = \sqrt{\frac{2K \cdot E}{m}}$

Substituting this in equation (i)

$\lambda = \frac{h}{m}\sqrt{\frac{m}{2K \cdot E}}$

$\lambda = h\sqrt{\frac{1}{2m(K \cdot E)}}$

i.e. $\lambda \propto \frac{1}{\sqrt{K \cdot E}}$

∴ when KE become 4 times wavelength become $1/2$.

Question196

The electrons identified by quantum numbers n and ℓ :
(A) n = 4, l = 1
(B) n = 4, l = 0

(C) n = 3, l = 2
(D) n = 3, l = 1
can be placed in order of increasing energy as:
[2012]

Options:

A. (C) < (D) < (B) < (A)

B. (D) < (B) < (C) < (A)

C. (B) < (D) < (A) < (C)

D. (A) < (C) < (B) < (D)

Answer: B

Solution:

(A)4p (B) 4s
(C) 3d (D) 3p Accroding to Bohr Bury's $(n+l)$ rule, increasing order of energy (D) < (B) < (C) < (A).
Note: If the two orbitals have same value of $(n+l)$ then the orbital with lower value of n will be filled first.

Question197

The increasing order of the ionic radii of the given iscelectronic species is :
[2012]

Options:

A. Cl , Ca^{2+}, K^{+}, S^{2}

B. S^{2}, Cl^{-}, Ca^{2+}, K^{+}

C. Ca^{2+}, K^{+}, Cl^{-}, S^{2}

D. K^{+}, S^{2}, Ca^{2+}, Cl^{-}

Answer: C

Solution:

Among isoelectronic species ionic radii increases as the negatives charge increases.
Order of ionic radii $Ca^{2+} < K^{+} < Cl^{-} < S^{2-}$
The number of electrons remains the same but nuclear charge increases with increase in the atomic number causing decrease in size.

Question198

The following sets of quantum numbers represent four electrons in an atom.
(i) n = 4, l = 1
(ii) n = 4, l = 0
(iii) n = 3, l = 2
(iv) n = 3, l = 1
The sequence representing increasing order of energy, is
[Online May 26, 2012]

Options:

A. (iii) < (i) < (iv) < (ii)

B. (iv < (ii) < (iii) < (i)

C. (i) < (iii) < (ii) < (iv)

D. (ii) < (iv) < (i) < (iii)

Answer: B

Solution:

(i) 4p (ii) 4s
(iii) 3d (iv) 3p
According to Bohr Bury's $(n+1)$ rule, increasing order of energy will be (iv) <(ii) < (iii) < (i).
Note: If the two orbitals have same value of $(n+1)$ then the orbital with lower value of n will be filled first.

Question199

The frequency of light emitted for the transition n = 4 to n = 2 of the He^+ is equal to the transition in H atom corresponding to which of the following ?
[2011RS]

Options:

A. n = 2 to n = 1

B. n = 3 to n = 2

C. n = 4 to n = 3

D. n = 3 to n = 1

Answer: A

Solution:

For He^+,

$$v = RZ^2\left(\frac{1}{2^2} - \frac{1}{4^2}\right) Hz$$

For H,

$$v = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) Hz$$

For same frequency,

$$Z^2\left(\frac{1}{2^2} - \frac{1}{4^2}\right) = \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

Since, $Z = 2$

$$\therefore \quad \frac{1}{n_1^2} - \frac{1}{n_2^2} = \frac{1}{1^2} - \frac{1}{2^2}$$

$\therefore n_1 = 1 \; \& \; n_2 = 2$

Question200

The energy required to break one mole of Cl − Cl bonds in Cl_2 is $242 kJ\ mol^{-1}$. The longest wavelength of light capable of breaking a single Cl − Cl bond is

$(c = 3 \times 10^8 ms^{-1}$ and $N_A = 6.02 \times 10^{23} mol^{-1})$
[2010]

Options:

A. 594nm

B. 640nm

C. 700nm

D. 494nm

Answer: D

Solution:

Energyrequired to break single $Cl - Cl$ bond

$$= \frac{242 \times 10^3}{6.023 \times 10^{23}} = \frac{hc}{\lambda}$$

$$= \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{\lambda}$$

$$\therefore \lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8 \times 6.023 \times 10^{23}}{242 \times 10^3}$$

$$= 0.4947 \times 10^{-6} m = 494.7 nm$$

Question201

Ionisation energy of He^+ is $19.6 \times 10^{-18} J\ atom^{-1}$. The energy of the first stationary state (n = 1) of Li^{2+} is
[2010]

Options:

A. $4.41 \times 10^{-16} J\ atom^{-1}$

B. $-4.41 \times 10^{-17} J\ atom^{-1}$

C. $-2.2 \times 10^{-15} J\ atom^{-1}$

D. $8.82 \times 10^{-17} J\ atom^{-1}$

Answer: B

Solution:

I. $E = \frac{Z^2}{n^2} \times 13.6 eV$

or $\frac{I_1}{I_2} = \frac{Z_1^2}{n_1^2} \times \frac{n_2^2}{Z_2^2}$

Given $I_1 = -19.6 \times 10^{-18} J/atom$, $Z_1 = 2$,
$n_1 = 1$, $Z_2 = 3$ and $n_2 = 1$

Substituting these values in equation (ii).

$-\frac{19.6 \times 10^{-18}}{I_2} = \frac{4}{1} \times \frac{1}{9}$

or $I_2 = -19.6 \times 10^{-18} \times \frac{9}{4}$

$= -4.41 \times 10^{-17} J/atom$

Question202

Calculate the wavelength (in nanometer) associated with a proton moving at $1.0 \times 10^3 ms^{-1}$. (Mass of proton $= 1.67 \times 10^{-27} kg$ and $h = 6.63 \times 10^{-34} J s$) [2009]

Options:

A. 0.40nm

B. 2.5nm

C. 14.0nm

D. 0.32nm

Answer: A

Solution:

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{1.67 \times 10^{-27} \times 1 \times 10^3}$$

$$= 3.97 \times 10^{-10} m = 0.397 nm.$$

Question203

In an atom, an electron is moving with a speed of 600m / s with an accuracy of 0.005%. Certainity with which the position of the electron can be located is ($h = 6.6 \times 10^{-34}$ kgm^2s^{-1}, mass of electron, $e_m = 9.1 \times 10^{-31} kg$):
[2009]

Options:

A. $5.10 \times 10^{-3} m$

B. $1.92 \times 10^{-3} m$

C. $3.84 \times 10^{-3} m$

D. $1.52 \times 10^{-4} m$

Answer: B

Solution:

According to Heisenberg uncertainty principle.

$\Delta xm \, \Delta v = \frac{h}{4\pi}; \Delta x = \frac{h}{4\pi m \, \Delta v}$

Here $\Delta v = \frac{600 \times 0.005}{100} = 0.03 m / s$

So,

$$\Delta x = \frac{6.6 \times 10^{-34}}{4 \times 3.14 \times 9.1 \times 10^{-31} \times 0.03}$$
$$= 1.92 \times 10^{-3} m$$

Question204

The ionization enthalpy of hydrogen atom is $1.312 \times 10^6 J \ mol^{-1}$. The energy required to excite the electron in the atom from n = 1 to n = 2 is
[2008]

Options:

A. $8.51 \times 10^5 J\ mol^{-1}$

B. $6.56 \times 10^5 J\ mol^{-1}$

C. $7.56 \times 10^5 J\ mol^{-1}$

D. $9.84 \times 10^5 J\ mol^{-1}$

Answer: D

Solution:

(ΔE), The energy required to excite an electron in an atom of hydrogen from $n = 1$ to $n = 2$ is ΔE (difference in energy E_2 and E_1)

Values of E_2 and E_1 are,

$$E_2 = \frac{-1.312 \times 10^6 \times (1)^2}{(2)^2} = -3.28 \times 10^5 J\ mol^{-1}$$

ΔE is given by the relation,

$E_1 = -1.312 \times 10^6 J\ mol^{-1}$

$\therefore \Delta E = E_2 - E_1 = [-3.28 \times 10^5] - [-1.312 \times 10^6] J\ mol^{-1}$

$= (-3.28 \times 10^5 + 1.312 \times 10^6) J\ mol^{-1}$

$= 9.84 \times 10^5 J\ mol^{-1}$

Question205

Which one of the following constitutes a group of the isoelectronic species?
[2008]

Options:

A. $C_2^{2-}, O_2^{-}, CO, NO$

B. $NO^{+}, C_2^{2-}, CN^{-}, N_2$

C. $CN^{-}, N_2, O_2^{2-}, C_2^{2-}$

D. N_2, O_2^{-}, NO^{+}, CO

Answer: B

Solution:

Species having same number of electrons are isoelectronic. On calculating the number of electrons in each species given here, we get.

$CN^{-}(6+7+1=14)$; $N_2(7+7=14)$

$O_2^{2-}(8+8+2=18)$; $C_2^{2}(6+6+2=14)$

$O_2^{-}(8+8+1=17)$; $NO^{+}(7+8-1=14)$

$CO(6+8=14)$; $NO(7+8=15)$

From the above calculation we find that all the species listed in choice (b) have 14 electrons each so it is the correct answer.

Question206

Which of the following sets of quantum numbers represents the highest energy of an atom?
[2007]

Options:

A. $n=3, l=0, m=0, s=+\frac{1}{2}$

B. $n=3, l=1, m=1, s=+\frac{1}{2}$

C. $n=3, l=2, m=1, s=+\frac{1}{2}$

D. $n=4, l=0, m=0, s=+\frac{1}{2}$.

Answer: C

Solution:

(a) $n=3, l=0$ means $3s$ -orbital and $n+l=3$
(b) $n=3, l=1$ means $3p$ -orbital $n+l=4$
(c) $n=3, l=2$ means $3d$ -orbital $n+l=5$
(d) $n=4, l=0$ means $4s$ -orbital $n+l=4$
Increasing order of energy among these orbitals is
$3s<3p<4s<3d$
$\therefore 3d$ has highest energy.

Question207

According to Bohr's theory, the angular momentum of an electron in 5^{th} orbit is
[2006]

Options:

A. $10h/\pi$

B. $2.5h/\pi$

C. $25h/\pi$

D. $1.0h/\pi$

Answer: B

Solution:

Angular momentum of an electron in n^{th} orbital is given by, $mvr = \frac{nh}{2\pi}$
For $n = 5$, we have
Angular momentum of electron $= \frac{5h}{2\pi} = \frac{2.5h}{\pi}$

Question208

Uncertainty in the position of an electron (mass $= 9.1 \times 10^{-31}$kg) moving with a velocity $300ms^{-1}$, accurate upto 0.001% will be ($h = 6.63 \times 10^{-34}$J s)
[2006]

Options:

A. $1.92 \times 10^{-2}m$

B. $3.84 \times 10^{-2}m$

C. 19.2×10^{-2}m

D. 5.76×10^{-2}m

Answer: A

Solution:

Given $m = 9.1 \times 10^{-31}$kg
$h = 6.6 \times 10^{-34}$J s
$\Delta v = \dfrac{300 \times .001}{100} = 0.003ms^{-1}$
From Heisenberg's uncertainity principle
$$\Delta x = \frac{6.62 \times 10^{-34}}{4 \times 3.14 \times 0.003 \times 9.1 \times 10^{-31}}$$
$= 1.92 \times 10^{-2}$m

Question209

Which one of the following sets of ions represents a collection of isoelectronic species?
[2006]

Options:

A. $N^{3-}, O^{2-}, F^{-}, S^{2}$

B. $Li^{+}, Na^{+}, Mg^{2+}, Ca^{2+}$

C. $K^{+}, Cl^{-}, Ca^{2+}, Sc^{3+}$

D. $Ba^{2+}, Sr^{2+}, K^{+}, Ca^{2+}$

Answer: C

Solution:

(a) $N^{3-} = 7 + 3 = 10e^{-}, O^{2-} \longrightarrow 8 + 2 = 10c^{-}$
$F^{-} = 9 + 1 = 10c^{-}, S^{2-} \longrightarrow 16 + 2 = 18e^{-}$
(not isoelectronic)
(b) $Li^{+} = 3 - 1 = 2e, Na^{+} = 11 - 1 = 10e^{-}$
$Mg^{2+} = 12 - 2 = 10e^{-}$

$Ca^{2+} = 20 - 2 = 18c^-$
(not isoelectronic)
(c) $K^+ = 19 - 1 = 18e^-$, $Cl^- = 17 + 1 = 18e^-$
$Ca^{2+} = 20 - 2 = 18e^-$, $Sc^{3+} = 21 - 3 = 18e^-$
(isoelectronic)
(d) $Ba^{2+} = 56 - 2 = 54e^-$, $Sr^{2+} = 38 - 2 = 36e^-$
$K^+ = 19 - 1 = 18e^-$, $Ca^{2+} = 20 - 2 = 18e^-$
(not isoelectronic)

Question210

Of the following sets which one does NOT contain isoelectronic species?
[2005]

Options:

A. $BO_3^{3-}, CO_3^{2-}, NO_3^-$

B. $SO_3^{2-}, CO_3^{2-}, NO_3^-$

C. CN^-, N_2, C_2^{2-}

D. $PO_4^{3-}, SO_4^{2-}, ClO_4^-$

Answer: B

Question211

In a multi-electron atom, which of the following orbitals described by the three quantum members will have the same energy in the absence of magnetic and electric fields?
(A) n = 1, l = 0, m = 0
(B) n = 2, l = 0, m = 0
(C) n = 2, l = 1, m = 1

(D) $n = 3, l = 2, m = 1$
(E) $n = 3, l = 2, m = 0$
[2005]

Options:

A. (D) and (E)

B. (C) and (D)

C. (B) and (C)

D. (A) and (B)

Answer: A

Solution:

The energy of an orbital is given by $(n+1)$ rule. $(n+1)$ value for option (E) and (D) is $(3+2)=5$ hence they will have same energy, since their n values are also same.

Question212

The wavelength of the radiation emitted, when in a hydrogen atom electron falls from infinity to stationary state 1, would be (Rydberg constant $= 1.097 \times 10^7 m^{-1}$)

[2004]

Options:

A. 406nm

B. 192nm

C. 91nm

D. 9.1×10^{-8}nm

Answer: C

Solution:

$$\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$
$$\frac{1}{\lambda} = 1.097 \times 10^7\left(\frac{1}{1} - \frac{1}{\infty}\right) = 1.097 \times 10^7$$
$$\lambda = 91.15 \times 10^{-9} m \approx 91 nm$$

Question213

Which of the following sets of quantum numbers is correct for an electron in 4f orbital ?
[2004]

Options:

A. $n = 4, l = 3, m = +1, s = +1/2$

B. $n = 4, l = 4, m = -4, s = -1/2$

C. $n = 4, l = 3, m = +4, s = +1/2$

D. $n = 3, l = 2, m = -2, s = +1/2$

Answer: A

Solution:

The possible quantum numbers for $4f$ electron are
$n = 4, l = 3, m = -3, -2 - 1, 0, 1, 2, 3$ and $s = \pm \frac{1}{2}$ Of various possiblities only option (a) is possible.

Question214

Consider the ground state of Cr atom (X = 24). The number of electrons with the azimuthal quantum numbers, l = 1 and 2 are, respectively
[2004]

Options:

A. 16 and 4

B. 12 and 5

C. 12 and 4

D. 16 and 5

Answer: B

Solution:

Electronic configuration of Cr atom $(Z = 24)$
$= 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$
when $l = 1$, p -subshell,
Numbers of electrons $= 12$
when $l = 2$, d -subshell,
Numbers of electrons $= 5$

Question215

Which one of the following sets of ions represents the collection of isoelectronic species?
(Atomic nos. :
$F = 9, Cl = 17, Na = 11, Mg = 12, Al = 13, K = 19, Ca = 20, Sc = 21$)
[2004]

Options:

A. $K^+, Cl^-, Mg^{2+}, Sc^{3+}$

B. $Na^+, Ca^{2+}, Sc^{3+}, F^-$

C. $K^+, Ca^{2+}, Sc^{3+}, Cl$

D. $Na^+, Mg^{2+}, Al^{3+}, Cl$

Answer: C

Solution:

${}_{19}K^+, {}_{20}Ca^{2+}, {}_{21}Sc^{3+}, {}_{17}Cl$
each contains 18 electrons.

Question216

The orbital angular momentum for an electron revolving in an orbit is given by $\sqrt{l(l+1)} \cdot \frac{h}{2\pi}$. This momentum for an s -electron will be given by
[2003]

Options:

A. zero

B. $\frac{h}{2\pi}$

C. $\sqrt{2} \cdot \frac{h}{2\pi}$

D. $+\frac{1}{2} \cdot \frac{h}{2\pi}$

Answer: A

Solution:

For s -electron, $l = 0$
∴ Orbital angular momentum
$= \sqrt{0(0+1)} \frac{h}{2\pi} = 0$

Question217

In Bohr series of lines of hydrogen spectrum, the third line from the red end corresponds to which one of the following inter-orbit jumps of the electron for Bohr orbits in an atom of hydrogen
[2003]

Options:

A. $5 \rightarrow 2$

B. $4 \rightarrow 1$

C. $2 \rightarrow 5$

D. $3 \rightarrow 2$

Answer: A

Solution:

The lines falling in the visible region comprise Balmer series. Hence the third line from red would be $n_1 = 2, n_2 = 5$ i.e. $5 \rightarrow 2$

Question218

The de Broglie wavelength of a tennis ball of mass 60g moving with a velocity of 10 metres per second is approximately Planck's constant, h = 6.63×10^{-34}J s [2003]

Options:

A. 10^{-31} metres

B. 10^{-16} metres

C. 10^{-25} metres

D. 10^{-33} metres

Answer: D

Solution:

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{60 \times 10^{-3} \times 10}$$
$$= 1.105 \times 10^{-33} \approx 10^{-33}\text{m}$$

Question219

The number of d -electrons retained in Fe^{2+} (At. no. of Fe=26) ion is [2003]

Options:

A. 4

B. 5

C. 6

D. 3

Answer: C

Solution:

$Fe^{++}(26 - 2 = 24) = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^0 3d^6$ hence no. of d electrons retained is 6 .
[Two $4s$ electron are removed]

Question220

Which one of the following groupings represents a collection of isoelectronic species?(At. nos. : Cs : 55, Br : 35)
[2003]

Options:

A. N^{3-}, F^-, Na^+

B. Be, Al^{3+}, Cl^-

C. Ca^{2+}, Cs^+, Br

D. Na^+, Ca^{2+}, Mg^{2+}

Answer: A

Solution:

N^{3-}, F^- and Na^+ contain 10 electrons each.

Question221

In a hydrogen atom, if energy of an electron in ground state is 13.6 . eV , then that in the 2 nd excited state is [2002]

Options:

A. 1.51eV

B. 3.4eV

C. 6.04eV

D. 13.6eV .

Answer: A

Solution:

2 nd excited state will be the 3 rd energy level

$E_n = \frac{13.6}{n^2} eV$

or $E_3 = \frac{13.6}{9} eV = 1.51 eV$.

Question222

Uncertainty in position of a minute particle of mass 25g in space is 10^{-5}m. What is the uncertainty in its velocity (in ms^{-1})?($h = 6.6 \times 10^{-34} J\ s$) [2002]

Options:

A. 2.1×10^{-34}

B. 0.5×10^{-34}

C. 2.1×10^{-28}

D. 0.5×10^{-23}.

Answer: C

Solution:

$\Delta x \cdot \Delta p = \frac{h}{4\pi}$; or $\Delta x, m, \Delta v = \frac{h}{4\pi}$

$\therefore \Delta v = \frac{6.62 \times 10^{-34}}{4 \times 3.14 \times 0.025 \times 10^{-5}} = 2.1 \times 10^{-28} ms^{-1}$
